



Addis Ababa University

College of Natural and Computational Sciences

Department of Computer Science

Ethiopian University Entrance Exam Preparation Platform

COSC 4122: Final Project I

By:-

- | | |
|--------------------|-------------|
| 1. Aman Atalay | UGR/4364/15 |
| 2. Asnake Mengesha | UGR/9465/15 |
| 3. Daniel Shitaye | NSR/9066/14 |
| 4. Fraol Dereje | UGR/6955/15 |

Advisor: Surafiel Habib

January 2025

Declaration

This is to declare that this project work is done under the supervision of Instructor **Mr. Surafiel Habib** and having the title “**Entrance Exam Preparation Platform**” is the sole contribution of **Aman Atalay, Asnake Mengesha, Daniel Shitaye, and Fraol Dereje**.

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Date: _____

Group Members:

Full Name	Signature
1. Aman Atalay	_____
2. Asnake Mengesha	_____
3. Daniel Shitaye	_____
4. Fraol Dereje	_____

Certificate

I certify that this BSc final project entitled “**Entrance Exam Preparation Platform**” by **Aman Atalay, Asnake Mengesha, Daniel Shitaye, and Fraol Dereje** is approved by me for submission. I certify further that, to the best of my knowledge, the report represents work carried out by the students.

Date

Name and Signature of Supervisor

Acknowledgment

First and foremost, we would like to thank Almighty God for giving us the strength, guidance, and patience to successfully complete this Final Project I.

We would like to express our sincere and heartfelt gratitude to our advisor and supervisor, **Mr. Surafiel Habib**, for his continuous guidance, encouragement, and support throughout the project work.

We are also thankful to our instructors and the **Department of computer science (AAU)** for providing the necessary knowledge, resources, and academic environment that contributed to the successful completion of this project.

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CHAPTER THREE-SYSTEM DESIGN

CHAPTER ONE - INTRODUCTION

1.1 Overview

The Ethiopian higher education university entrance examination is a highly competitive exam given each year in order to admit students for universities.

In recent years the Ethiopian higher education university entrance examination pass rate is becoming as low as 3–8%. The rate implies that of 100 students who take the entrance examination only 3-8 students are scoring the passing mark to join universities. ¹

The lack of organized study materials and exam preparation supporting tools, difficulty in understanding course content, burnout and frustration due to subject's content load and being unfamiliar with the structure as well as the content of the entrance examination can be the root causes.

The current product under development addresses the issues developing a platform that supports student's preparation and increases their chances of passing.

1.2 Statement of the Problem

The current approaches students follow seems unorganized, destructive and ultimately ineffective which results in scoring low. Although students try to utilize existing digital learning resources such as YouTube, searching and organizing contents can be so difficult and a waste of time. The following listed problems are the other core issues:

- Absence of structured and organized learning path that consist of concept explanation, set of working example problems, relevant exercise problems and their solved set that meets the course's objectives.
- Past year's university entrance examination problems and their respective solutions are not easily discoverable as a result familiarizing themselves with the structure and patterns of the examination becomes difficult.

¹ <https://shega.co/news/only-84-of-ethiopian-students-pass-university-entrance-exams>

- Lack of peer to peer collaborative learning leads to burnout, frustration and motivation.

Therefore, the current product solves the above stated core problems digitally by applying software engineering principles.

1.3 Objectives

1.3.1 General Objectives

Design and Implement an entrance exam preparation platform for Ethiopian students sitting to take the national entrance examination.

1.3.2 Specific Objectives

- Study the advantages and disadvantages of existing digital systems to identify the requirements for the proposed entrance exam preparation platform.
- Organize the identified requirements into functional and non-functional requirements for the proposed system.
- Describe the interactions within the proposed system and between the system and its environment.
- Design the proposed entrance exam preparation system according to the identified requirements.
- Implement the proposed system using a suitable and portable programming language.
- Test the proposed system with representative test cases and validate that the system meets the needs of the intended users which are students.

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1.4 Scope and Limitations of the Project

1.4.1 Scope

The Product is a mobile responsive web-based system that supports students sitting to take the Ethiopian higher education university entrance examination.

The system provides structured study resources, concept explanation with a set of working examples, curated list of interactive and solved problem sets, organized past year entrance examination problems, peer collaboration, admin and subject teacher management features.

1.4.2 Limitations

The platform is a fully web-based application; therefore internet access is a mandatory requirement to reap the full benefits.

Mobile Application version is out of scope. However, the backend supports easy integration of mobile application development tools such as flutter in order to build the mobile version in future.

1.5 System development methodology

The team applies object-oriented software development (OOP) approach and the waterfall software development model to deliver the final product.

1.5.1 Investigation (fact-finding methods)

- Interview with students sitting to take the national entrance examination on 2018 E.C in order to understand the underlying issues they are facing.
- Provide and collect a questionnaire in Google form format for current students preparing for the national entrance examination and students who have taken the exam previously for better problem understanding and solution suggestions based on their past experience and what they wish did better.

1.5.2 System development tools

- **Frontend Development:** React framework with Tailwind.
- **Backend Development:** SpringBoot framework Java.
- **Database Storage:** Mongo DB.
- **Code Development:** VS-Code Editor.
- **Code Collaboration and Version Controls:** Git and Github.
- **Deployment:** Virtual Private Server.
- **Document Preparation Tools:** Microsoft Word, UML, Drawio, Gantt-Chart.

1.6 Significance

Support students increase their exam preparation efficiency by minimizing course study duration, gain proper understanding and sufficient practices on complex topics on varies difficulty level while familiarizing themselves to actual exam problems, keeping them motivated to avoid burnout and frustration that comes with content coverage load these ultimately gives them an edge to score better results in the actual examination.

1.7 Beneficiaries

The product benefits Ethiopian students sitting to take the national entrance examination by providing them organized study resources and exam preparation paths.

1.8 Time Schedule (Gantt Chart)

The time schedule for entire components of the software engineering lifecycle.²

² <https://www.onlinegantt.com/#/>

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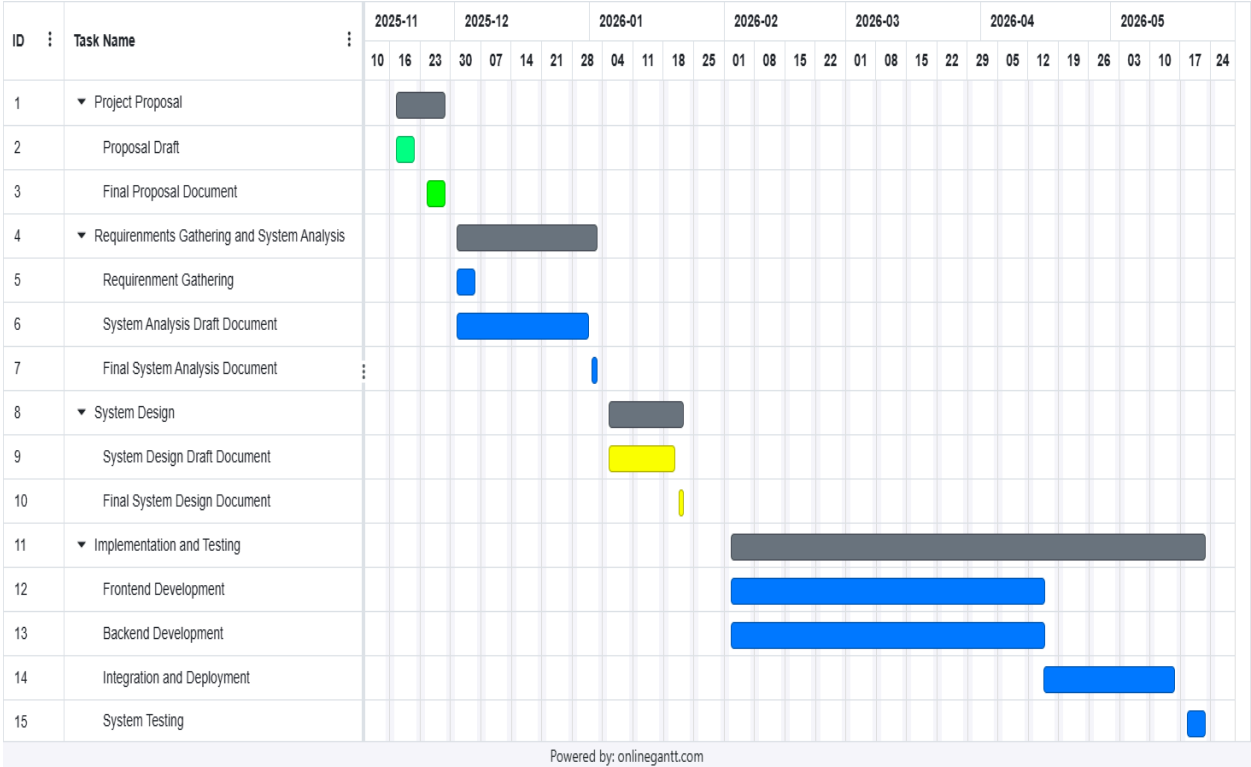


Figure 1-1 Project Time Schedule Gantt Chart

CHAPTER TWO – REQUIREMENT ANALYSIS

2.1 Introduction

This chapter describes the current system that students rely on when preparing for the national university entrance examination, such as reading textbooks, collecting and solving past years examination papers, watching random online videos, or studying through informal notes, and highlights the gaps and limitations of these methods.

It also explains the requirement-gathering process and presents the findings, the proposed solution, the functional and non-functional requirements, the system actors and their roles, and the diagrams that illustrate how the system operates and what it enables users to do.

2.2 Current System

Beyond standard classroom learning, students often demonstrate insufficient engagement in preparing for the national examination. In certain areas, poor instructional quality leads students to seek alternative study resources.

Their preparation typically involves reading textbooks, watching random online videos that are disorganized and not aligned with the curriculum, practicing exercise sets, and attempting a few previous-year exam papers. Nevertheless, these strategies often fail to adequately assess their performance, identify knowledge gaps, or ensure readiness for the examination.

2.2.1 Problems in the existing system

Students face several challenges in preparing for the national examination. They often rely on multiple study resources textbooks, online videos, and other materials chosen based on their learning style or to clarify difficult topics. However, these resources are frequently inconsistent, poorly organized, and not aligned with the subject objectives or the official exam syllabus. If students fail to understand a topic from textbooks, they may turn to other books or videos, but these alternatives often fail to meet the subject objectives. This unguided browsing wastes time, causes frustration, and reduces the efficiency of their preparation.

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Another key challenge is that students are often unaware of which chapters or topics carry more weight in the examination according to the syllabus. As a result, they cannot prioritize their study time effectively, leaving them unprepared for high-impact topics. Students also commonly review content without completing sufficient practice exercises or previous-year exam papers, limiting their ability to identify question patterns, evaluate their performance, and measure their readiness.

Although students are expected to revise subject content from Grade 9, Grade 10, Grade 11, and Grade 12, many postpone preparation until the final weeks before the exam. This inconsistent and unguided approach results in an overwhelming study load, wasted time, stress, reduced motivation, and ultimately, poor exam readiness.

2.3 Requirement Gathering

2.3.1 Requirement Gathering Methods

To collect insights from the target students, a questionnaire has been distributed across educational channels on Telegram. ³

Google Forms was used to design and administer the questionnaire; ensuring responses could be collected efficiently and systematically. ⁴

2.3.2 Results Found

Based on the responses collected, we gained an understanding of how students approach their preparation and identified the primary challenges they face when preparing for the national examination.

During the requirement-gathering phase, students were presented with a set of proposed requirements to address common issues and were asked to select the ones they considered most important.

³ https://t.me/Ethiopian_Minister_Of_Education/40536

⁴https://docs.google.com/forms/d/1iWa3xJLv2o0CRtDE8hUwnhPRTJSEvqT_NAZSL3_XMcI/edit

The aggregated selection data was then used to identify the system's functional requirements. In addition, further requirements were proposed based on the analysis of the proposed solution.

2.4 Proposed System

2.4.1 Overview

The proposed “**Entrance Exam Preparation Platform**” provides students with an organized and structured web-based platform for university entrance exam preparation. It offers access to curated learning resources, including videos; exercises aligned with subject objectives, quizzes, and solved previous-year exam problems. This ensures that students can learn effectively, even when textbooks alone may not be sufficient. Sometimes a video explanation can clarify a concept more clearly and help students master it.

Teachers can structure subjects by adding chapters and topics, manage content under each topic, and create quizzes, ensuring that all learning objectives are met. They can upload materials in video, text, and document formats, making content easily navigable for students.

Students can browse subject content, watch videos, complete exercises, take quizzes to evaluate understanding, and track their progress in each subject. They can also access previous-year exam problems to assess preparation, identify patterns, and build confidence for the exam. The system allows students to bookmark content for easy access, ask questions under each topic, and join peer community groups for collaborative learning. Students also receive notifications about subject updates and milestones.

Administrators can add new subjects, update subject information, assign teachers to subjects, and manage user accounts, including activating or deactivating students' and teachers' accounts. They can also monitor and address student-reported issues.

In conclusion, organized and curated resources that align with subject objectives, highlight chapter weightings based on the syllabus, and provide structured practice opportunities can help students optimize study time, track progress, and enhance comprehension and confidence

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for the examination. It balances student-centered learning with teacher and admin management, ensuring alignment with subject objectives and exam syllabus.

2.4.2 Functional Requirements

Functional requirements describe the functionalities that are to be expected from the system being designed.

The functional requirements of the proposed system are listed below:

- **FR-01:** The system shall allow students, teachers, and administrators to create or have a user account.
- **FR-02:** The system shall allow students, teachers, and administrators to view and update their account details.
- **FR-03:** The system shall provide a login option for students, teachers, and administrators with a registered account
- **FR-04:** The system shall provide a logout option for students, teachers, and administrators who are logged in.
- **FR-05:** The system shall allow students, teachers, and administrators to reset their passwords.
- **FR-06:** The system shall allow students to view and access all content uploaded under each subject.
- **FR-07:** The system shall allow students and teachers to access, search and filter previous-year exam problems by subject, chapter, by topic and exam year.
- **FR-08:** The system shall provide students with a numerical progress indicator for subject completion and notify them when they reach 25%, 50%, 75%, and 100% milestones.
- **FR-09:** The system shall display students' progress and results on the student's dashboard.
- **FR-10:** The system shall allow students to join peer community groups via Telegram.

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- **FR-11:** The system shall notify students via in-app notifications and email about subject updates.
- **FR-12:** The system shall allow students to navigate subjects and search for topics by title.
- **FR-13:** The system shall provide interactive validations when students attempt exercises, quizzes, or previous-year questions under each topic.
- **FR-14:** The system shall allow students to submit issues such as wrong answers, missing course materials, and related platform bugs.
- **FR-15:** The system shall allow students to bookmark content for easy access.
- **FR-16:** The system shall allow students to ask questions under each topic, and teachers can provide answers.
- **FR-17:** The system shall allow a teacher to structure a subject by adding chapters and topics
- **FR-18:** The system shall allow a teacher to add and manage content under each topic, including videos, concept pages, exercises, and previous-year exam questions.
- **FR-19:** The system shall allow teachers to create and manage quizzes under each topic.
- **FR-20:** The system shall allow the administrator to create a subject.
- **FR-21:** The system shall allow the administrator to update subject information.
- **FR-22:** The system shall allow the administrator to assign a teacher to a subject.
- **FR-23:** The system shall allow the administrator to manage roles and activate or deactivate students' and teachers' accounts.
- **FR-24:** The system shall display student reported issues such as wrong answers and missing subject content on the administrator and teacher dashboards and notify students of the outcomes.

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2.4.3 Non-Functional Requirement

Nonfunctional requirements describe aspects of the system that are not directly related to the functional behavior of the system. Nonfunctional requirements include a broad variety of requirements that apply to many different aspects of the system, from the system, from usability to performance [1].

The following sections outline the non-functional requirements of the proposed system:

2.4.3.1 User effects and Human factors

The system shall have a minimalistic, user-friendly interface, with clear and meaningful texts, buttons, form elements, page navigation, and validation messages, enabling users to perform tasks without confusion or overwhelm.

2.4.3.2 Documentation

The system shall provide complete and clear documentation of workflows, internal processes, and all development phases from requirement gathering and analysis to design, implementation, and testing enabling end users, developers, and stakeholders to understand its operation.

2.4.3.3 Hardware Consideration

The system shall run on Android or iOS mobile devices, as well as computers, provided the user has a compatible mobile or desktop browser, regardless of the operating system.

2.4.3.4 Performance Characteristics

The system shall ensure stability and performance under high traffic by handling concurrent user access, executing heavy tasks (e.g., database operations and email sending) asynchronously, and utilizing server-side caching for frequently requested data.

2.4.3.5 Error Handling and Extreme Conditions

The system shall validate and sanitize all end-user inputs, handle exceptions from internal processes (e.g., database operations and external API calls), and log errors to enable efficient bug identification and resolution.

2.4.3.6 Quality Issues

The system shall implement and validate all functional requirements to satisfy end-user needs and ensure product quality.

2.4.3.7 System Modification

The system shall be easily extendable to incorporate new functional requirements and enhance existing ones.

2.4.3.8 Physical Environment

The system shall operate on existing operating systems and be accessible through mobile and desktop browsers, given an active internet connection and a functioning server.

2.4.3.9 Security Issues

The system shall enforce role-based access control and secure user account management, including password hashing and encryption of sensitive user data.

The system shall restrict excessive or suspicious requests to prevent account breaches and mitigate system overload or other security attacks.

2.4.3.10 Resource Management

The system shall store only essential data in the database and return API responses containing only necessary data sets to optimize resource utilization.

2.5 Constraints / Pseudo Requirements

The system shall ensure fast and efficient responses, implement caching for frequently accessed data, comply with data privacy regulations, and follow industry-standard coding practices with proper documentation.

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2.6 System Model

2.6.1 Scenario

A scenario is a narrative description of what people do and experience as they try to make use of computer systems and applications [2].

The following tables present scenarios of the proposed system:

Table 2-0-1 Description of Create Account Scenario

SCENARIO ID	SC_001
SCENARIO NAME	Create Account
PARTICIPATING ACTORS	Student (Abebe)
FLOW OF EVENTS	1. Abebe opens the system's landing page. 2. The landing page displays two options: Login and Create Account. 3. Abebe selects the Create Account Option. 4. The system presents a registration form requesting Abebe's personal information. 5. Abebe enters the required details, including his email address and a secure password. 6. The system verifies that the provided email is not already associated with an existing student account.

SCENARIO ID	SC_001
	7. After successful validation, the system saves Abebe's information in the database, sends him a welcome email and redirects him to the student dashboard.

Table 2-2 Description of Login Scenario

SCENARIO ID	SC_002
SCENARIO NAME	Login
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)
FLOW OF EVENTS	1. The actor navigates to the system's landing page. 2. The landing page displays the Login and Create Account option. 3. The actor clicks the Login option. 4. The system displays a login form requesting an email and password. 5. The actor enters credentials and submits the form. 6. The system verifies that the email exists and the password matches the stored credentials. 7. If the credentials are valid, the system creates a session for the actor and redirects him to their respective dashboard(student, Administrator, Teacher)

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Table 2-0-2 Description of Logout Scenario

SCENARIO ID	SC_003
SCENARIO NAME	Logout
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)
FLOW OF EVENTS	1. The actor is currently Logged In. 2. The actor clicks the Logout button from the navigation bar. 3. The system removes the actor's session and clear session related data. 4. The system redirects the actor to the landing page with a confirmation message "You have been logged out successfully".

Table 2-3 Description of Password Reset Scenario

SCENARIO ID	SC_004
SCENARIO NAME	Password Reset
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)

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FLOW OF EVENTS	1. The actor navigates to the system's Login page. 2. The actor clicks on the "Forget Password?" link. 3. The system displays a password reset form asking to enter a registered email address. 4. The actor enters their registered email address and submits the password reset request form. 5. The system validates if there is a registered account associated with the provided email in the database. 6. If the validation is correct, the system generates a password reset token and sends a password reset link to the actor's email address. 7. The actor clicks on the password reset link. 8. The system verifies the token and displays a form to create a new password the actor enters a new and confirmed password then submits the form. 9. The system updates the password and confirms the password reset was successful and redirects the actor back to the login page to sign in with the new password.
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Table 2-0-4 Description of Update Profile Scenario

SCENARIO ID	SC_005
SCENARIO NAME	Update Profile
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)

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FLOW OF EVENTS	1. The actor logs in and navigates to their dashboard page. 2. The actor clicks on the “Profile” option. 3. The system displays the actor’s profile information. 4. The actor clicks on the “Update Profile” button. 5. The system displays editable fields to update their profile information. 6. The actor modifies the desired profile information. 7. The system validates the updated profile information. 8. The system saves the updated profile information to the database confirmation message “Your profile has been updated”.
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Table 2-5 Description of Create Subject Scenario

SCENARIO ID	SC_006
SCENARIO NAME	Create Subject
PARTICIPATING ACTORS	Administrator (Ayele)

FLOW OF EVENTS	1. Ayele logs into the system and navigates to the Administrator Dashboard. 2. He selects the “Manage Subjects” option from the menu options. 3. The system displays existing courses and a “Create Subject” button. 4. Ayele clicks on the “Create Subject” button. 5. The system displays a form prompting him to enter subject details such as: • Subject Title • Grade level (9, 10, 11, 12) • Subject Description 6. Ayele fills the required information and submits the form. 7. The system validates the input data. 8. If input is valid and no such existing subject available, the system saves the new subject in the database. 9. The system displays a confirmation message “New subject has been created”. 10. Ayele is redirected to the updated “Manage Subjects” page where the newly created subject appears in the list.
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Table 2-0-6 Description of Assign Teacher to Subject Scenario

SCENARIO ID	SC_007
SCENARIO NAME	Assign Teacher to Subject
PARTICIPATING ACTORS	Administrator (Ayele), Teacher (Ayalew)
FLOW OF EVENTS	1. Ayele navigates to the “Manage Subjects” page. 2. The system displays a list of subjects and their assigned teachers. 3. Ayele selects a subject that needs a teacher assignment. 4. Ayele enters ayalew’s email address. 5. The system assigns ayalew to the subject and sends an email link to Ayalew. 6. Ayalew follows the email link and creates a teacher account with a valid password and profile details. 7. The system links Ayalew’s teacher account to the selected subject. 8. Both Ayele and Ayalew receive notifications confirming the assignment.

Table 2-0-7 Description of Teacher Structures Subject Scenario

SCENARIO ID	SC_008
SCENARIO NAME	Structures Subject
PARTICIPATING ACTORS	Teacher (Ayalew)
FLOW OF EVENTS	1. Ayalew logs into the system using his teacher account and navigates to the “My Subjects” section. 2. The system displays a list of subjects assigned to ayalew. 3. Ayalew selects a subject to structure. 4. The system displays the subject details along with existing chapters and topics if any available. 5. Ayalew clicks “Add Chapter” and provides chapter name. 6. The system saves the chapter and displays under the subject. 7. Ayalew again clicks “Add Topic” under a chapter and provides the topic title and learning objectives. 8. The system saves the topic and displays it under each chapter. 9. The system confirms the successful creation of chapters and topics and makes the subject available for student access.

Table 2-0-8 Description of Teacher Uploads Subject Content Scenario

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SCENARIO ID	SC_009
SCENARIO NAME	Uploads Subject Content
PARTICIPATING ACTORS	Teacher (Ayalew)
FLOW OF EVENTS	1. Ayalew logs into the system using his teacher account. 2. The system displays the teacher dashboard with an option to manage subjects and topics. 3. Ayalew selects a subject he is assigned to and navigates to the topic where content needs to be added. 4. The system displays options to add subject content such as videos, concept pages, exercises, and past year exam problems. 5. Ayalew clicks the “Add Content” button. 6. The system presents a form to upload content and enter details such as title, description and file type. 7. Ayalew fills the form and uploads the selected content. 8. The system validates the uploaded files. 9. Once validation is completed the system stores the content in the database and makes it available under the corresponding subject and topic. 10. The system shows a confirmation message indicating the successful upload of subject content.

Table 2-0-9 Description of Teacher Creates and Manages Quizzes Scenario

SCENARIO ID	SC_010
SCENARIO NAME	Creates and Manages Quizzes
PARTICIPATING ACTORS	Teacher (Ayalew)
FLOW OF EVENTS	1. Ayalew logs into the system using his teacher account. 2. The system displays the teacher dashboard with assigned subjects and topics. 3. Ayalew selects a subject and navigates to the topic where he wants to create a quiz. 4. The system displays an option to create or manage quizzes for that topic. 5. Ayalew clicks the “Create Quiz” button. 6. The system presents a form to enter quiz details such as title, instructions, number of questions, question type, and time limit. 7. Ayalew adds questions, answers, and marks for each question. 8. The system validates the quiz details and questions, stores the quiz in a database and associates it with the selected topic.

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	9. The system displays a confirmation message indicating successful quiz creation and notifies students. 10. Ayele can edit or delete existing quizzes using the “Manage Quiz” option.
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Table 2-0-10 Description of Access and View Subject Content Scenario

SCENARIO ID	SC_011
SCENARIO NAME	Access and View Subject
PARTICIPATING ACTORS	Student (Abebe)
FLOW OF EVENTS	1. Abebe logs in and navigates to the subject's list. 2. The system displays all subjects available to Abebe. 3. Abebe selects a subject to view. 4. The system displays chapters and topics within the subject. 5. Abebe selects a topic to view its content, including videos, concept pages, exercises, and previous-year exam problems. 6. The system allows Abebe to browse, read, and watch the materials.

Table 2-0-11 Description of Attempts Exercises, Quizzes and Previous Year Exam Problems Scenario

SCENARIO ID	SC_012
SCENARIO NAME	Attempts Exercises, Quizzes and Previous Year Exam Problems
PARTICIPATING ACTORS	Student (Abebe)
FLOW OF EVENTS	1. Abebe logs in and navigates to a subject topic. 2. The system displays exercises, quizzes, and previous-year exam problems for that topic. 3. Abebe selects an exercise, quiz, or exam problem to attempt. 4. The system presents the questions with interactive validations. 5. Abebe submits responses. 6. The system evaluates the answers, displays the score, and updates Abebe's progress.

Table 2-0-12 Description of Searches and Filters Previous Year Exam Problems Scenario

SCENARIO ID	SC_013
SCENARIO NAME	Searches and Filters Previous Year Exam Problems

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PARTICIPATING ACTORS	Student (Abebe), Teacher (Ayele)
FLOW OF EVENTS	1. Abebe logs in and navigates to the “Previous-Year Exam Problems” section. 2. The system displays available subjects and exams. 3. Abebe enters search criteria, such as subject, chapter, topic, or exam year. 4. The system filters and displays the matching exam problems. 5. Abebe selects problems to view and attempt. 6. The system provides options to attempt, bookmark problems for later review.

Table 2-0-13 Description of View Progress and Results Scenario

SCENARIO ID	SC_014
SCENARIO NAME	View Progress and Results
PARTICIPATING ACTORS	Student (Abebe)

FLOW OF EVENTS	1. Abebe logs in and navigates to the “Progress & Results” section. 2. The system retrieves and displays his progress for each subject, chapter, and topic. 3. The system shows completed exercises, quizzes, and previous-year exam problems, along with scores or completion percentage. 4. Abebe reviews the displayed results to identify strengths and areas needing improvement. 5. The system allows Abebe to navigate from progress indicators to corresponding exercises, quizzes, or topics for further study.
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Table 2-0-14 Description of Student Joins Peer Community Groups

SCENARIO ID	SC_015
SCENARIO NAME	Join Peer Community Groups
PARTICIPATING ACTORS	Student (Abebe)

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FLOW OF EVENTS	1. Abebe navigates to the “Community Groups” section on the dashboard. 2. The system displays available peer groups by Subject and Grade level 9, 10, 11 and 12. 3. Abebe clicks on a selected group. 4. The system redirects Abebe to the corresponding Telegram group.
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Table 2-0-15 Description of Student Receives and View Notifications Scenario

SCENARIO ID	SC_016
SCENARIO NAME	Receives and View Notifications
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)
FLOW OF EVENTS	1. Ayalew updates course content or adds new material, or Administrator Ayele updates Abebe’s account status. 2. The system generates a notification corresponding to the action. 3. Abebe receives the notification via the in-app notification system and email. 4. Abebe views the notification to stay informed about course updates or account status changes.

Table 2-0-16 Description of Student Asks Questions

SCENARIO ID	SC_017
SCENARIO NAME	Ask Questions
PARTICIPATING ACTORS	Student (Abebe), Teacher (Ayalew)
FLOW OF EVENTS	1. Abebe logs into his student account. 2. Abebe navigates to a subject and selects a specific topic. 3. The system displays the topic main page and an “Ask Question” section. 4. Abebe heads to the “Ask Question” section, clicks on “New Question button” , types his question related to the topic and clicks the “Submit Question” button. 5. The system saves the question and posts it under the “Ask Question” section and confirms when this question has been submitted. 6. Ayalew receives a notification that a new student has submitted a question under the topic.

Table 2-0-17 Description of Teacher Answer Questions

SCENARIO ID	SC_018
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SCENARIO NAME	Answer Question
PARTICIPATING ACTORS	Teacher (Ayalew), Student (Abebe)
FLOW OF EVENTS	1. Ayalew logs into his teacher account and navigates to the subject he teaches. 2. The system displays a list of student questions marked as Pending Response. 3. Ayalew selects Abebe’s question from the list. 4. The system shows the question details and a response input box. 5. Ayalew writes the answer and clicks the “Submit Answer” button. 6. The system saves the response and marks the question as Answered and notifies Abebe that his question has been answered.

Table 2-0-18 Description of Student Bookmarks Content Scenario

SCENARIO ID	SC_019
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SCENARIO NAME	Bookmarks Content
PARTICIPATING ACTORS	Student (Abebe)
FLOW OF EVENTS	1. Abebe logs into the system and navigates to a subject or resources he wants to save. 2. The system displays the content along with the “Bookmark” button. 3. Abebe clicks the “Bookmark” button. 4. The system stores the bookmarked content under Abebe’s Profile and Abebe can later access the bookmarks from his dashboard.

Table 2-0-19 Description of Student Submit an Issue Scenario

SCENARIO ID	SC_020
SCENARIO NAME	Submit Issue
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)

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FLOW OF EVENTS	1. Abebe navigates to a specific resource page, such as a subject topic, exercise, quiz, or a previous-year question. 2. Abebe notices an issue (e.g., confusing explanation, broken video link, incorrect answer key). 3. Abebe clicks the “Submit Issue” icon displayed on the resource page. 4. The system opens a Report Issue form, automatically attaching resource identifiers. 5. Abebe writes a short description of the issue and submits the form. 6. The system validates the form input, saves the data in the database and notifies the subject assigned teacher Ayalew and the administrator Ayele.
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Table 2-0-20 Description of Provide Response to Student Submitted Issue Scenario

SCENARIO ID	SC_021
SCENARIO NAME	Respond to Issue
PARTICIPATING ACTORS	Student (Abebe), Administrator (Ayele), Teacher (Ayalew)

FLOW OF EVENTS	1. Abebe identifies an issue in a resource (video, exercise, quiz, or concept page) and clicks the “Report Issue” button. 2. The system collects issue details and notifies Ayalew (Teacher) and Ayele (Admin). 3. Ayalew or Ayele reviews the issue. Ayalew handles subject related issues whereas Ayele handles system bugs. 4. The actor writes a response, updates the issue states to Resolved and the system notifies Abebe.
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2.6.2 Use Case Model

A use case is a list of actions or event steps typically defining the interactions between an actor and a system to achieve a goal. The actor can be a human or other external system [1].

The following use case diagram presents the actors and their interaction with the system;

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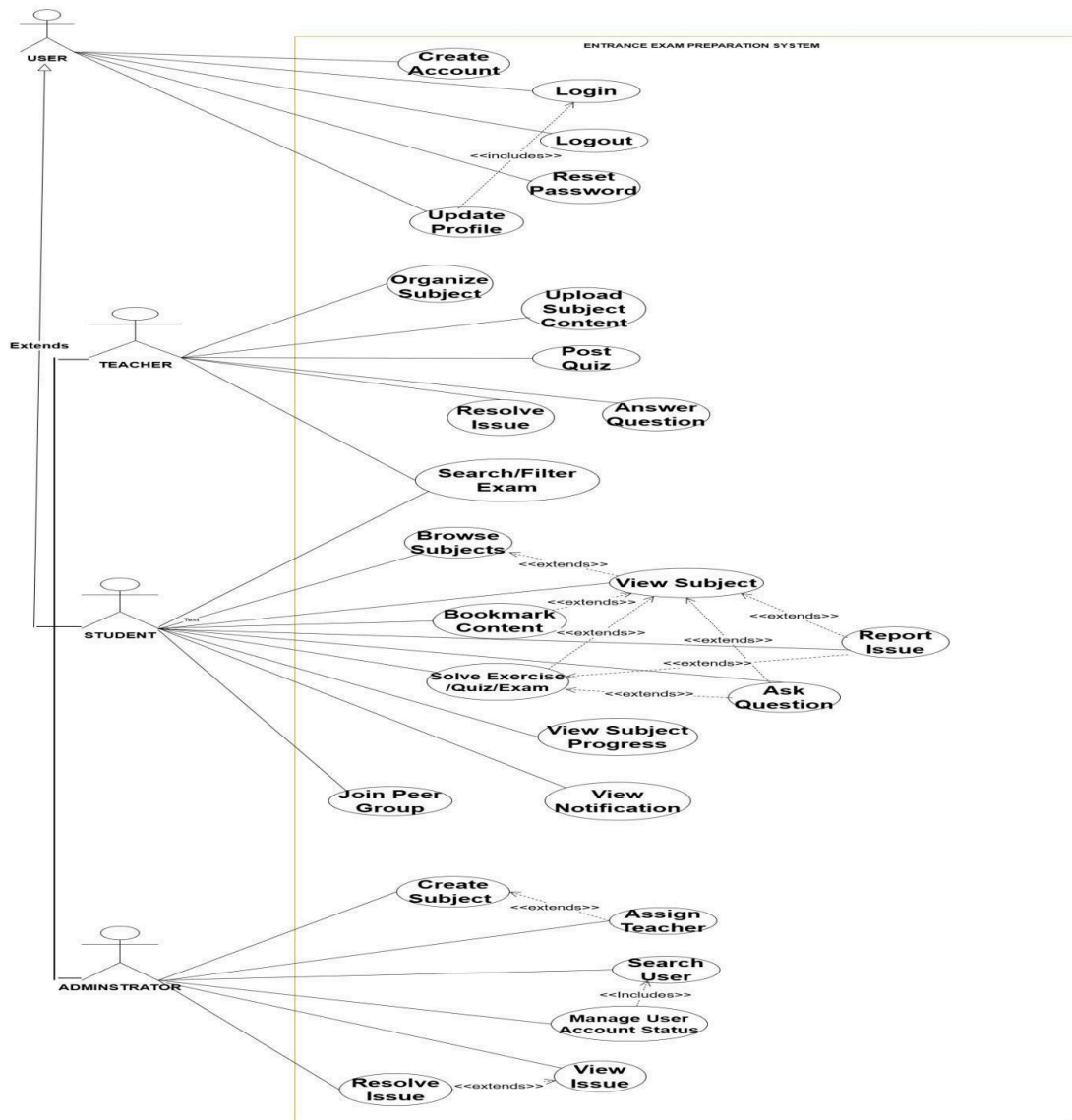


Figure 2-0-1 Use Case Diagram

Note: While the *Login* use case is implicitly required for all other use cases, it is intentionally represented as a separate use case to improve the readability and clarity of the diagram.

2.6.2.2 Description of Selected Use Case Model

Table 1-0-21 Description of Login Use Case

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Use Case ID	UC001
Use Case Name	Login
Goal	To allow a registered user to authenticate into the system and access their respective dashboard based on their role.
Participating Actor	Student, Teacher, Administrator
Pre-conditions	• The user has a registered account in the system. • The user is not currently logged in.
Flow of Events	1. The user navigates to the system landing page. 2. The system displays the Login option. 3. The user selects the Login option. 4. The system displays a login form requesting email and password. 5. The user enters valid credentials and submits the form. 6. The system verifies the credentials against stored records. 7. The system creates a session for the user. 8. The system redirects the user to the appropriate dashboard based on role (student, teacher, or administrator).
Alternative Flow	4a. Cancel Login • The user closes the login page.
Exception Flow	6a. User Account does not exist • The system displays an error message: “Account not found.” 6b. Password is incorrect • The system displays an error message: “Invalid credentials.” 6c. User account is deactivated

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	The system blocks login and displays: “Account is inactive. Contact administrator.”
Post-conditions	On success: - The user is authenticated. - A session is created. - The user is redirected to the correct dashboard. On failure: - The user is not logged in .No session is created.
Business Rule	A user must have a registered account.

Table 0-22 Description of View and Access Subject Content Use Case

Use Case ID	UC002
Use Case Name	View and Access Subject Content
Goal	To allow a student to browse subjects and access learning resources such as videos, concept pages, exercises, quizzes, and previous-year exam problems.
Participating Actor	Student
Pre-conditions	- The student is successfully logged into the system - Subjects and related content are already created and published by the teacher - The student account is active

Flow of Events	9. The student navigates to the subjects section from the dashboard 10. The system displays a list of available subjects 11. The student selects a subject 12. The system displays chapters and topics under the selected subject 13. The student selects a topic 14. The system displays available content including videos, concept notes, exercises, quizzes, and previous-year exam problems 15. The student views and interacts with the selected content
Alternative Flow	4a. No Topic • The system displays a message indicating that no topics are currently available for the selected subject
Exception Flow	2a. Student account is inactive • The system denies access and displays an account status notification 6a. Content temporarily unavailable • The system displays an error message and prompts the student to try again later
Post-conditions	On success: • The selected subject content is displayed and accessed by the student On failure: • No content is displayed and the system state remains unchanged
Business Rule	• Only active students can access subject content.

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	Students can only view content that has been published by assigned teachers.
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Table 2-0-23 Description of Attempt Exercises, Quizzes, and Exam Problems Use Case

Use Case ID	UC003
Use Case Name	Attempt exercises, quizzes, and previous-year exam problems
Goal	To allow a student to attempt practice exercises, quizzes, and previous-year exam problems in order to assess understanding and reinforce learning
Participating Actor	Student
Pre-conditions	- The student is logged into the system - The student has access to at least one subject and topic - Exercises, quizzes, or previous-year exam problems exist for the selected topic
Flow of Events	- The student navigates to a subject and selects a topic - The system displays available exercises, quizzes, and previous-year exam problems - The student selects one item to attempt - The system presents the questions with appropriate input controls - The student answers the questions and submits the attempt - The system evaluates the responses - The system displays the score, feedback, or correct answers - The system updates the student's progress record

Alternative Flow	4a. Timer • The system displays a countdown timer • when time expires, the system automatically submits the attempt 7a. Feedback • The system displays instant feedback after each question
Exception Flow	3a. No available items • The system displays a message indicating that no exercises or quizzes are available for the selected topic 5a. Incomplete submission • The system prompts the student to answer all mandatory questions before submission
Post-conditions	On success: • The student's attempt is recorded • Scores and feedback are stored and linked to the student profile On failure: • No attempt data is saved • The system state remains unchanged
Business Rule	• Quizzes may have a defined time limit • Each attempt is associated with one student and one topic • Scoring logic must follow predefined marking schemes

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Table 2-0-24 Description of Search and Filter Previous-Year Exam Problems Use Case

Use Case ID	UC004
Use Case Name	Search and Filter Previous-Year Exam Problems Use Case
Goal	Students find relevant exam problems efficiently.
Participating Actor	Student
Pre-conditions	• student is logged in
Flow of Events	1. student navigates to exam problems section 2. system displays search and filter options 3. student applies filters (subject, year, topic) 4. system displays matching problems 5. student selects a problem to view or attempt
Alternative Flow	3a. More Filter • The student may change filters to refine results. 4a. Bookmark • The student may bookmark problems.
Exception Flow	4a. No results found • The system displays “no matching records”
Post-conditions	• filtered exam problems are displayed
Business Rule	• filters must match existing metadata

Table 2-0-25 Description of Ask Questions under Topic Use Case

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Use Case ID	UC005
Use Case Name	Ask Questions Under Topic
Goal	Allow students to ask questions
Participating Actor	Student
Pre-conditions	• Student is logged in • Topic exists
Flow of Events	1. The student navigates to a topic 2. The student selects “ask question” 3. The student submits question 4. The system saves and posts the question 5. The Teacher receives notification
Alternative Flow	3a. Edits • The student edits the question before submission.
Exception Flow	3a. empty question • The system rejects submission
Post-conditions	• The question is available for teacher response.
Business Rule	• Questions must relate to the selected topic.

Table 2-0-26 Description of Bookmark Content Use Case

Use Case ID	UC006
Use Case Name	Bookmark Content
Goal	To allow the student to save content for later review

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Participating Actor	Student
Pre-conditions	• The student is logged in. • Content is available.
Flow of Events	1. The student views learning content. 2. The system displays a bookmark option. 3. The student bookmarks the content. 4. The system saves it under the student's profile.
Alternative Flow	3a. Existing bookmark • The student removes an existing bookmark.
Exception Flow	3b. Bookmarked Content • If the content is already bookmarked, the system prevents duplication.
Post-conditions	• The content is saved or removed from bookmarks.
Business Rule	• Each content item can be bookmarked only once per student.

Table 2-0-27 Description of Create Subject Use Case

Use Case ID	UC007
Use Case Name	Create Subject
Goal	To allow the administrator to create a new subject so that it can be assigned to teachers and accessed by students.
Participating Actor	Administrator
Pre-conditions	• The administrator is successfully logged into the system.

	The administrator has subject management privileges.
Flow of Events	- The administrator navigates to the “manage subjects” section. - The system displays a list of existing subjects and a “create subject” option. - The administrator selects “create subject.” - The system displays a subject creation form. - The administrator enters subject title, grade level, and description. - The administrator submits the form. - The system validates the input data. - The system saves the new subject and displays a success message.
Alternative Flow	5a. The administrator decides to cancel subject creation. The system returns to the subject list without saving data.
Exception Flow	7a. The system detects missing or invalid input fields. - The system displays validation error messages and requests correction. 7b. The system detects a duplicate subject for the same grade. - The system rejects the request and displays an error message.
Post-conditions	On success: - The subject is stored in the system and available for teacher assignment. On failure: - No subject data is saved.

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Business Rule	• A subject title must be unique per grade level. • Grade level must be between 9 and 12.
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Table 2-0-28 Description of Assign Teacher to Subject Use Case

Use Case ID	UC008
Use Case Name	Assign Teacher to Subject
Goal	To assign a teacher to a subject so that the teacher can manage and deliver the subject content.
Participating Actor	Administrator
Pre-conditions	• The administrator is logged in. • The subject exists in the system.
Flow of Events	1. The administrator navigates to the “manage subjects” page. 2. The system displays subjects and their assigned teachers. 3. The administrator selects a subject without an assigned teacher. 4. The administrator enters the teacher’s email address. 5. The system sends an account invitation email to the teacher. 6. The teacher creates a teacher account using the provided link. 7. The system links the teacher account to the selected subject.
Alternative Flow	6a. The teacher already has an existing account. • The system directly assigns the teacher to the subject.
Exception Flow	4a. The entered email is invalid. • The system displays an error message.

	6b. The teacher does not complete account creation. • The subject remains unassigned.
Post-conditions	On success: • The subject is assigned to the teacher. On failure: • No assignment is made.
Business Rule	• One subject can have only one assigned teacher. • A teacher can be assigned to multiple subjects.

Table 2-0-29 Description of Structure Subject Use Case

Use Case ID	UC009
Use Case Name	Structure Subject
Goal	To allow a teacher to organize a subject into chapters and topics for learning delivery.
Participating Actor	Teacher
Pre-conditions	• The teacher is logged in. • The teacher is assigned to the subject.
Flow of Events	1. The teacher navigates to “my subjects.” 2. The system displays assigned subjects. 3. The teacher selects a subject. 4. The teacher adds a chapter title. 5. The system saves the chapter.

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	6. The teacher adds topics under the chapter. 7. The system saves the topics and confirms completion.
Alternative Flow	6a. The teacher edits an existing chapter or topic. • The system updates the structure.
Exception Flow	4a. The chapter title is empty. • The system displays a validation error.
Post-conditions	On success: • The subject structure is saved and visible to students. On failure: • No changes are stored.
Business Rule	• A chapter must contain at least one topic. • Topic titles must be unique within a chapter.

Table 2-0-30 Description of Upload Subject Content Use Case

Use Case ID	UC010
Use Case Name	Upload Subject Content
Goal	To allow a teacher to upload learning materials under a specific topic.
Participating Actor	Teacher
Pre-conditions	• The teacher is logged in. • The subject and topic exist.

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Flow of Events	1. The teacher selects a subject and topic. 2. The system displays available content options. 3. The teacher selects “add content.” 4. The teacher uploads files or enters content details. 5. The system validates the content. 6. The system stores the content and confirms success.
Alternative Flow	4a. The teacher saves content as a draft. • The content is not visible to students.
Exception Flow	5a. Unsupported file type detected. • The system rejects the upload.
Post-conditions	On success: • Content is available to students. On failure: • Content is not stored.
Business Rule	• Maximum file size limits apply. • Only assigned teachers can upload content.

Table 2-0-31 *Description of Create and Manage Quizzes Use Case*

Use Case ID	UC011
Use Case Name	Create and Manage Quizzes
Goal	To allow a teacher to create, edit, and manage quizzes for student assessment

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Participating Actor	Teacher
Pre-conditions	• The teacher is logged in. • The topic exists under the subject.
Flow of Events	1. The teacher selects a subject topic. 2. The system displays quiz management options. 3. The teacher selects “create a quiz.” 4. The teacher enters quiz details and questions. 5. The system validates the quiz. 6. The system saves the quiz and makes it available to students.
Alternative Flow	4a. The teacher edits an existing quiz. • The system updates the quiz.
Exception Flow	5a. The quiz contains no questions. • The system prevents saving.
Post-conditions	On success: • Quiz is available for student attempts. On failure: • The quiz is not published.
Business Rule	• A quiz must contain at least one question. • Time limits must be greater than zero.

2.6.3 Object Model

2.6.3.1 Data dictionary

Table 3-32 *USER MODEL*

Field	Description	Data Type	Length	Example
<i>User_ID</i>	Unique user identifier	UUID	36	550e8400-e29b-41d4-a716-446655440000
<i>First_Name</i>	First name of the user	String	30	Yonas
<i>Middle_Name</i>	Middle name of the user	String	30	Alemu
<i>Last_Name</i>	Last name of the user	String	30	Habte
<i>Email</i>	Email address	String	255	yonasalemu@gmail.com
<i>Phone_Number</i>	User's phone number	String	255	+251-9-37-33-33-33
<i>Password</i>	Hashed password	String	255	\$2y\$10\$Ka3/TxAu3UrGX4E8su
<i>Role</i>	Role in the system	Enum	-	Student
<i>Status</i>	Account status	Enum	-	Active
<i>Profile_Image</i>	Profile photo Url	String	255	https://storage.platform.com/users/gryonas_profile_image.jpg
<i>Created_At</i>	Account creation date	DateTime	-	2025-12-20

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<i>Updated_At</i>	Account update date	DateTime	-	2025-12-03
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Table 2-0-32 *SUBJECT MODEL*

Field	Description	Data Type	Length	Example
<i>Subject_ID</i>	Unique subject Identifier	UUID	36	184fa9a3-f967-4a98-9d8f-57152e7cbe64
<i>Name</i>	Name	String	30	Mathematics
<i>Description</i>	Description	String	255	Grade 12 Mathematics
<i>Grade</i>	Grade	Integer	-	12

Table 2-0-33 CHAPTER MODEL

Field	Description	Data Type	Length	Example
<i>Chapter_ID</i>	Unique chapter Identifier	UUID	36	123e4567-e89b-12d3-a456-426614174000
<i>Title</i>	Name	String	50	Introduction to sequence and series
<i>Description</i>	Description	String	255	Theoretical and practical applications of arithmetic and geometric sequence and series
<i>Exam_Weight</i>	Exam coverage in %	Integer	-	25

Table 2-0-34 TOPIC MODEL

Field	Description	Data Type	Length	Example
<i>Topic_ID</i>	Unique topic identifier	UUID	36	8b8907ce-6ea1-4505-a016-f34959e6dd7a
<i>Title</i>	Topic name	String	50	Introduction to Arithmetic Sequence
<i>Description</i>	Description	String	255	Understanding the natures of arithmetic sequences
<i>Objectives</i>	Expected objectives to meet	List	-	[‘Define Arithmetic sequences, ‘Identify Arithmetic Sequences’, Explain the properties of Arithmetic sequences’]

Table 2-0-35 VIDEO MODEL

Field	Description	Data Type	Length	Example
<i>Video_ID</i>	Unique video Identifier	UUID	36	3efe9292-21c2-4421-ada0-e17ac8bb8b1b
<i>Video_Url</i>	Link to video source	String	255	https://www.youtube.com/watch?v=_cooC3yG_p0

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<i>Duration</i>	Length in minutes	Integer	-	7
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Table 2-0-36 CONCEPT MODEL

Field	Description	Data Type	Length	Example
<i>Concept_Page_ID</i>	Unique concept page identifier	UUID	36	19350fc3-a11d-422f-ad7b-2e671084e60e
<i>Text</i>	Concept Explanation	Text	-	An arithmetic sequence is a list of numbers where the difference between any two consecutive terms is always the same constant value, called the common difference, found by adding or subtracting it to get the next number in the series
<i>Reading_Duration</i>	Duration estimate to complete reading in minutes	Integer	-	60

Table 2-0-37 EXERCISE MODEL

Field	Description	Data Type	Length	Example
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exerciseId	Unique exercise identifier	UUID	36	c2a3a4cd-df81-4e70-a067-5685e131e48c
numberOfProblems	Set of exercise problems in an exercise	Int	-	15

Table 2-0-38 EXERCISE_PROBLEM MODEL

Field	Description	Data Type	Length	Example
<i>Exercise_ID</i>	Unique exercise identifier	UUID	36	e3316ba0-82cd-4993-864d-784d3f336435
<i>Question</i>	Exercise question	Text	-	An arithmetic sequence has a first term of 3 and a common difference of 5. What is the 6th term of the sequence?
<i>Choices</i>	Multiple choices	Dictionary	-	{‘A’: 23 , ‘B’: 25, ‘C’: 28, ‘D’: 30}
<i>Correct_Answer</i>	Correct answers	List	-	[‘C’]

CHAPTER THREE-SYSTEM DESIGN

<i>Answer_Explanation</i>	Textual solution explanation	Text	-	In an arithmetic sequence, each term is found by adding the common difference to the previous term. First term $a=3$ Common difference $d=5$ The 6th term is: $a_6 = a + (6 - 1)d = 3 + 5 \times 5 = 3 + 25 = 28$ So, the answer is 28 (C).
---------------------------	------------------------------	------	---	--

Table 2-0-39 EXAM_PAPER MODEL

Field	Description	Data Type	Length	Example
examPaperID	Unique exam paper identifier	UUID	36	c2a3a4cd-df81-4e70-a067-5685e131e48c
examYear	Year examination taken	Date	-	2017

Table 2-0-40 EXAM_PROBLEM MODEL

Ethiopia Entrance Exam Preparation Platform

Field	Description	Data Type	Length	Example
<i>Question_ID</i>	Unique question identifier	UUID	36	ef73337a-8d03-41cb-97a7-baea2039e441
<i>Question</i>	Exam question	Text	-	If the 6th term and the 11th term of an arithmetic sequence are 35 and 75 respectively, then what is the sum of the first 31 terms of the sequence?
<i>Choices</i>	Multiple Choices	Dictionary	-	{ 'A': 2145, 'B': 724, 'C': 3565, 'D': 846 }
<i>Correct_Answer</i>	Correct answers	List	-	['C']
<i>Answer_Explanation</i>	Textual Solution Explanation	Text	-	The difference between the 11th and 6th terms is 40, which corresponds to five common differences, so the common difference is 8. Using this value, the first term is -5. Applying the sum formula for 31 terms gives a total of 3565.

Table 2-0-41 QUIZ MODEL

Field	Description	Data Type	Length	Example
<i>Quiz_ID</i>	Unique quiz identifier	UUID	36	f7f84774-bce7-4827-9975-9e32808d087e
<i>Total_Weight</i>	Total mark	Integer	-	10

CHAPTER THREE-SYSTEM DESIGN

<i>Time_Limit</i>	Time limit in minutes	Integer	-	60
<i>Created_At</i>	Date quiz posted	DateTime	-	2025-05-10 12:00

Table 2-0-42 QUIZ_PROBLEM MODEL

Field	Description	Data Type	Length	Example
<i>Question_ID</i>	Unique question identifier	UUID	36	661d702d-e781-4b09-a0a0-7c11d26e72e6
<i>Question_Text</i>	Quiz question in text format	Text	9-	What is the sum of the following arithmetic sequence 1, 3, 5, 7
<i>Question_Url</i>	Quiz question in an image format	String	255	https://storage.platform.com/quiz/grade9_math_arithmetic_2001.jpg
<i>Choices</i>	Available answer choices	Dictionary	-	{ 'A': 2145, 'B': 724, 'C': 3565, 'D': 846 }
<i>Correct_Answer</i>	Correct answer	List	-	['A']

Table 2-0-43 QUIZ_SCORE MODEL

Field	Description	Data Type	Length	Example
<i>Attempt_ID</i>	Unique quiz attempt identifier	UUID	36	e2deabe1-860e-4d8f-a602-57fe97ad2565

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<i>Quiz_ID</i>	Reference to quiz	UUID	36	f7f84774-bce7-4827-9975-9e32808d087e
<i>status</i>	Quiz completion status	Enum	-	COMPLETED
<i>Score</i>	Achieved score	Integer	-	10
<i>Attempt_Date</i>	Submission date	DateTime	-	2025-05-02 12:36

Table 2-0-44 *TOPIC PROGRESS MODEL*

Field	Description	Data Type	Length	Example
<i>Progress_ID</i>	Unique progress identifier	UUID	36	6022a60e-ba00-42d3-a03d-8a5c1898a22d
<i>CompletedAt</i>	Date Topic is completed	DateTime	-	5-23-2025

Table 2-0-45 *BOOKMARK MODEL*

Field	Description	Data Type	Length	Example
<i>Bookmark_ID</i>	Unique bookmark identifier	UUID	36	29f9726e-74a0-4a53-909b-5f63d7cc0c30
<i>Content_ID</i>	Short description about the bookmark	String	255	I need to revise exercise 3 & 5
<i>Created_At</i>	Date bookmarked	DateTime	-	2025-05-07 12:00

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Table 2-0-46 QUESTION MODEL

Field	Description	Data Type	Length	Example
<i>Question_ID</i>	Unique question identifier	UUID	36	a51eaeaa-f95d-495c-9761-25555d609c44
<i>Question_Text</i>	Question in Text format	Text	-	Find the sum of the first 20 terms of an arithmetic sequence whose first term is 3 and common difference is 5.
<i>Question_Image_Url</i>	Question uploaded in an image format, holds the image URL	String	255	https://storage.platform.com/questions/grade9_math_arithmetic_2001.jpg
<i>Created_At</i>	Date question posted	DateTime	-	2025-05-05 13:35
<i>Updated_At</i>	Date question edited	DateTime	-	2025-05-06 12:00

Table 2-0-47 ANSWER MODEL

Field	Description	Data Type	Length	Example
<i>Answer_ID</i>	Unique answer identifier	UUID	36	dc07c8fa-bec4-477a-8358 6a798f897870

Ethiopia Entrance Exam Preparation Platform

<i>Answer_Text</i>	Textual answer	Text	-	The correct answer is 1200
<i>Answer_Image_Url</i>	Answer in an image format	String	255	https://storage.platform.com/answers/grade9_math_arithmetic_2001.jpg
<i>Created_At</i>	Date answer provided	DateTime	-	2025-05-05 11: 35
<i>Updated_At</i>	Date answer updated	DateTime	-	2025-05-06 12:35

Table 2-0-48 **ISSUE MODEL**

Field	Description	Data Type	Length	Example
<i>Issue_ID</i>	Unique issue identifier	UUID	36	5350fd30-8b4f-45ec-9a0e-d03e63d46c3e
<i>Issue_Type</i>	Type of issue	Enum	36	Wrong Answer
<i>Issue_Description</i>	Issue description	Text	36	The provided answer is incorrect
<i>Response</i>	Given response	Text	36	The answer has been corrected
<i>Status</i>	Issue status	Enum	36	Closed
<i>CreatedAt</i>	Date issue submitted	DateTime	-	1-24-2026
<i>UpdatedAt</i>	Date issue updated	DateTime	-	1-25-2026

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Table 2-0-49 NOTIFICATION MODEL

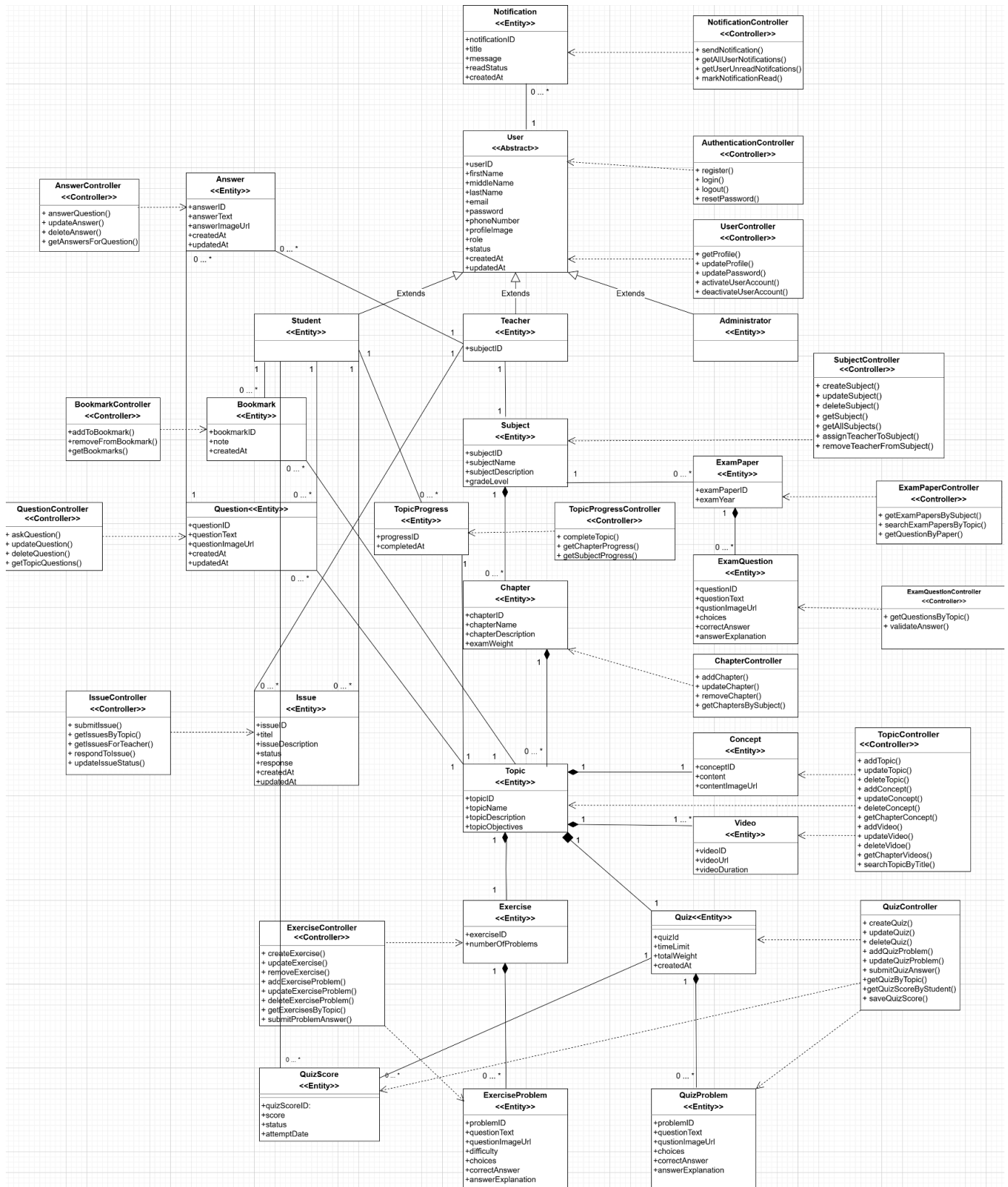
Field	Description	Data Type	Length	Example
<i>Notification_ID</i>	Unique notification identifier	UUID	36	c2a3a4cd-df81-4e70-6474g474g474
<i>Message</i>	Actual message sent	Text	-	A new quiz has been posted on introduction to sequences.
<i>readStatus</i>	Shows whether in app notification is read or not	Enum	-	READ
<i>SentAt</i>	Date notification sent	DateTime	-	2025-11-02

2.6.3.2 Class Diagram

A class diagram is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (methods) and the relationships among objects.

The class diagram of our system is presented below.

Ethiopia Entrance Exam Preparation Platform



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Figure 2-0-2 Class Diagram

2.6.4 Dynamic Modeling

2.6.4.1 Sequence Diagram

A sequence diagram is an interaction diagram that shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations [1].

Accordingly, some selected to show sequence diagrams of our proposed system are as follows.

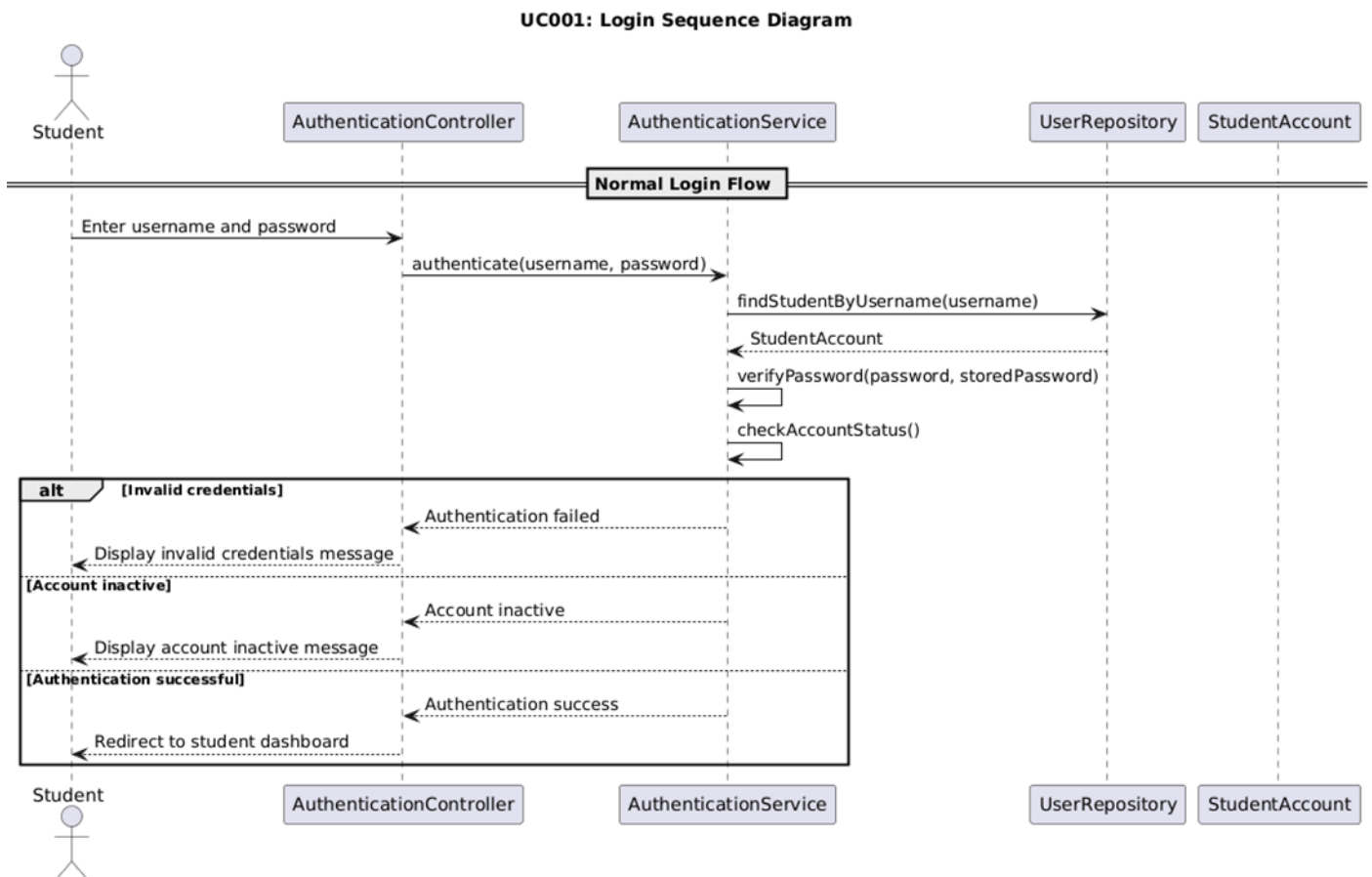
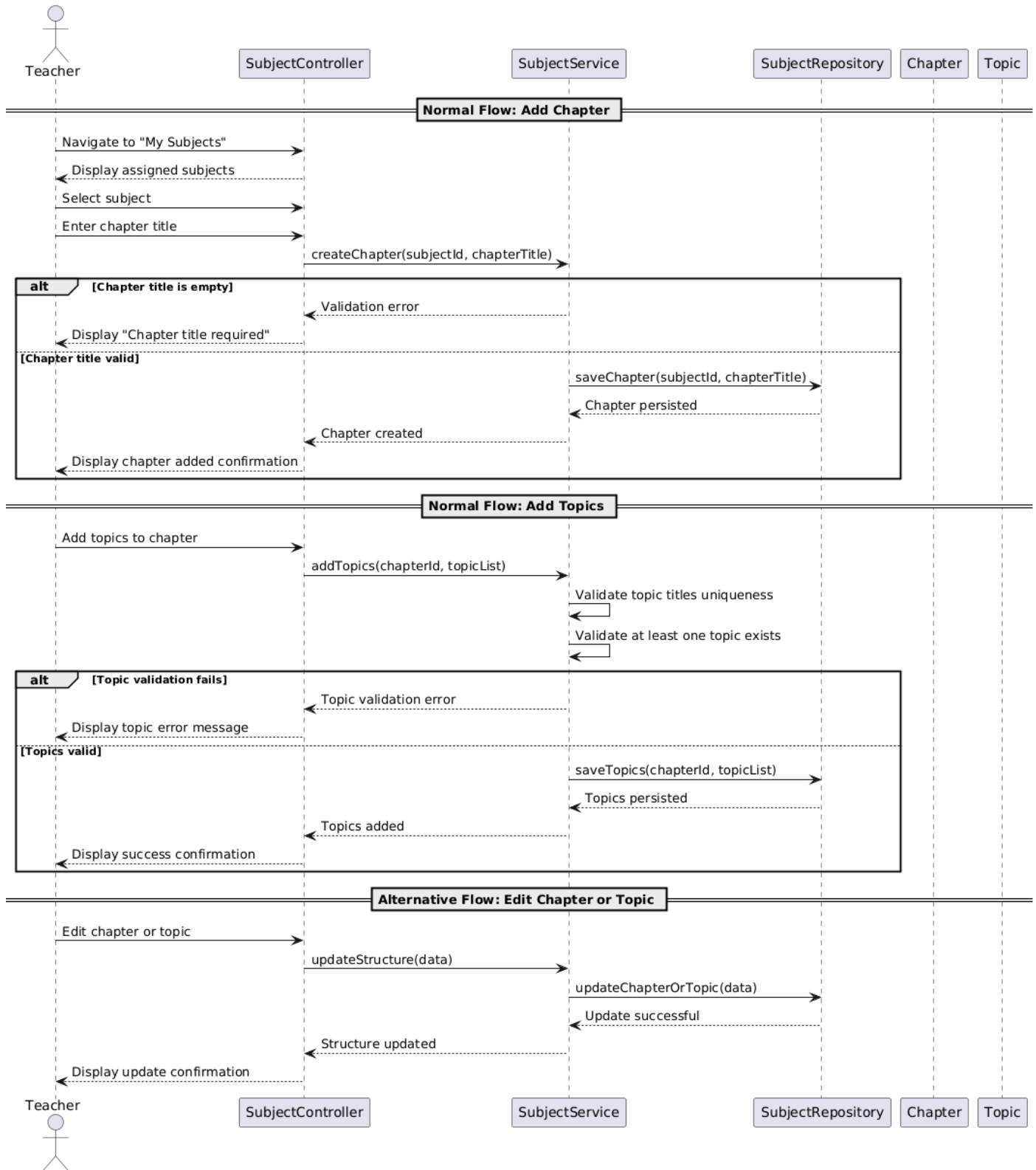


Figure 2-0-3 Login Sequence Diagram

Ethiopia Entrance Exam Preparation Platform

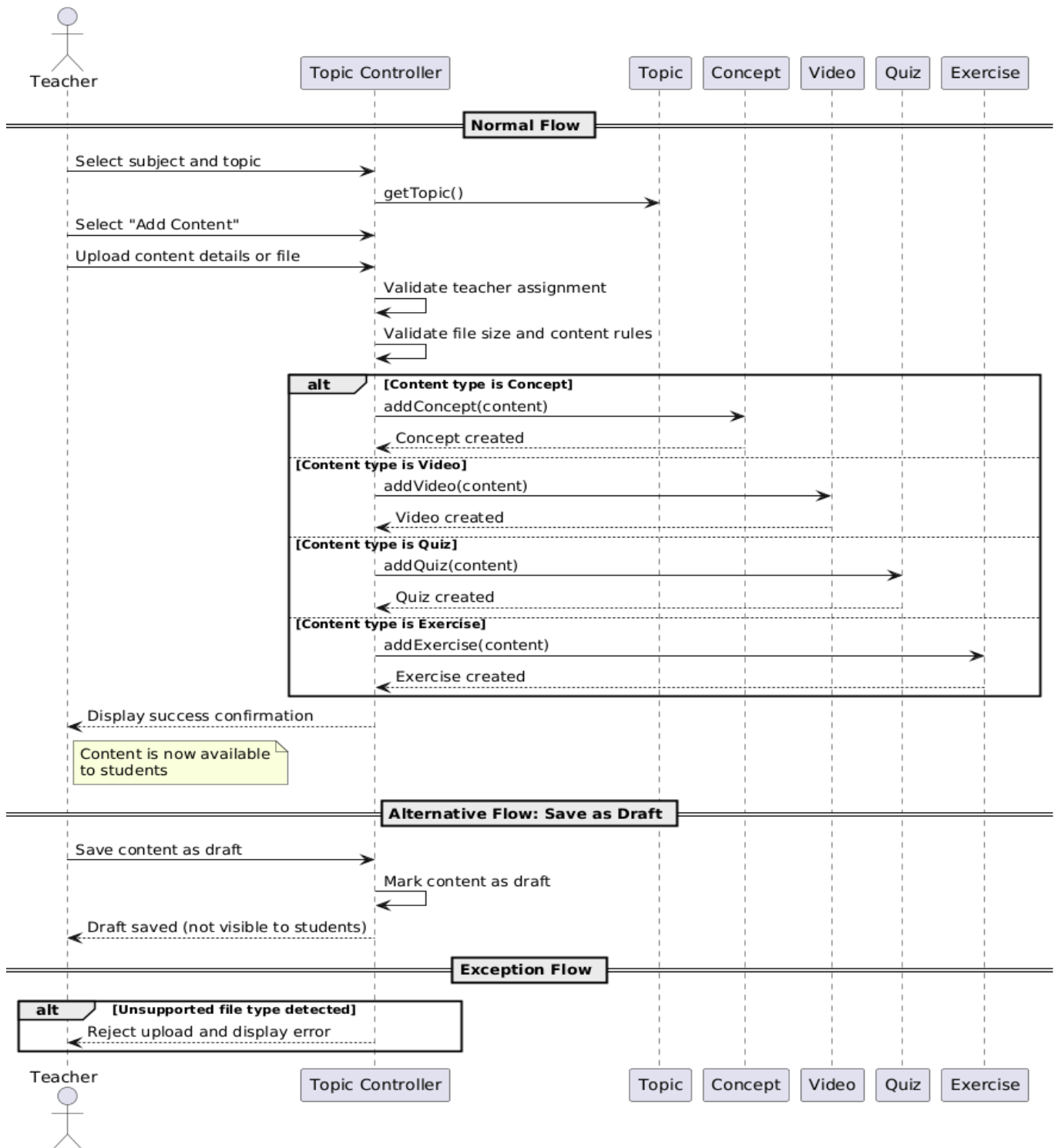
UC009: Structure Subject Sequence Diagram



CHAPTER THREE-SYSTEM DESIGN

Figure 2-0-4 Structure Subject Sequence Diagram

UC010: Upload Subject Content Sequence Diagram



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Figure 2-0-5 Upload Subject Content

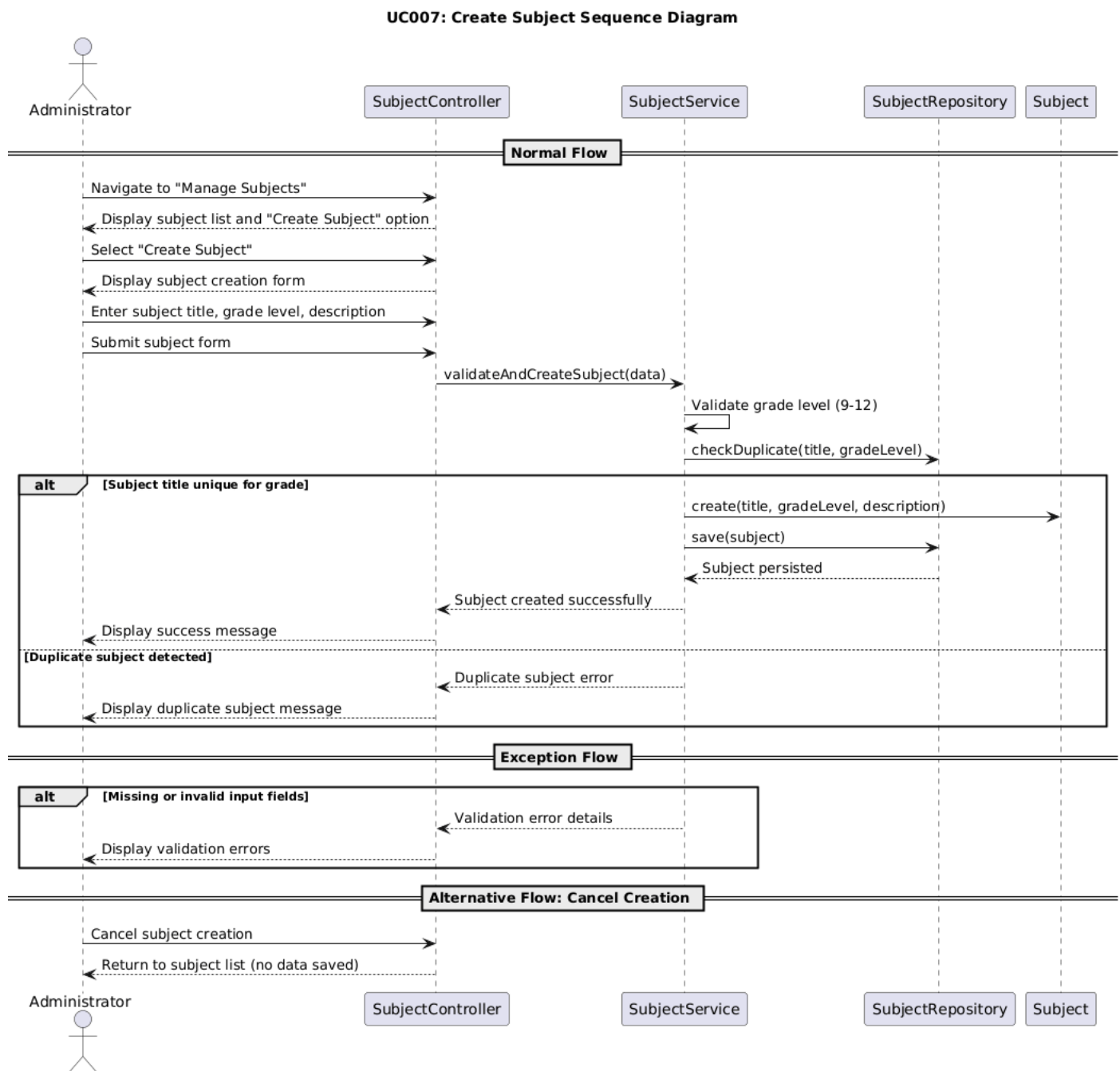


Figure 2-0-6 Create Subject Sequence Diagram

Ethiopia Entrance Exam Preparation Platform

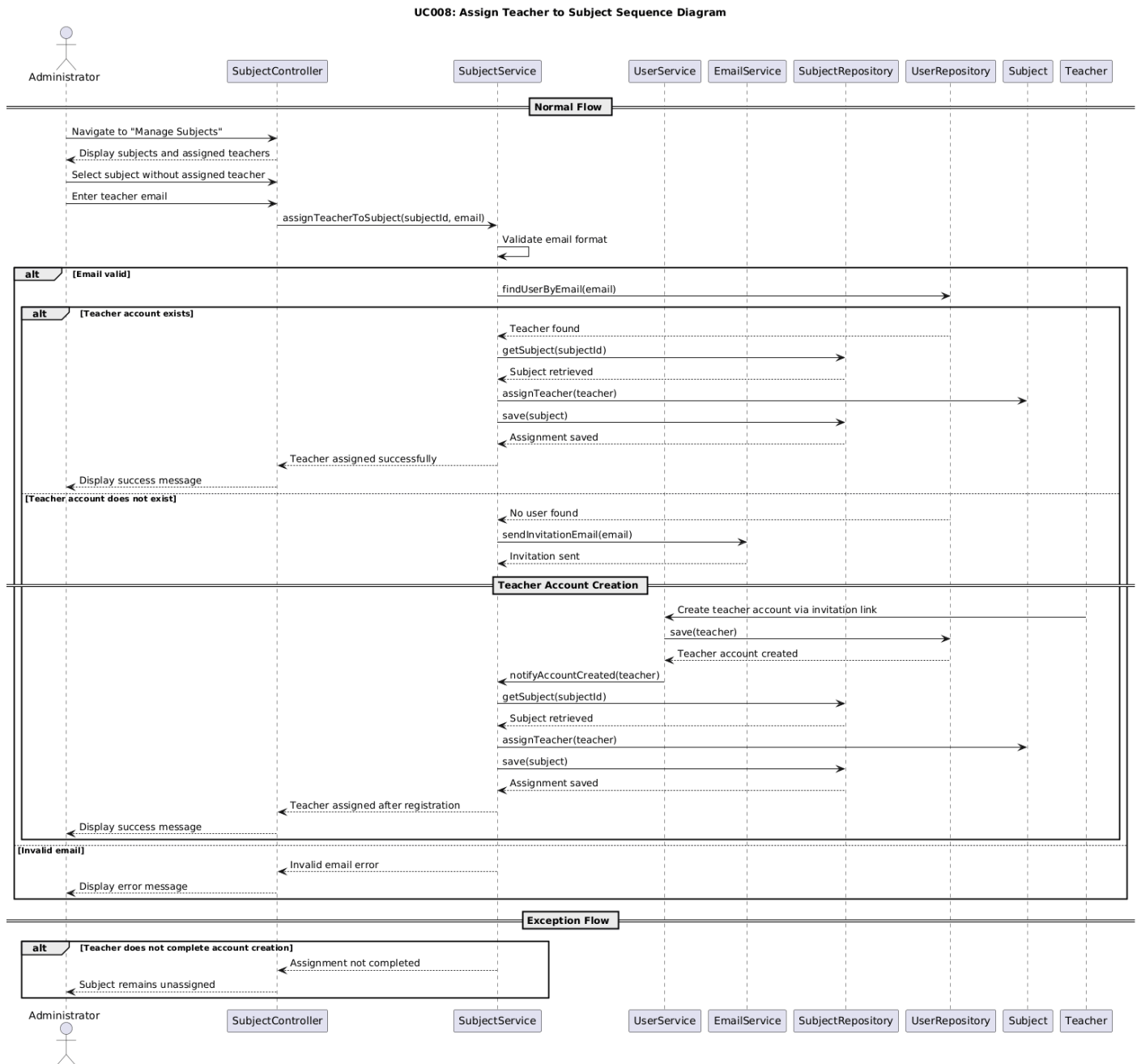


Figure 2-0-7 Assign Teacher to Subject

CHAPTER THREE-SYSTEM DESIGN

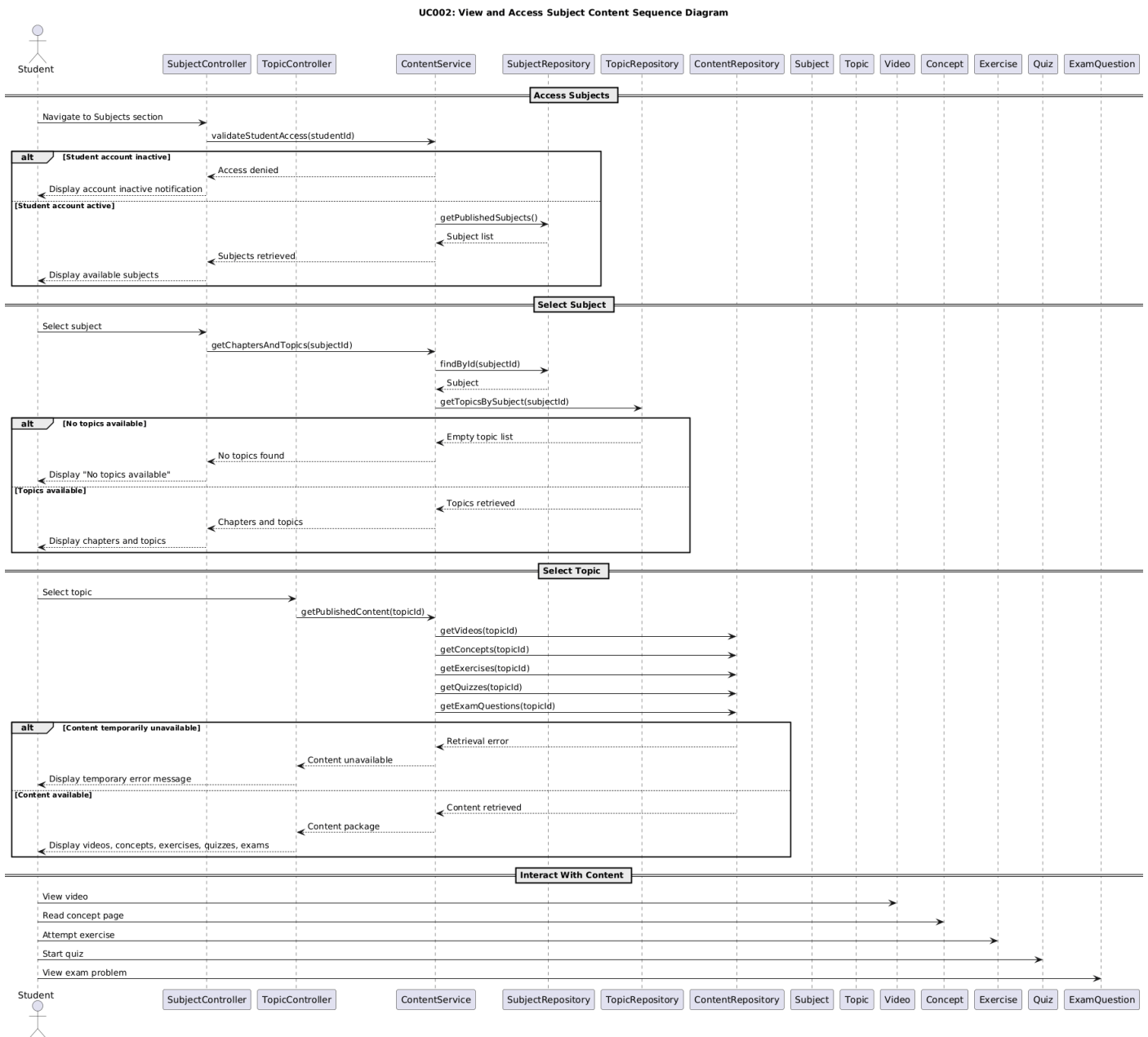


Figure 2-0-8 View and Access Subject Content Sequence Diagram

Ethiopia Entrance Exam Preparation Platform

Search and Filter Previous-Year Exam Problems Sequence Diagram

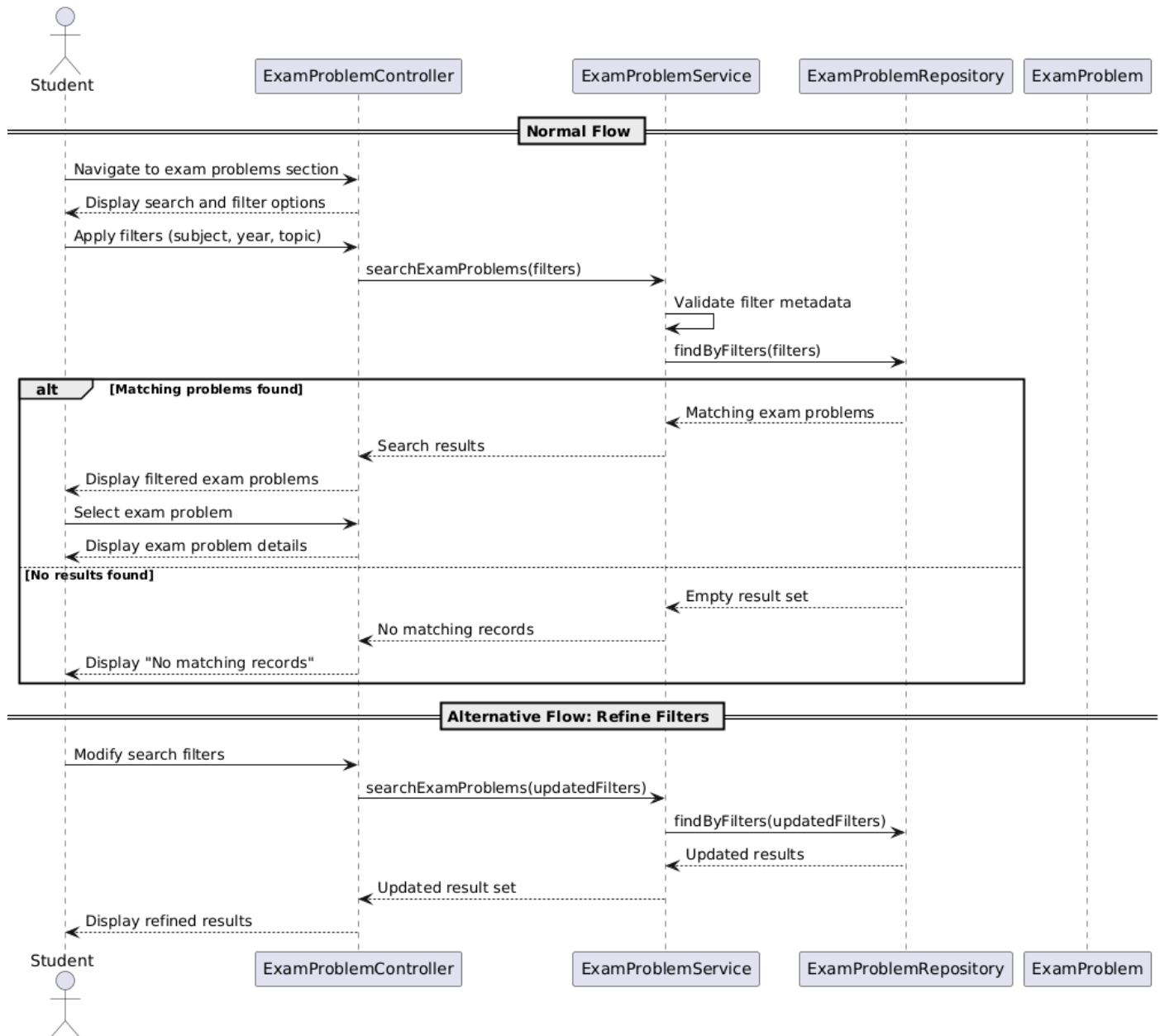


Figure 2-0-9 Search Previous Year Exam Problems Sequence Diagram

CHAPTER THREE-SYSTEM DESIGN

2.6.4.2 Activity Diagram

The activity diagram describes the sequencing of activities, with support for both conditional and parallel behavior [3]

Accordingly some selected activity diagrams of our proposed system are presented as follows.⁵

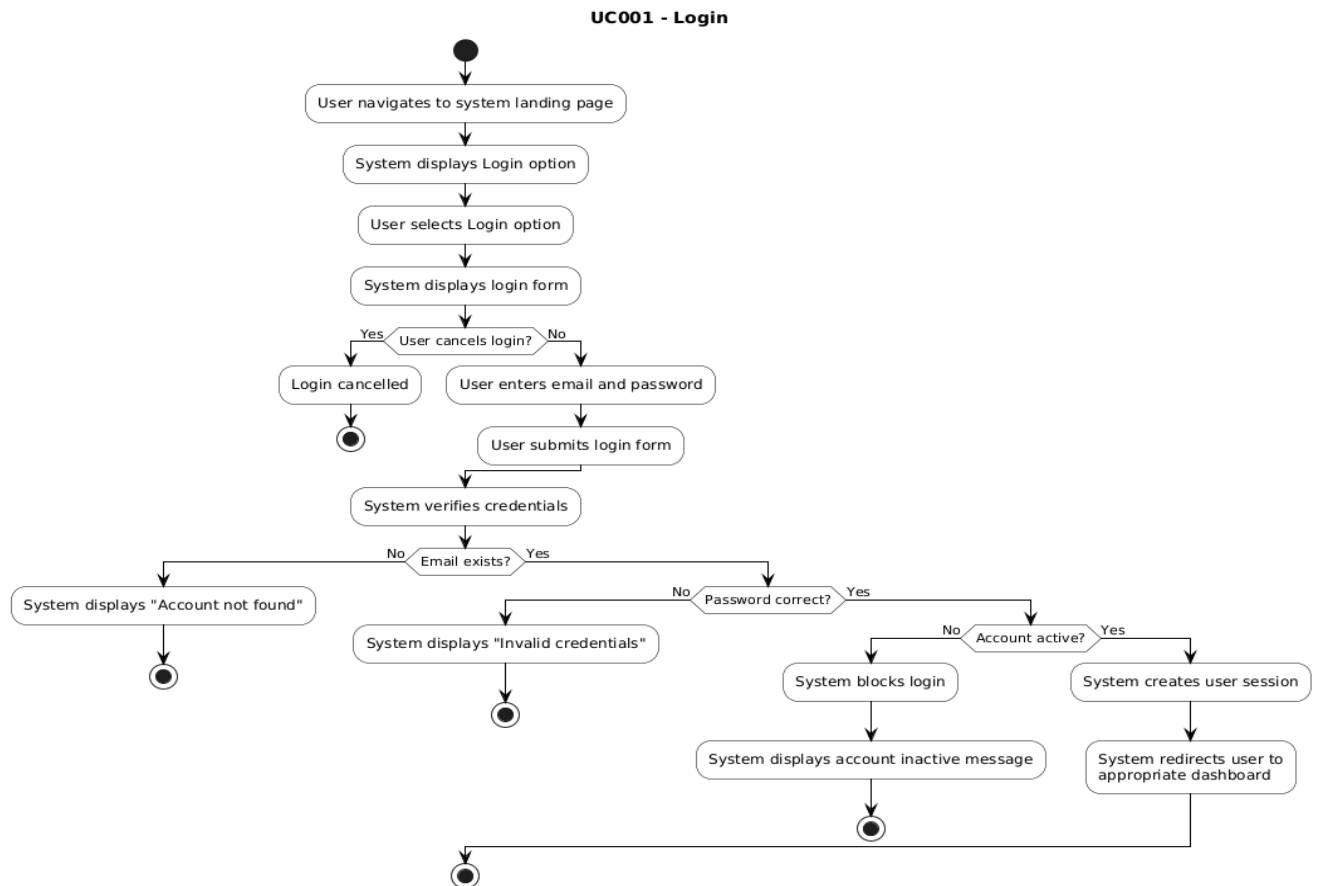


Figure 2-0-10 *User Login Activity Diagram*

⁵ <https://plantuml.com/>

UC002 - View and Access Subject Content

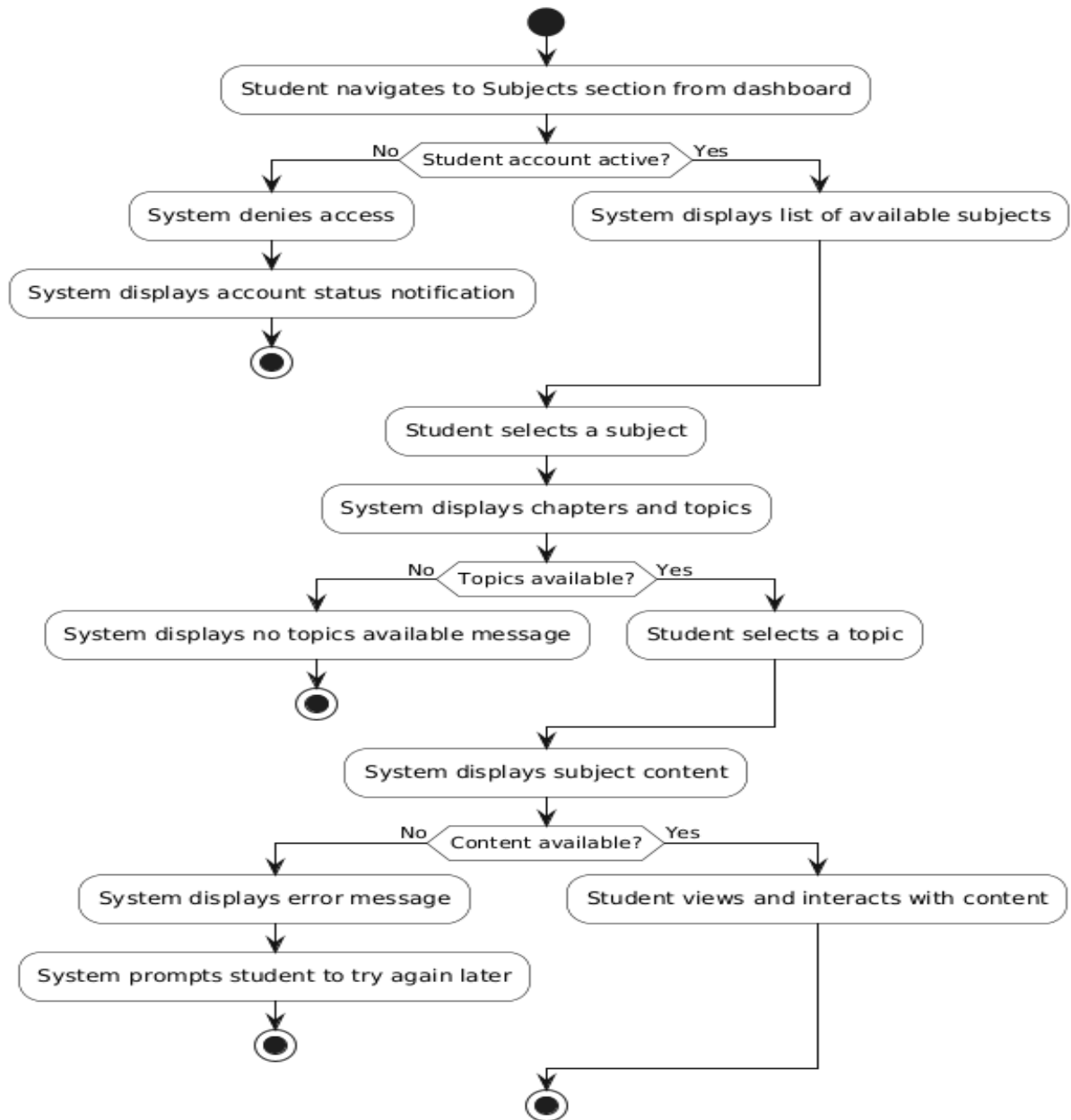


Figure 2-0-11 *Student View and Access Subject Content*

CHAPTER THREE-SYSTEM DESIGN

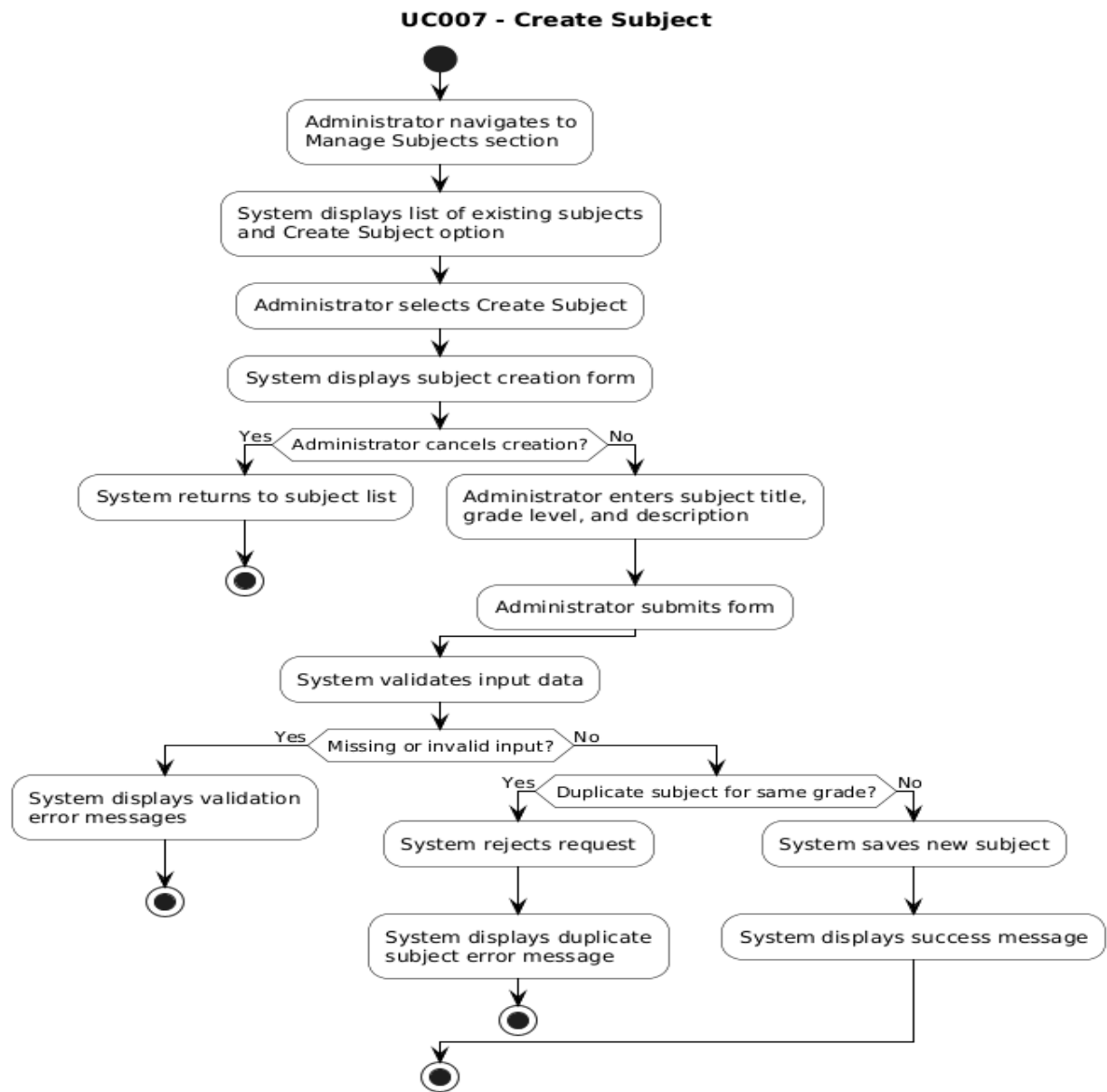


Figure 2-0-12 Create Subject Activity Diagram

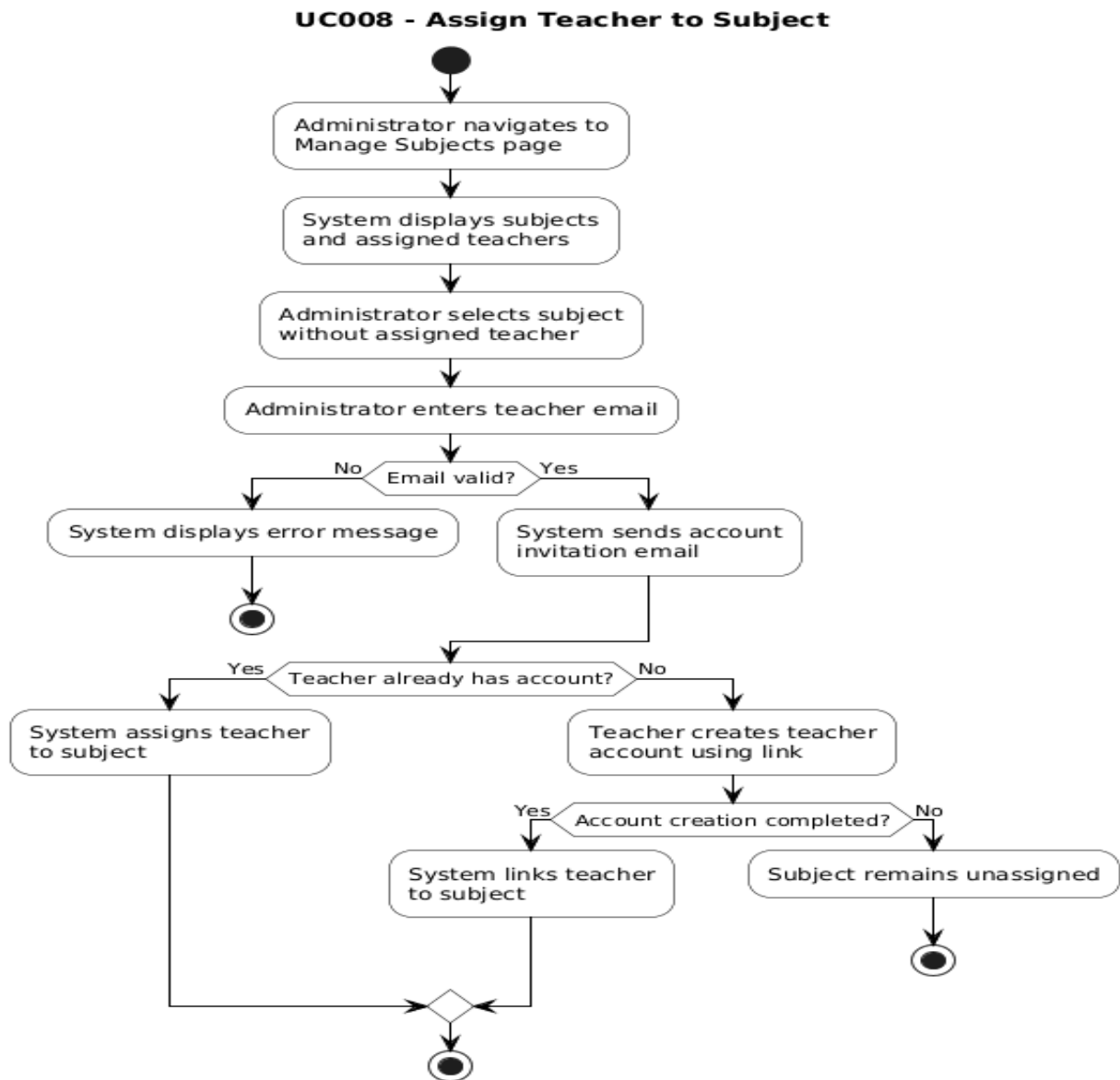


Figure 2-0-13 *Assign Subject Teacher Activity Diagram*

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UC010 - Upload Subject Content

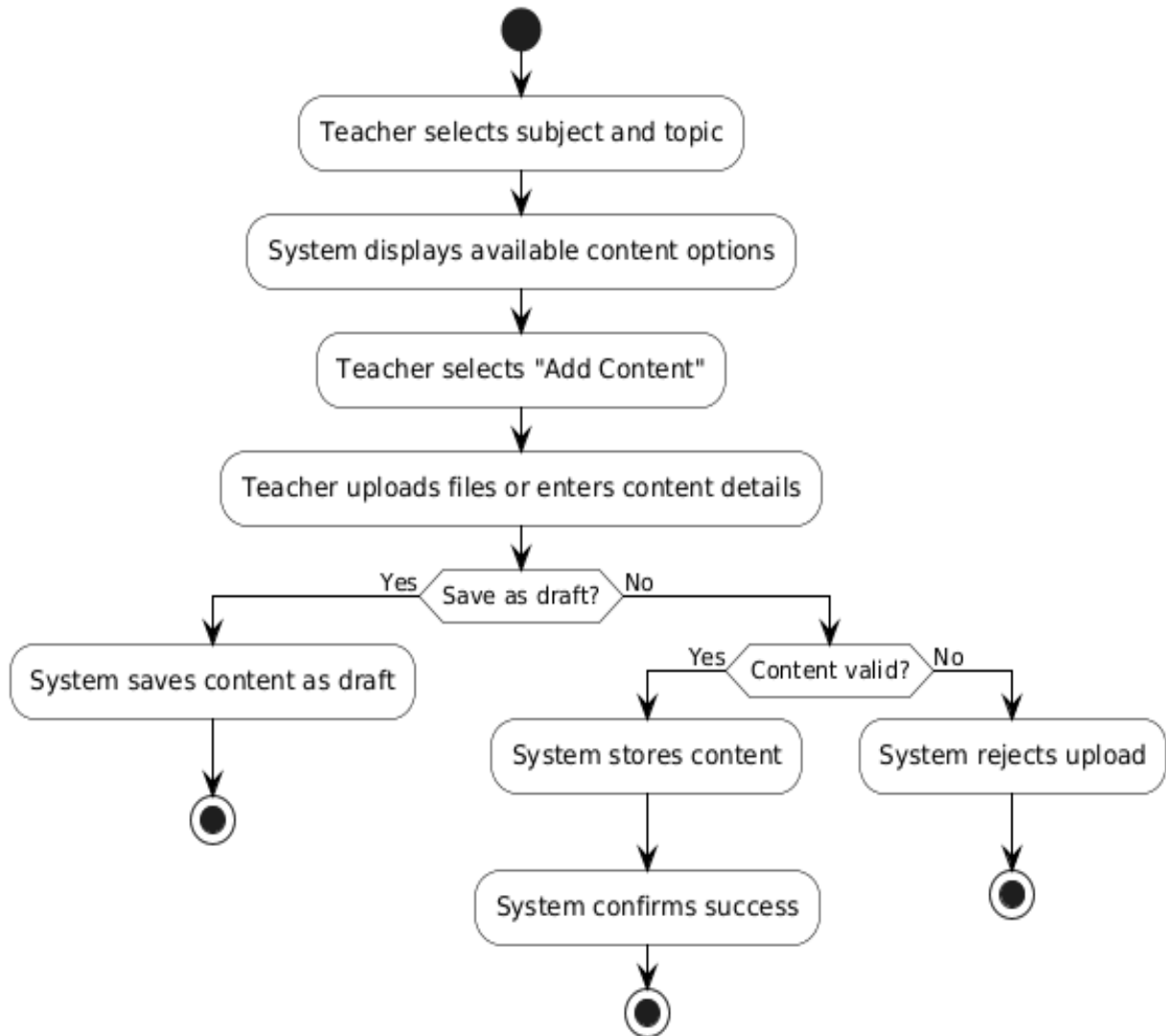


Figure 2-0-14 *Upload Subject Content Activity Diagram*

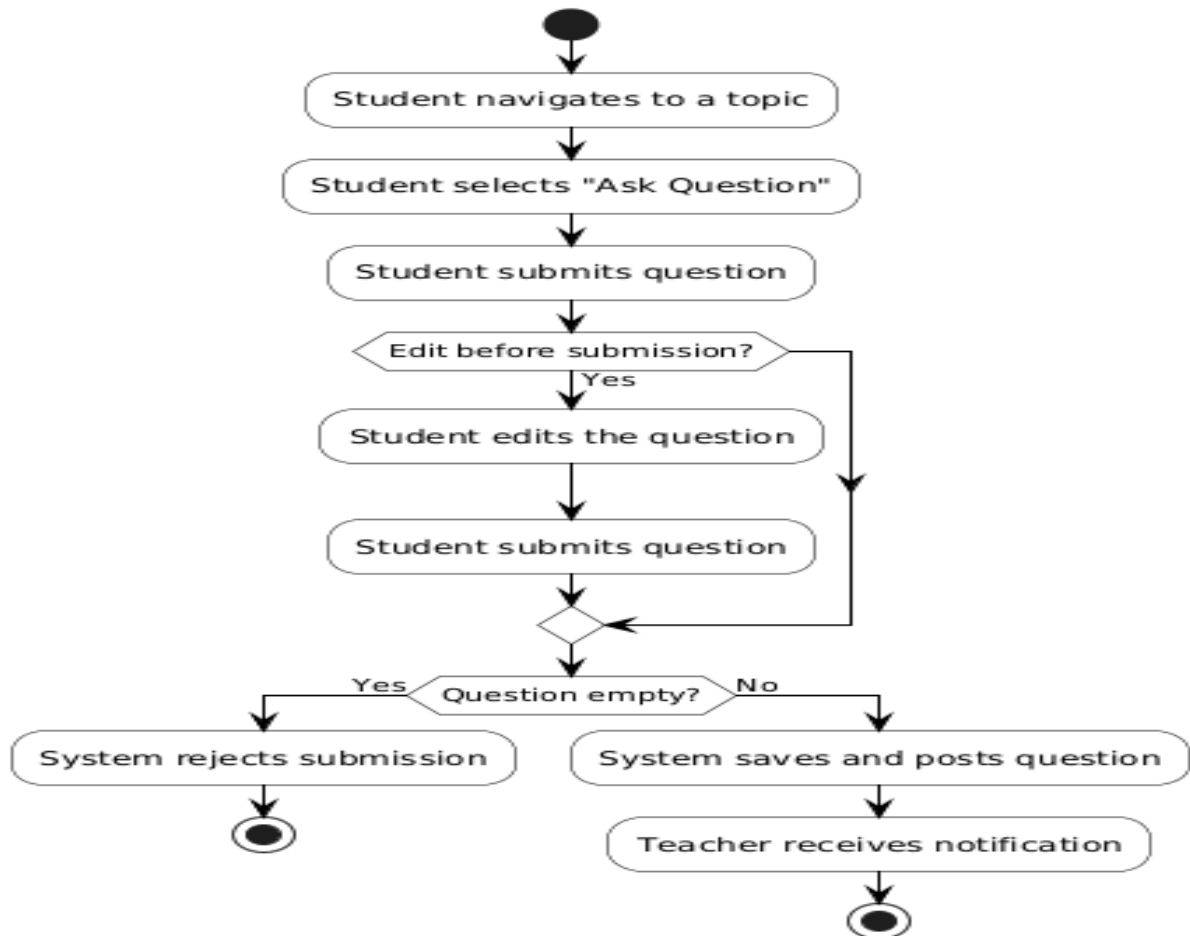
UC005 - Ask Questions Under Topic

Figure 2-0-15 Ask Question Activity Diagram

2.6.4.3 State Chart Diagrams

A state chart diagram describes a state machine. A state machine can be defined as a machine which defines different states of an object and these states are controlled by external or internal events. State machine diagrams represent behavior from the perspective of a single object. Viewing behavior from the perspective of each object enables the developer to build a more formal description of the object [1].

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Some selected to show state diagrams of the proposed system is as follows:

User_Account_Lifecycle_State_Diagram

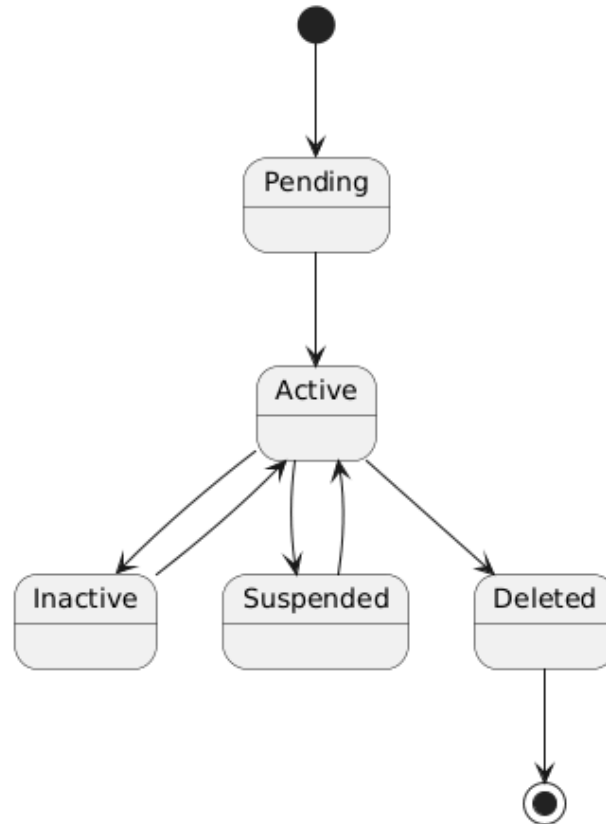


Figure 2-0-16 User Account Lifecycle State Diagram

Subject_Content_Lifecycle_State_Diagram

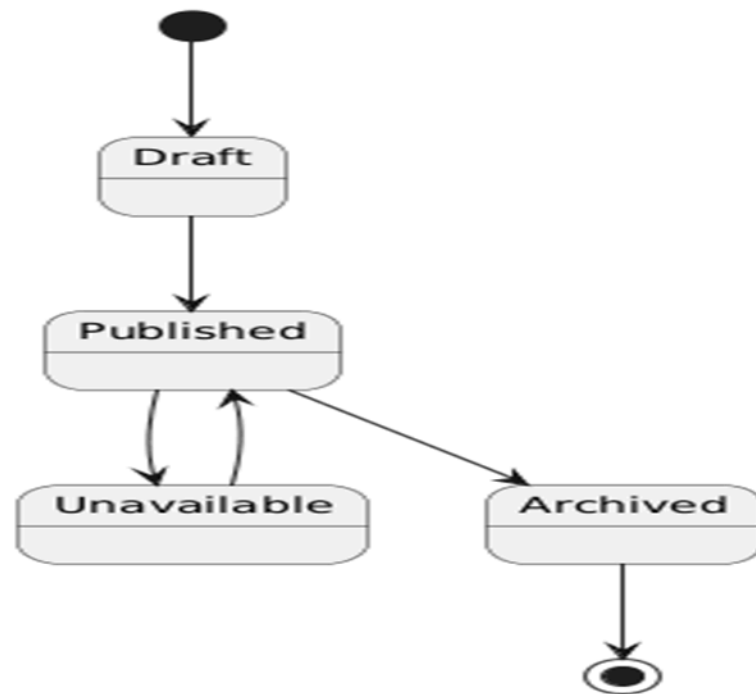
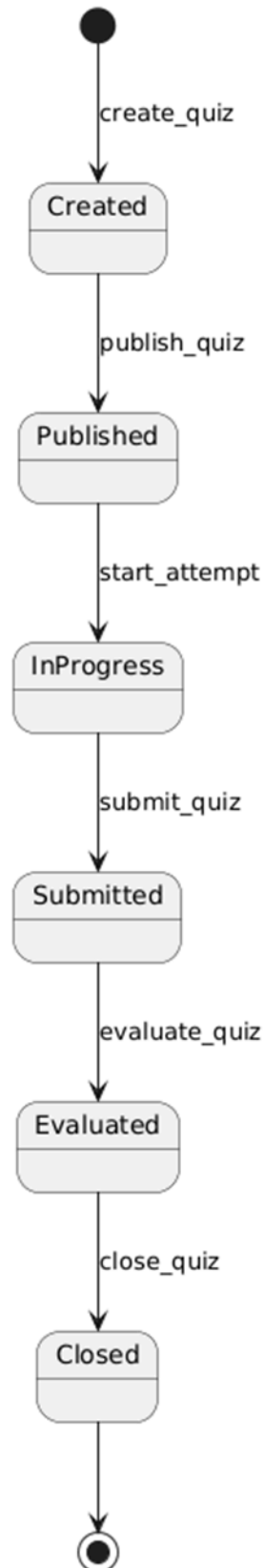


Figure 2-0-17 User Account Lifecycle State Diagram

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Quiz_Assessment_Lifecycle_State_Diagram



Student_Question_Lifecycle_State_Diagram

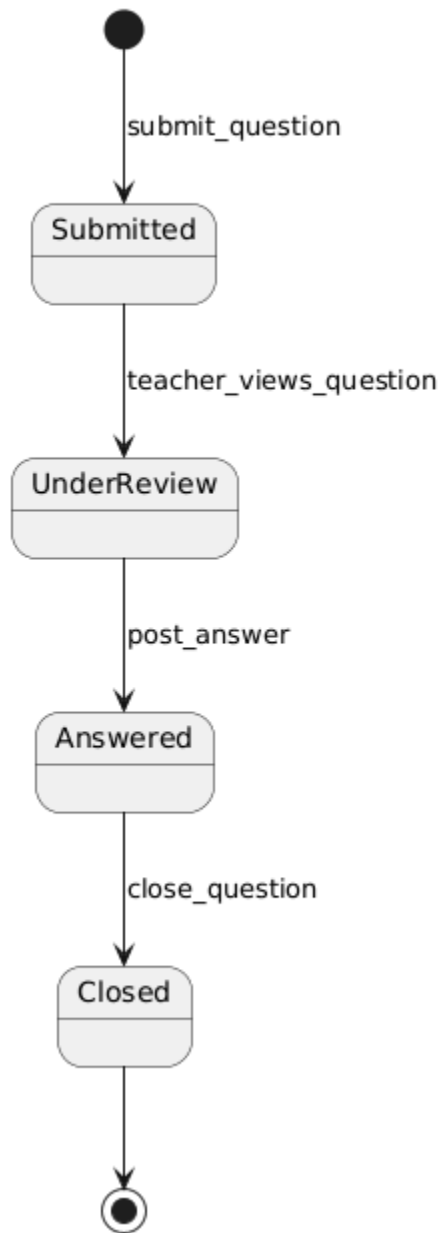


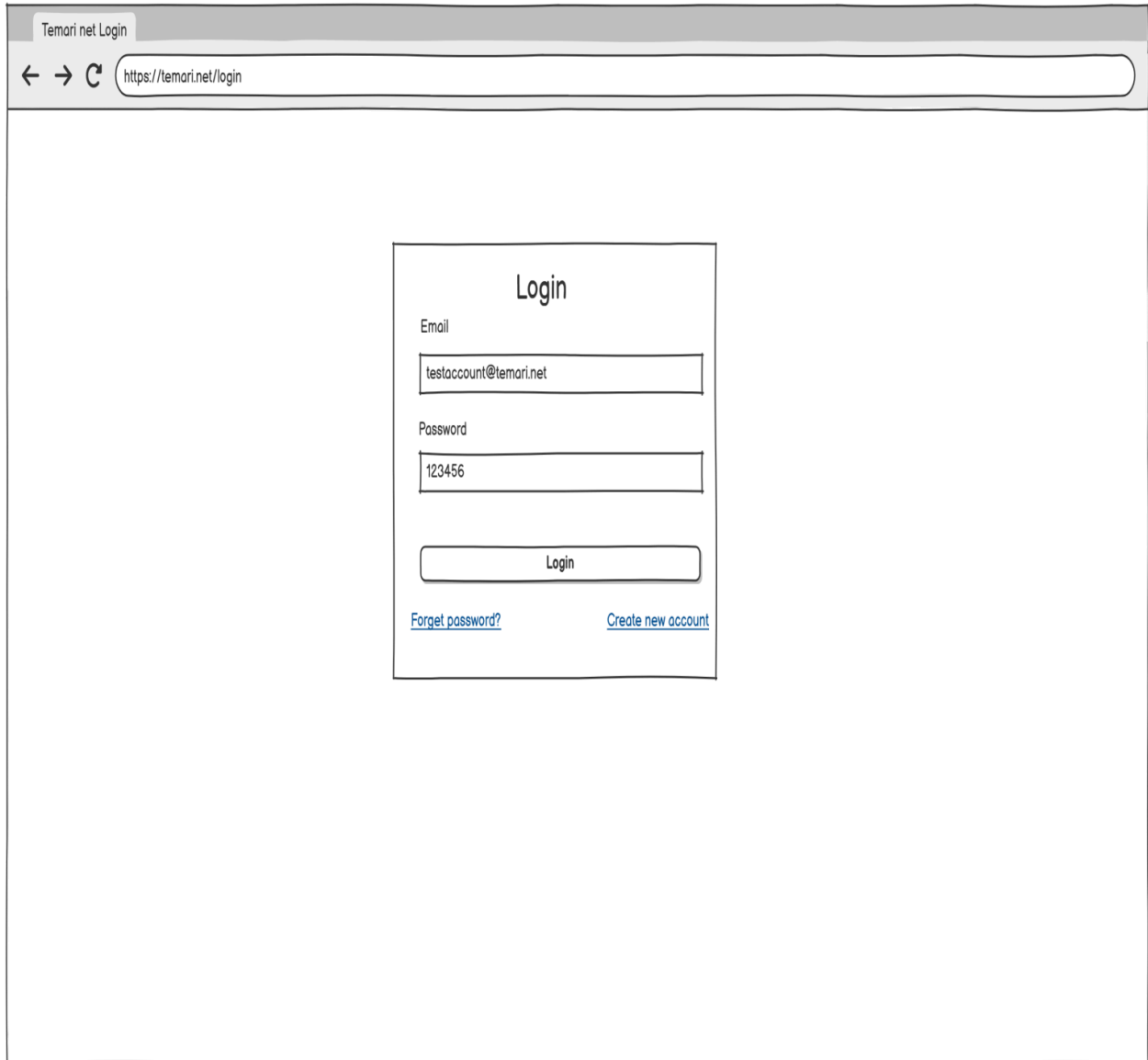
Figure 3-0-18 User Account Lifecycle State Diagram

CHAPTER THREE-SYSTEM DESIGN

2.6.5 User Interface

As per stated in the non-functional requirement the system implements an easy to use intuitive user interface.

The following selected user interface mockups show the screen mockups the system implements.⁶



The mockup shows a web browser window with the title 'Temari net Login' and the URL 'https://temari.net/login'. The main content area contains a centered login form. The form has a title 'Login', an 'Email' field with the value 'testaccount@temari.net', a 'Password' field with the value '123456', a 'Login' button, and two links: 'Forget password?' and 'Create new account'.

Temari net Login

← → ↻ https://temari.net/login

Login

Email

testaccount@temari.net

Password

123456

Login

[Forget password?](#) [Create new account](#)

Figure 2-0-19 User Login Page Mockup

⁶ <https://balsamiq.com/>

Ethiopia Entrance Exam Preparation Platform

Mathematics Grade 12

[←](#)
[→](#)
[↺](#)

https://temari.net/mathematics/12/chapter1/arithmetic/1.1

1.1 Introduction To Arithmetic Sequences

Arithmetic Sequences

▶

⏮

⏭

⏸

⏪

⏩

View Asked Questions

Ask Your Teacher

By the end of this topic, students will be able to:

- Define an arithmetic sequence
- Identify the first term and common difference
- Determine whether a given sequence is arithmetic
- Find the next terms of an arithmetic sequence
- Represent arithmetic sequences using simple notation
- Apply arithmetic sequences to basic real-life situations

Arithmetic Sequence – Concept Overview

An **arithmetic sequence** is a list of numbers where the **difference between consecutive terms is constant**.

Key Idea

If you subtract any term from the one after it, you always get the **same value**, called the **common difference (d)**.

Example

Sequence:

2, 5, 8, 11, 14

Common difference:

$d = 3$

General Form

An arithmetic sequence can be written as:

$a, a + d, a + 2d, a + 3d, \dots$

1.1 Introduction To Arithmetic Sequences

Introduction to Arithmetic Sequences

⏮

▶

⏭

Examples of Arithmetic Sequences

⏮

▶

⏭

Exercises>

[Have you found any issues on this page ? Report](#)

My Subjects

My Progress

Previous Year Exams

My Peer Group

Figure 3-0-20 Student Content Page (Concept Page) Mockup

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Mathematic Grade 12 Exercises

← → ↻ https://temari.net/mathematics/12/chapter1/arithmetic/exercises/1.1

[My Subjects](#)

[My Progress](#)

[Previous Year Exams](#)

[My Peer Group](#)

1 Which of the following is an arithmetic sequence?

☐ A. 3, 6, 12, 24

☒ B. 5, 8, 11, 14

☐ C. 3, 6, 18, 36

☐ D. 6, 12, 36, 46

Your answer is correct !

Explanation:

Only option B has a constant difference ($d = 3$) between consecutive terms.

2 Which of the following is an arithmetic sequence?

☒ A. 3, 6, 12, 24

☐ B. 5, 8, 11, 14

☐ C. 3, 5, 7, 10

☐ D. 2, 3, 5, 6

Your answer is wrong !

Try again !

Please refer the concept page and working examples.

[Have you found any issues on this page ? Report](#)

[Exam Questions >](#)

Figure 3-21 Student Content Page (Exercise) Mockup

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Mathematics Teacher

← → ↺

https://temari.net/mathematics/12/chapter1/arithmetic/content

👤

🔔

🏠

My Subjects

❓

New Questions

⊗

Reports

📄

Manage Quizzes

✎

Previous Year Exams

Create Topic Content

Study Resources

Insert Video Url

https://www.youtube.com/watch?v=XZJdyPkCxuE

https://www.youtube.com/watch?v=XZJdyPkCxuE

https://www.youtube.com/watch?v=XZJdyPkCxuE

+

Insert Concept Explanation

Arithmetic Sequence — Concept Overview

An **arithmetic sequence** is a list of numbers where the **difference between consecutive terms is constant**.

If you subtract any term from the one after it, you always get the **same value**, called the **common difference (d)**.

Example

Sequence:

2, 5, 8, 11, 14

Common difference:

$d = 3$

Insert Resource Files

Browse...

NEXT >

Figure 3-23 Teacher Topic Content Structure Page Mockup

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CHAPTER THREE-SYSTEM DESIGN

3.1 Introduction

This chapter presents the architectural decisions made to implement the proposed “**Entrance Exam Preparation platform**”.

The chapter includes the design goals and strategies how to meet them, the decomposition of the proposed system into multiple subsystems; the descriptions of services provided by each subsystem, an access control list to manage privileges of actors, and a detailed class diagram all are based on what have been presented on the previous requirement analysis chapter.

3.2 Proposed Software Architecture

3.2.1 Design Goals

The design goals have been derived from the non-functional requirements of the proposed system.

1. Performance

Performance refers to the ability of the system to efficiently handle requests and maintain responsiveness under heavy load.

The following performance optimization strategies will be implemented to meet the current design goal:

- **Caching:** Frequently requested data is identified and stored in in-memory data stores such as Redis. This minimizes repetitive database read operations and enables the system to return responses in significantly reduced time with at least 60% of read requests served from cache and average response times reduced to fewer than 300 milliseconds for cached data.

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- **Database Indexing:** Efficient database indexing shall be applied to frequently queried attributes such as searching and filtering of past year exam papers to optimize query execution. Indexing shall reduce query execution time by at least 50% for indexed queries.
- **Asynchronous Processing:** Time consuming operations, such as database reads or external service calls, are handled asynchronously. This prevents blocking of the main execution flow and allows users to continue interacting with other system functionalities without delay ensuring that 95% of user requests complete within 300 milliseconds under normal operating conditions.
- **Background Job Processing:** Long running tasks such as sending registration confirmation emails, notifying subject updates and quiz publishing are processed in the background, outside the request response lifecycle with 95% of background jobs completed within 30 seconds and no noticeable impact on user facing response times. This ensures that the core components handling user requests remain responsive even during high traffic periods.

2. Availability

Availability refers to the ability of the system to remain operational and accessible to users for the majority of the time, including during periods of high traffic.

The following availability strategies will be implemented to meet the current design goal:

- **Deployment:** The system shall be deployed using multiple application server instances to eliminate single points of failure. If one instance becomes unavailable, incoming requests shall be automatically redirected to healthy instances, ensuring continuous service availability.
- **Monitoring:** The system shall implement continuous monitoring using tools to track the health and performance of critical components, including server availability, response times, and resource utilization. Critical failures shall be detected within 1 minute, and alerts shall be generated to enable rapid response

3. Reliability

Reliability refers to the ability of the system to consistently perform its intended functions correctly and accurately over time, even in the presence of errors or unexpected conditions.

The following reliability strategies will be implemented to meet the current design goal:

- **Error Handling:** The system shall implement structured error handling mechanisms to gracefully manage runtime errors and failures.
- **Logging:** The system shall maintain detailed logs of system activities and errors. 100% of critical system events shall be logged to support fault diagnosis.

4. Scalability

Scalability refers to the ability of the system to handle increasing workloads and user growth without degradation in performance or availability.

The following scalability strategies will be implemented to meet the current design goal:

- **Server Load Distribution:** The system shall maintain balanced request distribution as the number of users grows, ensuring that no individual server instance becomes a bottleneck as traffic increases

3.2.2 Architecture

The proposed “**Entrance Exam Preparation Platform**” is designed as a distributed web-based application. The system architecture follows a multi-tier Client-Server model and implements the **Model-View-Controller (MVC)** architectural pattern to ensure a clean separation between data management, business logic, and user presentation.

1. Client-Server Model

The system is divided into two primary logical components: the client-side interface and the server-side processing unit.

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- **The Client:** Developed using the React.js framework, the client-side application runs within the user's web browser. It is responsible for handling user interactions, managing local state, and rendering the platform's interface.
- **The Server:** The backend is powered by Spring Boot (Java) and is hosted on a central server. It serves as the core of the system, processing incoming requests from the client, executing business rules (such as exam grading), and communicating with the database.

2. Model-View-Controller (MVC) Pattern

Following the standards of modern software engineering, the system utilizes the MVC pattern to organize its internal structure:

- **View:** The View represents the visual layer that allows users (Students, Teachers, and Admins) to interact with the system. In this platform, the View is composed of React components that display study materials, practice exams, and progress dashboards.
- **Controller:** The Controllers are implemented as RESTful APIs within the Spring Boot framework. They act as the intermediary between the View and the Model. When a user performs an action, the Controller receives the request, calls the appropriate services for processing, and returns the results to the View in JSON format.
- **Model:** The Model layer consists of Java objects (Entities) and MongoDB collections that represent the system's data and business logic. This includes data structures for User Profiles, Exam Questions, Subject Content, and Performance Analytics.

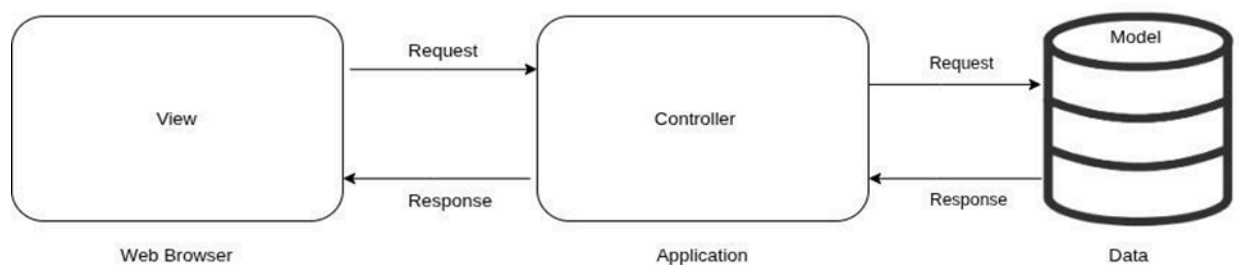


Figure 3-0-1 MVC Software Architecture Diagram

3.2.3 Subsystem Decomposition

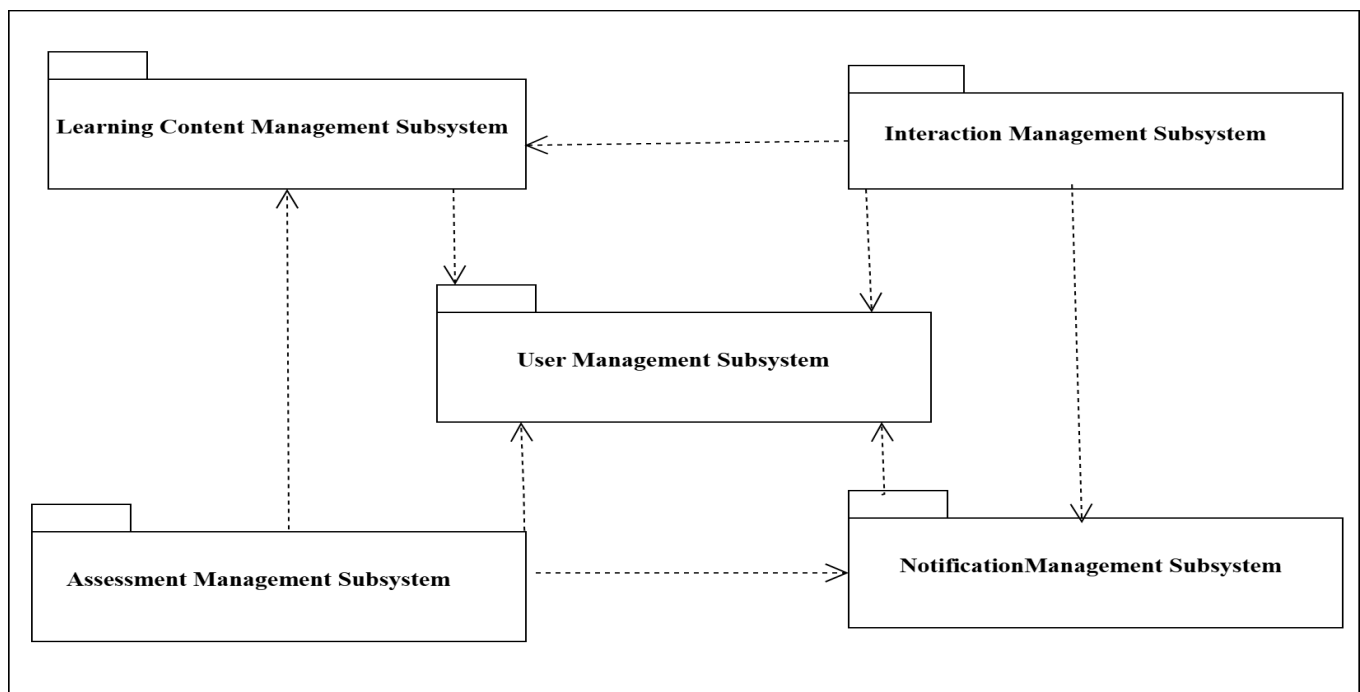
Subsystem decomposition is breaking down a complex system into smaller systems called subsystems. Subsystem decomposition helps to manage the implementation of the system.

The decomposition of the system “**Entrance Exam Preparation Platform**” has been made on the basis of functional responsibilities and grouping classes that satisfies a common functionality into a single subsystem.

This subsystems are the results of the decomposition process:

- User Management Subsystem
- Learning Content Management Subsystem
- Assessment Management Subsystem
- Interaction Management Subsystem
- Notification Management Subsystem

This system decomposition diagram shows the dependency between the above listed subsystems:



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Figure 3-0-2 System Decomposition Diagram

1. User Management Subsystem

The user management subsystem acts as the foundation for the rest of the subsystems and handles user registration, authentication, authorization, role and account life cycle, profile update and password management processes.

The following classes involved in this subsystem and their description provided below:

- **User:** an abstract class that acts as a base model containing shared attributes between Student, Teacher and Administrator classes.
- **Student:** represents learners who consume learning content.
- **Teacher:** represents educators who are responsible to manage subject content, provide assessment, and provide responses to student queries.
- **Administrator:** represents authority that has control over user's activity and learning content management.
- **AuthenticationController:** manages authentication flows such as login, logout, register, password reset.
- **UserController:** manages profile updates, password changes, user account activation and deactivation flows.

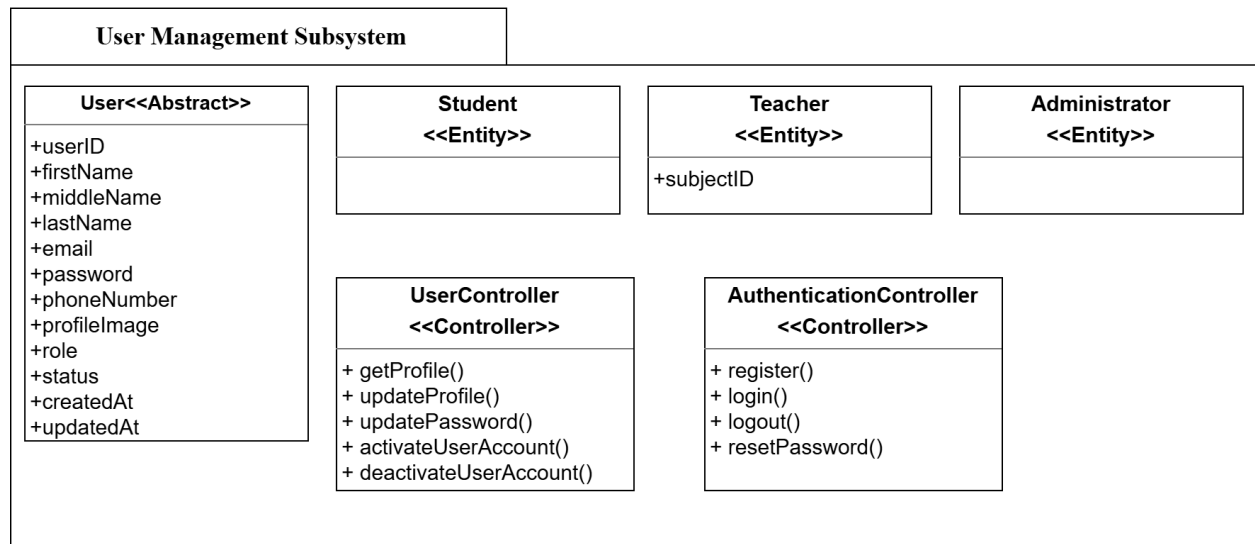


Figure 0-3 User Management Subsystem Diagram

2. Learning Content Management Subsystem

The learning content subsystem handles defining and managing subject structure and learning materials. Organization of subjects, chapters, topics and management of associated learning materials such as concept pages and videos.

The following classes involved in this subsystem and their description provided below:

- **Subject:** represents subjects under a grade level.
- **Chapter:** represents chapters under a subject.
- **Topic:** represents topics under a chapter.
- **ConceptPage:** represents textual concept explanation for a topic.
- **Video:** represents video resources associated with a topic.
- **SubjectController:** manages subject related operations such as creating, updating, deleting and assigning teachers.
- **ChapterController:** manages chapters under a subject.

CHAPTER THREE-SYSTEM DESIGN

- **TopicController:** manages topic related operations and their learning content such as concept explanations and videos.

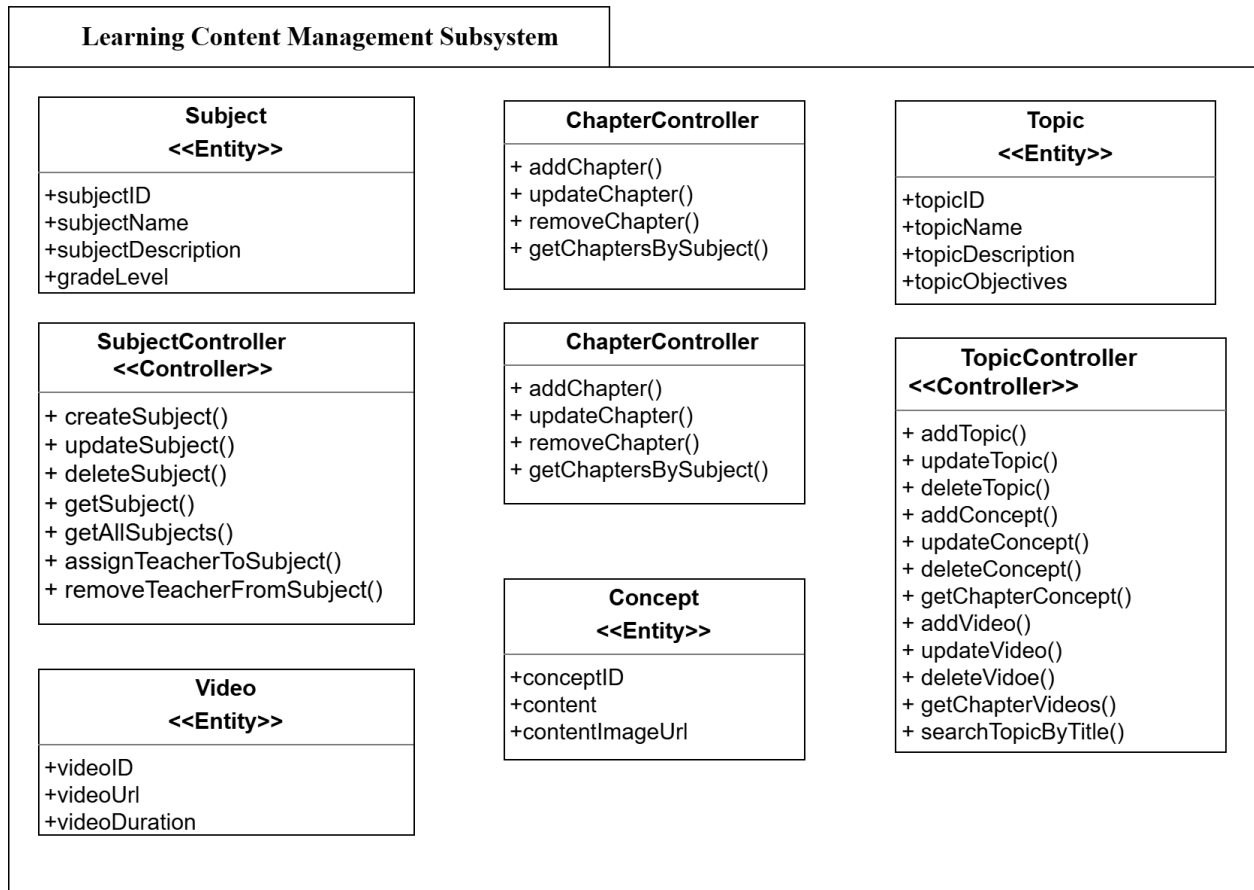


Figure 3-4 Learning Content Management Subsystem Diagram

3. Assessment Management Subsystem

The assessment subsystems are responsible for handling learning assessment and practice such as exercises, quizzes, past year examination problems, answer validation and feedback delivery to students.

The following classes involved in this subsystem and their description provided below:

- **Exercise:** represents practice problems under a topic.

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- **ExerciseProblem:** represents individual practice problems part of an exercise under a topic.
- **Quiz:** represents a quiz provided by a subject teacher for a topic.
- **QuizQuestion:** represents individual quiz questions under a quiz.
- **ExamPaper:** represents past year examination papers for each individual subject given in different years.
- **ExamQuestion:** represents the actual past year examination paper problems tagged by a topic.
- **ExcerciseController:** manages exercises and exercise problems operations.
- **QuizController:** manages quiz and quiz questions operations.
- **ExamPaperController:** manages fetching, searching and filtering of past year exam papers.
- **ExamQuestionController:** manages fetching of past year exams for an exam paper and performs answer validations.

CHAPTER THREE-SYSTEM DESIGN

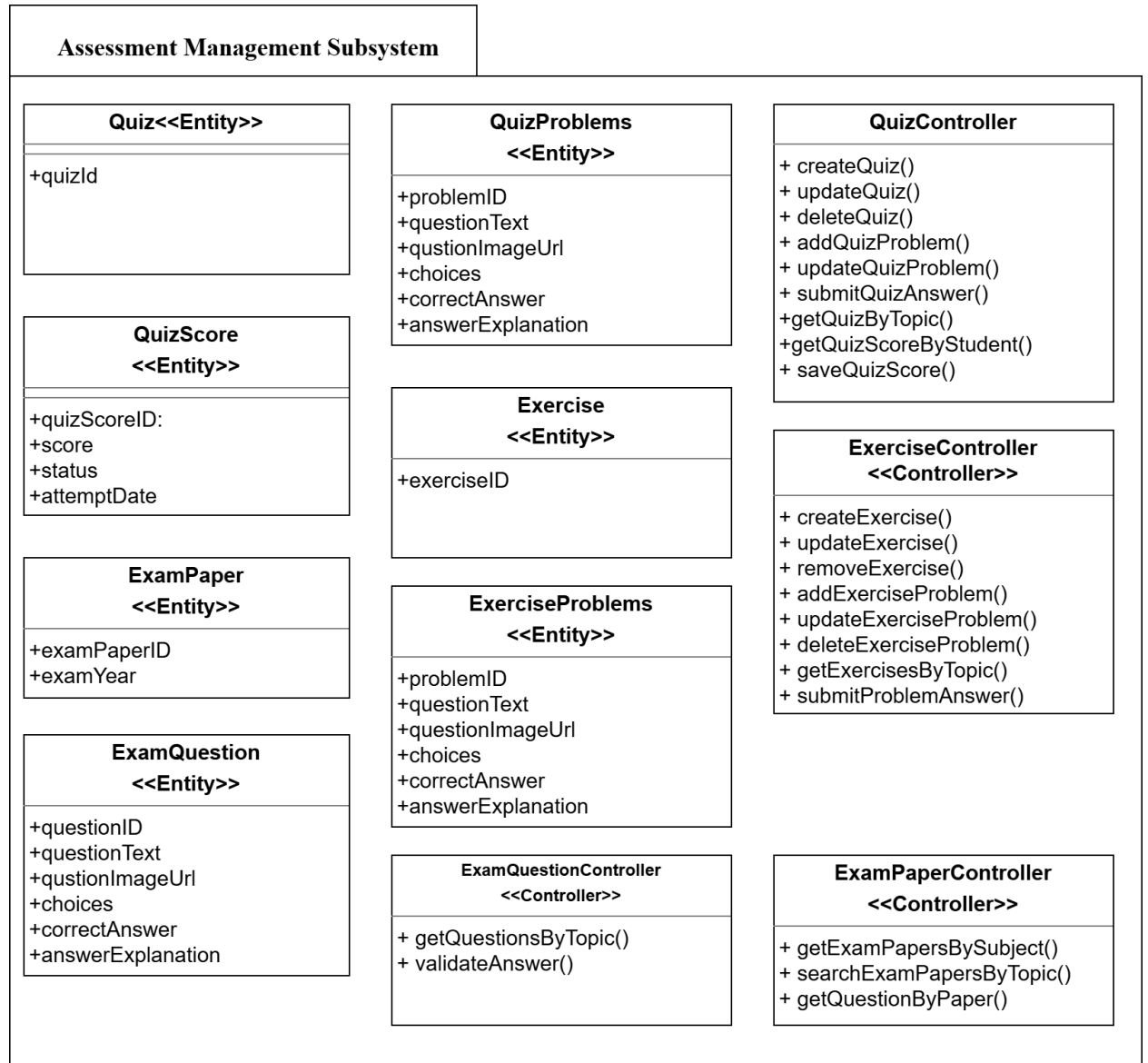


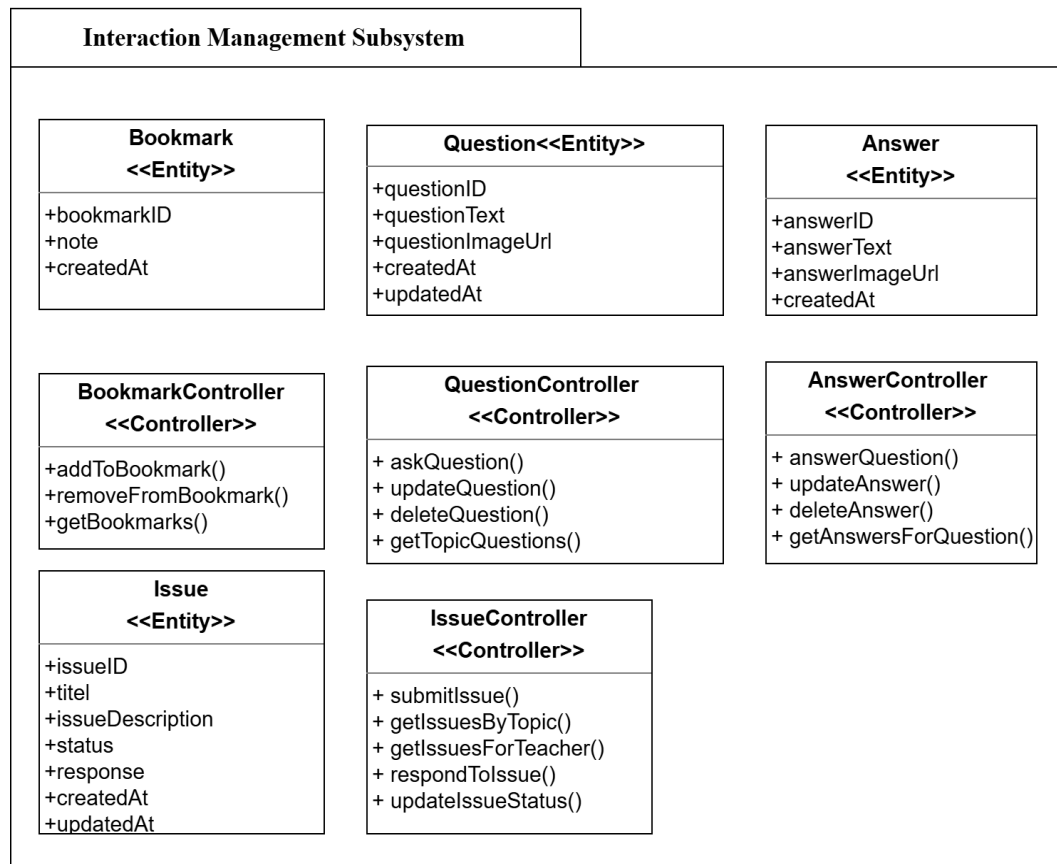
Figure 3-5 *Assessment Management Subsystem Diagram*

4. Interaction Management Subsystem

The interaction subsystem manages student driven interactions with the platform such as bookmarks, students ask questions and teacher provides answer, issue submission and response and topic completion tracking.

The following classes involved in this subsystem and their description provided below:

- **Issue:** represents issues submitted by a student.
- **IssueResponse:** represents responses given for submitted issues.
- **Question:** represents a student asked question under a topic discussion.
- **Answer:** represents an answer provided by a teacher for a student asked question under a topic discussion.
- **Bookmarks:** represents resources bookmarked by a student.
- **TopicProgress:** represents topic completion by a student.
- **IssueController:** manages issues submission and response flow.
- **QuestionController:** manages student's discussion question submission process.
- **AnswerController:** manages teacher discussion question answer process.
- **BookmarkController:** manages student's resource bookmark operations.
- **TopicProgressController:** manages student's progress under a subject, chapter and topic.



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Figure 3-6 Interaction Management Subsystem Diagram

5. Notification Management Subsystem

The notification subsystem handles delivery of system generated messages for users during registration, subject content updates, quiz posts and responses given to submitted issues.

The following classes involved in this subsystem and their description provided below:

- **Notification:** represents a message delivered to a user.
- **NotificationController:** sends and manages notifications triggered by system events.

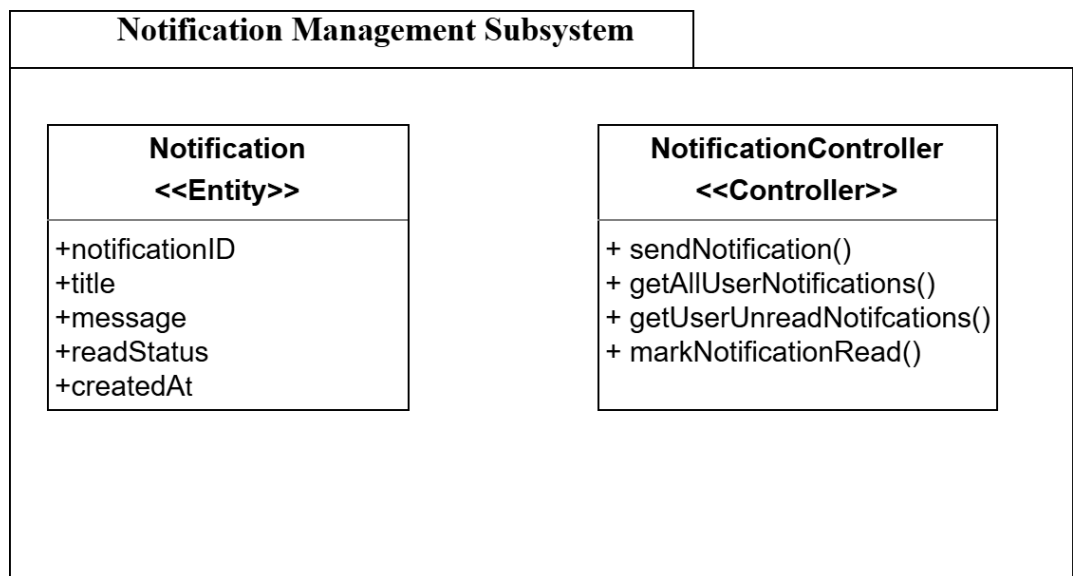


Figure 3-7 Notification Management Subsystem Diagram

3.2.4 Hardware/Software Mapping

The Hardware/Software mapping describes the physical configuration of the system and the allocation of software artifacts to the various hardware nodes.

Since the Entrance Exam Preparation Platform is a web-based system utilizing a client-server architecture, the subsystems are distributed across three primary physical locations: the Client Machine, the Application Server, and the Database Server

1. Client Machine

The client node represents the end-user's device, which can be a desktop computer, laptop, or mobile device (smartphone/tablet) equipped with a modern web browser.

- **Software Artifacts:** The React.js frontend bundle is downloaded and executed within the user's web browser
- **Role:** It is responsible solely for the User Interface Subsystem, handling user input and rendering data received from the server

2. Application Server (Web Server)

This node consists of a high-performance server (such as an AWS instance or a dedicated VPS) that remains in a listening state for client requests.

- **Software Artifacts:** It hosts the Spring Boot application (compiled as an executable JAR file).
- **Role:** It contains the core logic of the system, including the User Management, Assessment, and Content Management subsystems. It processes business rules and coordinates data exchange between the client and the database.

3. Database Server

The database server is a dedicated node optimized for persistent storage and rapid data retrieval.

- **Software Artifacts:** This node runs the MongoDB database engine.
- **Role:** It manages the Data Management Subsystem, storing persistent data such as student profiles, exam question banks, and media metadata. It ensures that data remains available even if the application server is restarted

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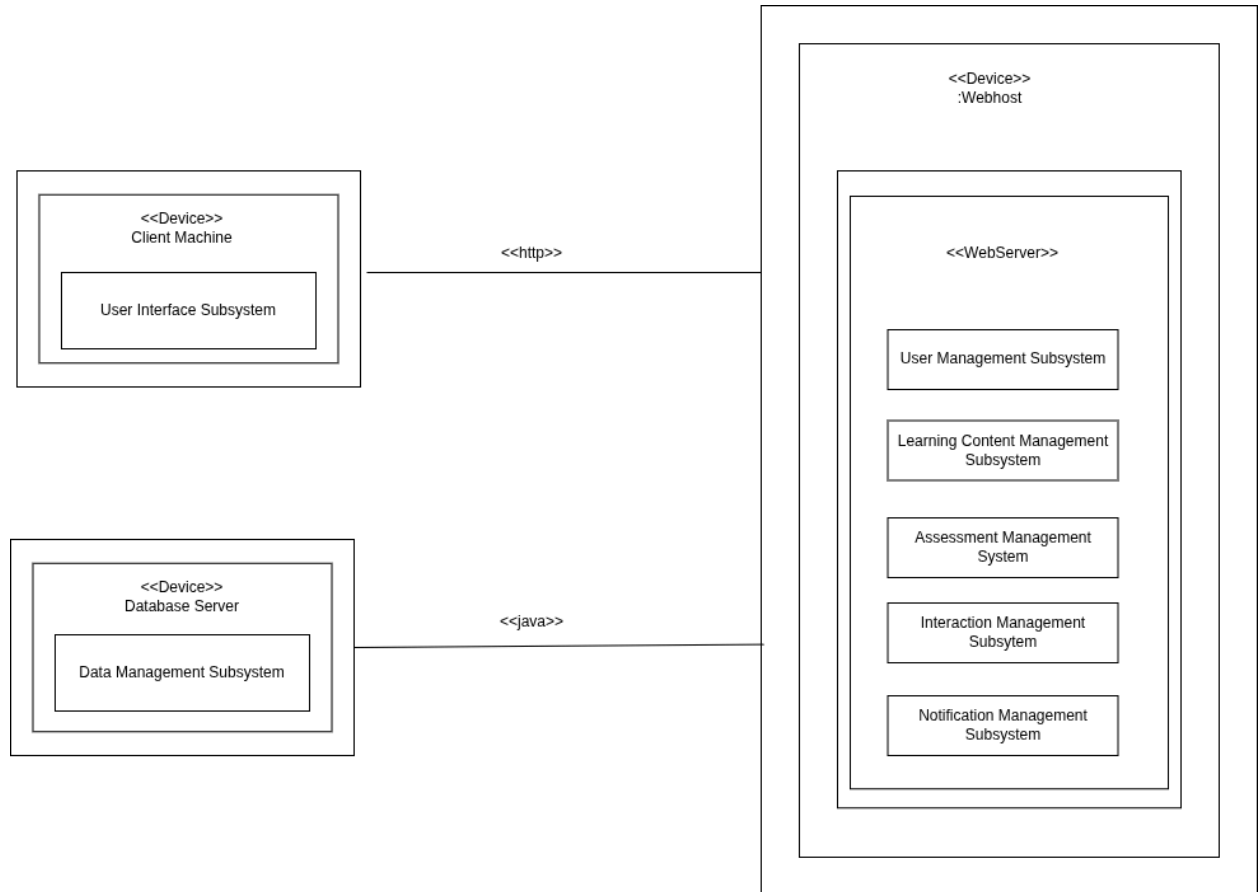


Figure 3-0-8 Hardware Software Mapping

3.2.5 Persistent Data Management

We have decided to use **MongoDB** to build the proposed “**Entrance Exam Preparation Platform**” because it efficiently handles flexible and hierarchical learning content, such as subjects, chapters, topics, videos, exercises, quizzes and past year exams, allowing us to easily add or modify fields as the platform evolves.

Its document-based, JSON-like structure naturally fits nested data, enabling faster reads and simpler queries compared to relational databases. MongoDB also scales horizontally to support growing numbers of students and handles read-heavy operations efficiently, which is ideal for content delivery.

The following tables present the mongoDB schema fields, their datatype and specify primary key, foreign key and indexed fields.

Table 3-0-1 User Entity Data Description**User Entity**

Field	Data Type	Constraints
userID	ObjectId	- Primary Key, - Auto-generated - Indexed
firstName	String	- Required
middelName	String	- Required
lastName	String	- Required
Email	String	- Unique - Indexed
Password	String	- Required - Hashed
phoneNumber	String	- Optional - Unique
profileImageUrl	String	- Optional
Status	Enum	- Required - ["STUDENT", "TEACHER", "ADMIN"]

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createdAt	Date	- Default now
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Table 3-0-2 Student Entity Data Description

Student Entity

Field	Data Type	Constraints
Stream	Enum	- Required - [“NATURAL” , “SOCIAL”]
userId	ObjectId	- Foreign Key to User Entity

Table 3-3 Teacher Entity Data Description

Teacher Entity

Field	Data Type	Constraints
userId	ObjectId	- Foreign Key to User Entity
subjectId	ObjectId	- Foreign Key to Subject Entity

Table 3-0-3 Administrator Entity Data Description

Administrator Entity

Field	Data Type	Constraints
userId	ObjectId	- Foreign Key to User Entity

Table 3-0-4 Subject Entity Data Description

Subject Entity

Field	Data Type	Constraints
subjectID	ObjectId	- Primary Key - Auto generated
subjectName	String	- Unique - Indexed
subjectDescription	String	- Required
gradeLevel	Number	- Required
Stream	Enum	- Required - [“ NATURAL” , “ SOCIAL”]

Table 3-0-5 *Chapter Entity Data Description***Chapter Entity**

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Field	Data Type	Constraints
chapterID	ObjectId	- Primary Key - Auto generated
chapterName	String	- Required
chapterDescription	String	- Required
examWeight	Number	- Required
subjectId	ObjectId	- Foreign Key to Subject Entity

Table 3-0-6 Entity Data Description

Topic Entity

Field	Data Type	Constraints
topicID	ObjectId	- Primary key - Auto generated
topicName	String	- Required
topicDescription	String	- Required
topicObjectives	Array<String>	- Required
chapterId	objectId	- Foreign Key to chapter

Table 3-0-7 TopicProgress Entity Data Description

TopicProgress Entity

Field	Data Type	Constraints
progressID	ObjectId	- Primary Key - Auto generated
completedAt	DateTime	- Required - default now
topicId	ObjectId	- Foreign Key to Topic Entity
studentId	ObjectId	- Foreign Key to Student Entity

Table 3-0-8 Concept Entity Data Description

Concept Entity

Field	Data Type	Constraints
conceptID	ObjectId	- Primary Key - Auto generated
Content	String	- Required
contentImageUrl	String	- Required
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-0-9 Video Entity Data Description

Video Entity

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Field	Data Type	Constraints
videoID	ObjectId	- Primary Key - Auto generated
videoUrl	String	- Required
videoDuration(min)	Number	- Required
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-10 Notification Entity Data Description

Notification Entity

Field	Data Type	Constraints
notificationID	ObjectId	- Primary Key - Auto generated
Title	String	- Required
Message	String	- Required
readStatus	Enum	- Required
createdAt	DateTime	- default now
userId	ObjectId	- Foreign Key to User

Table 3-0-11 Exercise Entity Data Description

Exercise Entity

Field	Data Type	Constraints
exerciseID	objectID	- Primary Key - Auto generated
description	String	- Required
numberOfProblems	DateTime	- Required
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-12 ExerciseProblem Entity Data Description

ExerciseProblem Entity

Field	Data Type	Constraints
questionID	ObjectId	- Primary Key - Auto generated
questionText	String	- Required
questionImageUrl	String	- Optional
choices	Array<String>	- Required
correctAnswer	Array<String>	- Required
answerExplanation	String	- Required
exerciseId	ObjectId	- Foreign Key to Exercise Entity

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Table 3-13 Quiz Entity Data Description

Quiz Entity

Field	Data Type	Constraints
quizID	ObjectId	- Primary Key - Auto generated
description	String	- Required
numberOfProblems	Number	- Required
duration	Number	- Required
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-0-14 QuizProblem Entity Data Description

QuizProblem

Field	Data Type	Constraints
questionID	ObjectId	- Primary Key - Auto generated
questionText	String	- Required
questionImageUrl	String	- Optional
choices	Array<String>	- Required
correctAnswer	String	- Required

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answerExplanation	String	- Required
quizId	ObjectId	- Foreign Key to Quiz Entity

Table 3-0-15 QuizScore Entity Data Description

QuizScore

Field	Data Type	Constraints
quizScoreId	ObjectId	- Primary Key - Auto generated
score	Number	- Required
status	Enum	- ["Complete", "Incomplete"]
attemptDate	DateTime	- Default now
studentId	ObjectId	- Foreign Key to Student Entity
quizId	ObjectId	- Foreign Key to Quiz Entity

Table 3-0-16 ExamPaper Entity Data Description

ExamPaper

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Field	Data Type	Constraints
examPaperID	ObjectId	- Primary Key - Auto generated
examYear	DateTime	- Default now - Indexed
subjectId	ObjectId	- Foreign Key to Subject Entity

Table 3-0-17 ExamQuestion Entity Data Description

ExamQuestion Entity

Field	Data Type	Constraints
questionID	ObjectId	- Primary Key - Auto generated
topic	String	- Required
difficulty	Enum	- Required - ["EASY", "MEDIUM", "DIFFICULT"]
questionText	String	- Required
questionImageUrl	String	- Optional
choices	Array<String>	- Required
correctAnswer	String	- Required

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answerExplanation	String	- Required
examPaperId	ObjectId	- Foreign Key to ExamPaper Entity

Table 3-0-18 Bookmark Entity Data Description

Bookmark

Field	Data Type	Constraints
bookmarkID	ObjectID	- Primary Key - Auto generated
note	String	- Optional
createdAt	DateTime	- Default now
studentId	ObjectId	- Foreign Key to Student Entity
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-0-19 Question Entity Data Description

Question

Field	Data Type	Constraints
questionID	ObjectID	- Primary Key - Auto generated
questionText	String	- Required

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questionImageUrl	String	- Optional
createdAt	DateTime	- Default now
updatedAt	DateTime	- Default now
topicId	ObjectId	- Foreign Key to Topic Entity

Table 3-0-20 Answer Entity Data Description

Answer

Field	Data Type	Constraints
answerID	ObjectID	- Primary Key - Auto generated
answerText	String	- Required
answerImageUrl	String	- Optional
createdAt	DateTime	- Default now
updatedAt	DateTime	- Default now
questionId	ObjectId	- Foreign Key to Question Entity

Table 3-0-21 *Issue Entity Data Description*

Issue

Field	Data Type	Constraints
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issueID	ObjectId	- Primary Key - Auto generated
title	String	- Required
issueDescription	String	- Required
status	Enum	- [“OPEN”, “CLOSED”]
response	String	- Required
createdAt	DateTime	- Default now
updatedAt	DateTime	- Default now
topicId	ObjectId	- Foreign to Topic Entity
studentId	ObjectId	- Foreign Key to Student Entity

3.2.6 Access Control and Security

Access Control List or ACL provides the privileges each actor of the system has over services provided in each subsystem. ACL can be used to determine which actor can access which functionalities of the system.

Proper hashing algorithms such as **bcrypt** will be used to hash the user’s password and store in the database in the hashed format rather than as a plain text to enforce security.

Json web tokens (**JWT**) in combination with session based authentication will be used to authenticate users and manage user account status.

The following table lists the actors of the system and their respective privileges over a service in a subsystem.

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Subsystem	Classes	Operations	Users		
			Student	Teacher	Administrator
User Management Subsystem	<i>UserController</i>	getProfile()	•	•	•
		updateProfile()	•	•	•
		updatePassword()	•	•	•
		activateUserAccount() ()			•
		deactivateUserAccount() unt()			•
	<i>AuthenticationController</i>	register()	•	•	
		login()	•	•	•
		logout()	•	•	•
		resetPassword()	•	•	•

Learning Content Management Subsystem	<i>SubjectController</i>	createSubject()			•
		updateSubject()			•
		deleteSubject()			•
		getSubject()	•	•	•
		getAllSubject()	•	•	•
		getChapters()	•	•	•
		assignTeacherToSubject()			•
		removeTeacherFromSubject()			•
	<i>ChapterController</i>	createChapter()		•	
		updateChapter()		•	
		deleteChapter()		•	
		getChaptersTopics()	•	•	•
	<i>TopicController</i>	addTopic()		•	

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		updateTopic()		•	
		deleteTopic()		•	
		addConcept()		•	
		updateConcept()		•	
		deleteConcept()		•	
		getChapterConcept()	•	•	•
		addVideo()		•	
		updateVideo()		•	
		deleteVideo()		•	
		getTopicVideos()	•	•	•
		SearchTopicByTitle()	•	•	•
Assessment Management Subsystem	<i>ExerciseController</i>	createExercise()		•	
		updateExercise()		•	

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		deleteExercise()		•	
		addExerciseProblem() ()		•	
		updateExerciseProblem() em()		•	
		deleteExerciseProblem() em()		•	
		getExercisesByTopic() c()	•	•	•
		validateAnswer()	•	•	•
	<i>QuizController</i>	createQuiz()		•	
		updateQuiz()		•	
		deleteQuiz()		•	
		addQuizProblem()		•	
		updateQuizProblem() ()		•	

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Interaction Management Subsystem	getQuizByTopic()		•	•	•
	validateAnswer()		•	•	•
	<i>ExamPaperController</i>	getExamPapersBySubject()	•	•	•
		searchExamPaperByTopic()	•	•	•
		getExamQuestionByPaper()	•	•	•
	<i>ExamQuestionController</i>	getExamQuestionsByTopic()	•	•	•
		validateAnswer()	•	•	•
	<i>BookmarkController</i>	addToBookmark()	•		
		removeFromBookmark()	•		
		getBookmarks()	•		
	<i>TopicProgressController</i>	completeTopic()	•		

		getChapterProgress()			
		getSubjectProgress()	•		
	<i>QuestionController</i>	askQuestion()	•		
		updateQuestion()	•		
		deleteQuestion()	•		
		getTopicQuestions()	•	•	•
	<i>AnswerController</i>	answerQuestion()		•	
		updateAnswer()		•	
		deleteAnswer()		•	
	<i>IssueController</i>	submitIssue()	•		
		getIssuesByTopic()		•	•
		respondToIssue()		•	•
		updateIssueStatus()		•	•

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Notification Management Subsystem	<i>NotificationController</i>	sendNotification()	•	•	•
		getAllUserNotifications()	•	•	•
		getAllUserUnreadNotifications()	•	•	•
		markNotificationRead()	•	•	•

Table 3-0-22 Access Control List

3.2.7 Subsystem Services

This section holds a detailed description of services provided by each class in each subsystem.

Table 3-23 *Subsystem Services*

Subsystem	Classes	Operations	Description
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User Management Subsystem	UserController	getProfile()	Retrieves the authenticated user's profile information.
		updateProfile()	Retrieves the authenticated user's profile information.
		updatePassword()	Updates editable user profile fields (e.g., name, email, contact info).
		activateUserAccount()	Enables a user account, allowing authentication and system access.
		deactivateUserAccount())	Enables a user account, allowing authentication and system access.
	AuthenticationController	register()	Creates a new user account with a specific user role.

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		login()	Authenticates a user using credentials and establishes a session.
		logout()	Terminates the user's active session.
		resetPassword()	Resets a user's password through a recovery process

Learning Content Management Subsystem	SubjectController	createSubject()	Creates a new subject in the academic structure.
		updateSubject()	Updates subject details such as name and description.
		deleteSubject()	Removes a subject and its associations

			from the system.
		getSubject()	Retrieves details of a specific subject.
		getAllSubject()	Retrieves a list of all available subjects.
		getChapters()	Retrieves all chapters under a subject.
		assignTeacherToSubject()	Associates a teacher with a subject for instructional responsibility.
		removeTeacherFromSubject()	Removes the teacher assignment from a subject.

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	<i>ChapterController</i>	createChapter()	Creates a new chapter under a subject.
		updateChapter()	Updates chapter metadata.
		deleteChapter()	Removes a chapter from a subject.
		getChaptersTopics()	Retrieves all topics associated with a chapter.
	<i>TopicController</i>	addTopic()	Creates a new topic under a chapter.
		updateTopic()	Updates topic details.

		deleteTopic()	Removes a topic from a chapter.
		addConcept()	Add a Concept page to a topic.
		updateConcept()	Updates a topic concept page.
		deleteConcept()	Removes a concept page from a topic.
		getChapterConcept()	Retrieves a topic concept Page
		addVideo()	Adds a video resource into a topic.
		updateVideo()	Updates a video resource under a topic.

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		deleteVideo()	Removes a video from a topic.
		getTopicVideos()	Retrieves all videos under a topic.
		searchTopicByTitle()	Searches for a topic given the topic title.

Assessment Management Subsystem	ExerciseController	createExercise()	Creates a new exercise set under a topic.
		updateExercise()	Updates exercise metadata
		deleteExercise()	Removes an exercise from the system.
		addExerciseProblem()	Adds a problem/question to an exercise.

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		updateExerciseProblem()	Updates an exercise problem.
		deleteExerciseProblem()	Removes a problem from an exercise.
		getExercisesByTopic()	Retrieves all exercises linked to a topic.
		validateAnswer()	Validates a student's submitted answer for an exercise problem.
	QuizController	createQuiz()	Creates a quiz under a topic.
		updateQuiz()	Updates quiz details.
		deleteQuiz()	Removes a quiz.

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		addQuizProblem()	Adds a question to a quiz.
		updateQuizProblem()	Updates a quiz question.
		getQuizByTopic()	Retrieves quizzes associated with a topic.
		validateAnswer()	Validates a student's answer for a quiz question
	ExamQuestionController	getExamQuestionsByTopic()	Retrieves past exam questions related to a topic
		validateAnswer()	Validates a student's answer for an exam question.
	ExamPaperController	getExamPapersBySubject()	Retrieves all exam papers for a subject.

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		searchExamPaperByTopic()	Searches exam papers that include questions from a specific topic.
		getExamQuestionByPaper()	Retrieves all questions belonging to a specific exam paper.

Interaction Management Subsystem	BookmarkController	addToBookmark()	Adds a topic to the student's bookmarks.
		removeFromBookmark())	Removes a topic from the student's bookmarks.
		getBookmarks()	Retrieves all bookmarked topics for a student.
	TopicProgressController	completeTopic()	Marks a topic as completed by a student.
		getChapterProgress()	Calculates student's completion percentage for a chapter
		getSubjectProgress()	Calculates & returns a student's completion percentage for a subject.

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	QuestionController	askQuestion()	Creates new student questions under a topic.
		updateQuestion()	Updates a student's question.
		deleteQuestion()	Deletes a student's question.
		getTopicQuestions()	Retrieves all questions associated with a topic
	AnswerController	answerQuestion()	Submits an answer to a question
		updateAnswer()	Updates an existing answer.
		deleteAnswer()	Deletes an answer.
	IssueController	submitIssue()	Submits an issue related to a topic.
		getIssuesByTopic()	Retrieves issues reported under a topic.
		respondToIssue()	Submits a response to an issue (teacher or admin).
		updateIssueStatus()	Updates the status of an issue (e.g., open, resolved).

NotificationController	sendNotification()	Creates and delivers a notification to a target user.
	getAllUserNotifications()	Retrieves all notifications for a user.
	getAllUserUnreadNotifications()	Retrieves only unread notifications for a user
	markNotificationRead()	Marks a specific notification as read.

3.3 Detailed Class Diagram

A detailed class diagram is used to identify missing attributes and operations. The pre and post conditions and invariants for some of selected classes are shown in the following diagrams.

User: this class is responsible for representing system users and maintaining consistent identity, authentication credentials, roles, and account status.

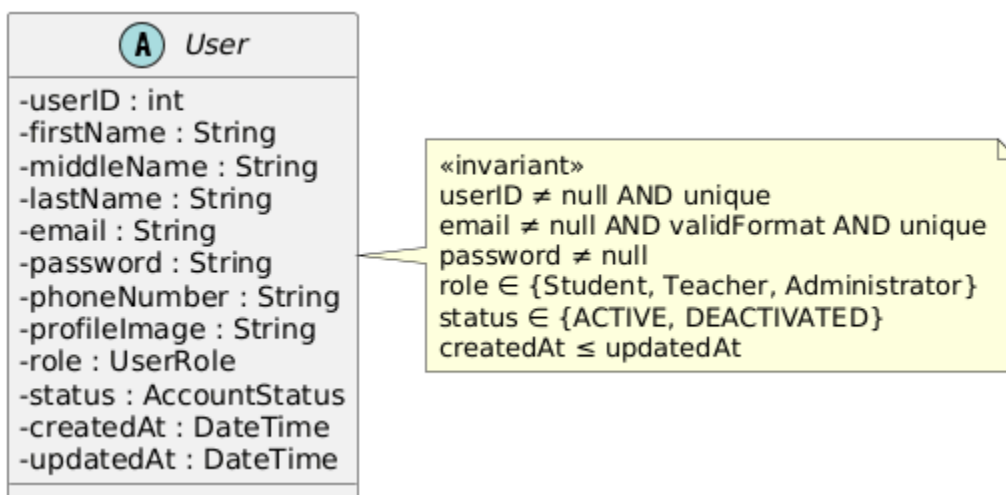


Figure 3-0-9 User Class Detailed Diagram

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AuthenticationController: this class is responsible for handling user authentication operations including registration, login, logout, and password reset.



Figure 3-10 AuthenticationController Class Detailed Diagram

Question: this class is responsible for storing and maintaining questions asked under a topic and ensuring their correct association with topics and creators.

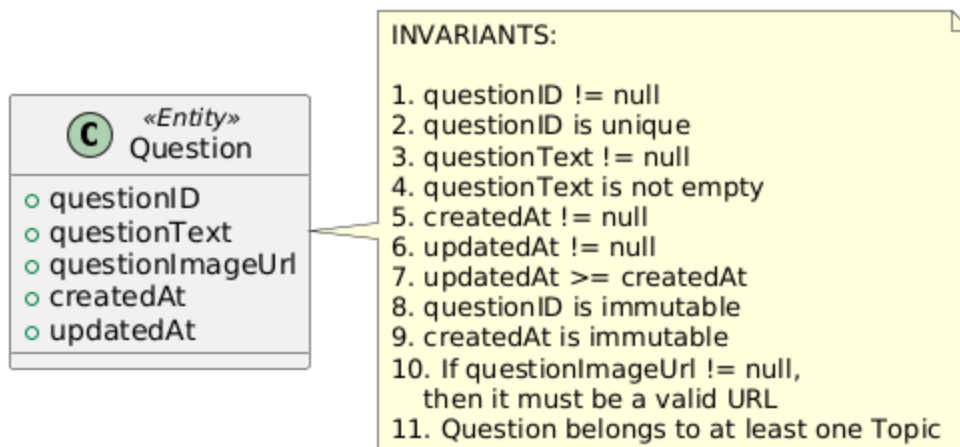


Figure 3-11 Question Class Detailed Diagram

QuestionController: this class is responsible for managing question-related operations within the system.

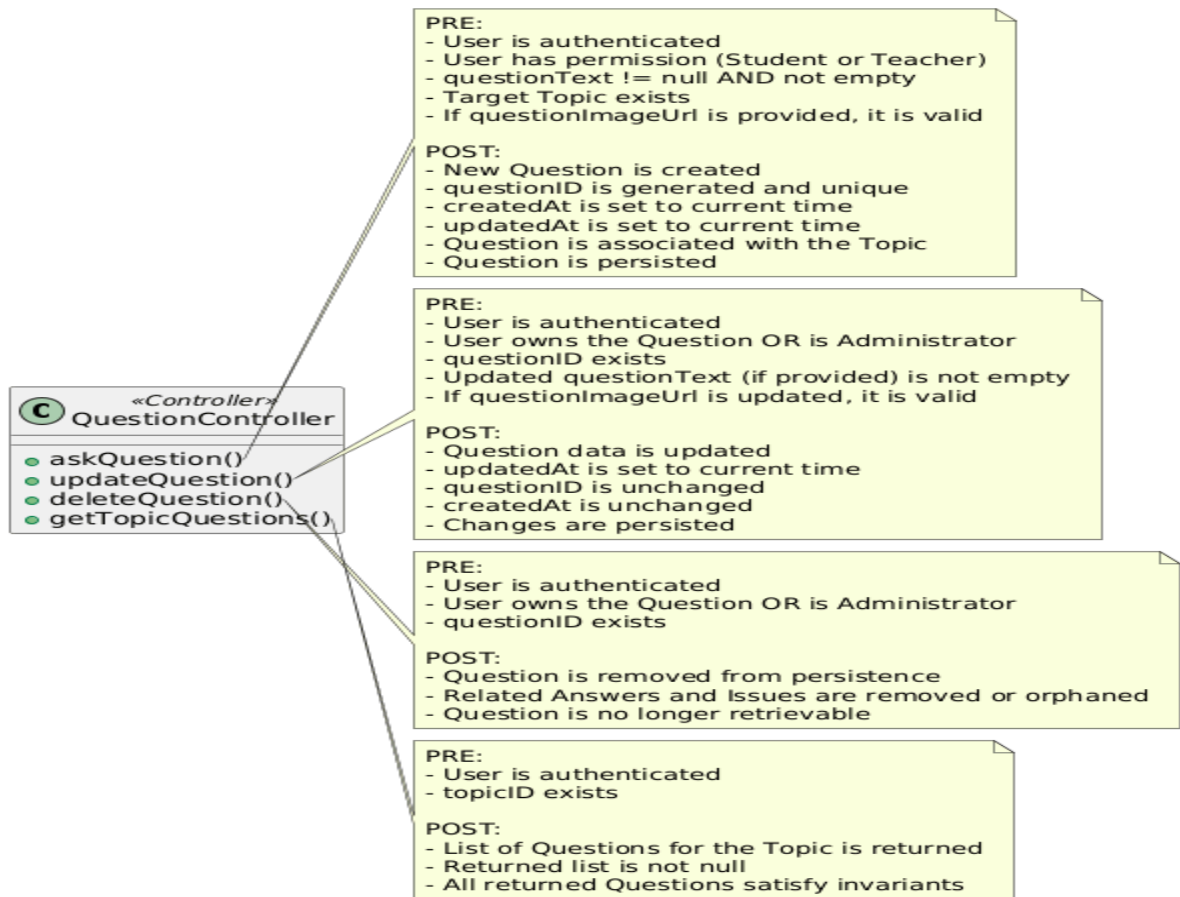


Figure 3-0-12 QuestionController Class Detailed Diagram

Quiz: this class is responsible for representing quizzes and maintaining valid assessment constraints.

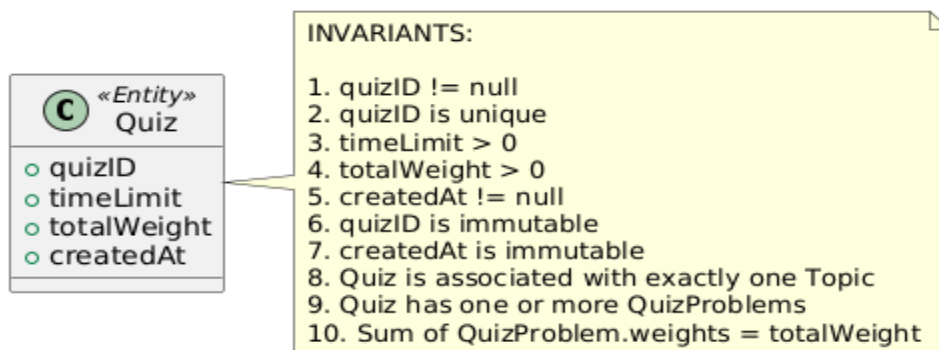


Figure 3-0-13 Quiz Class Detailed Diagram

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QuizController: this class is responsible for managing quiz creation, participation, and scoring.

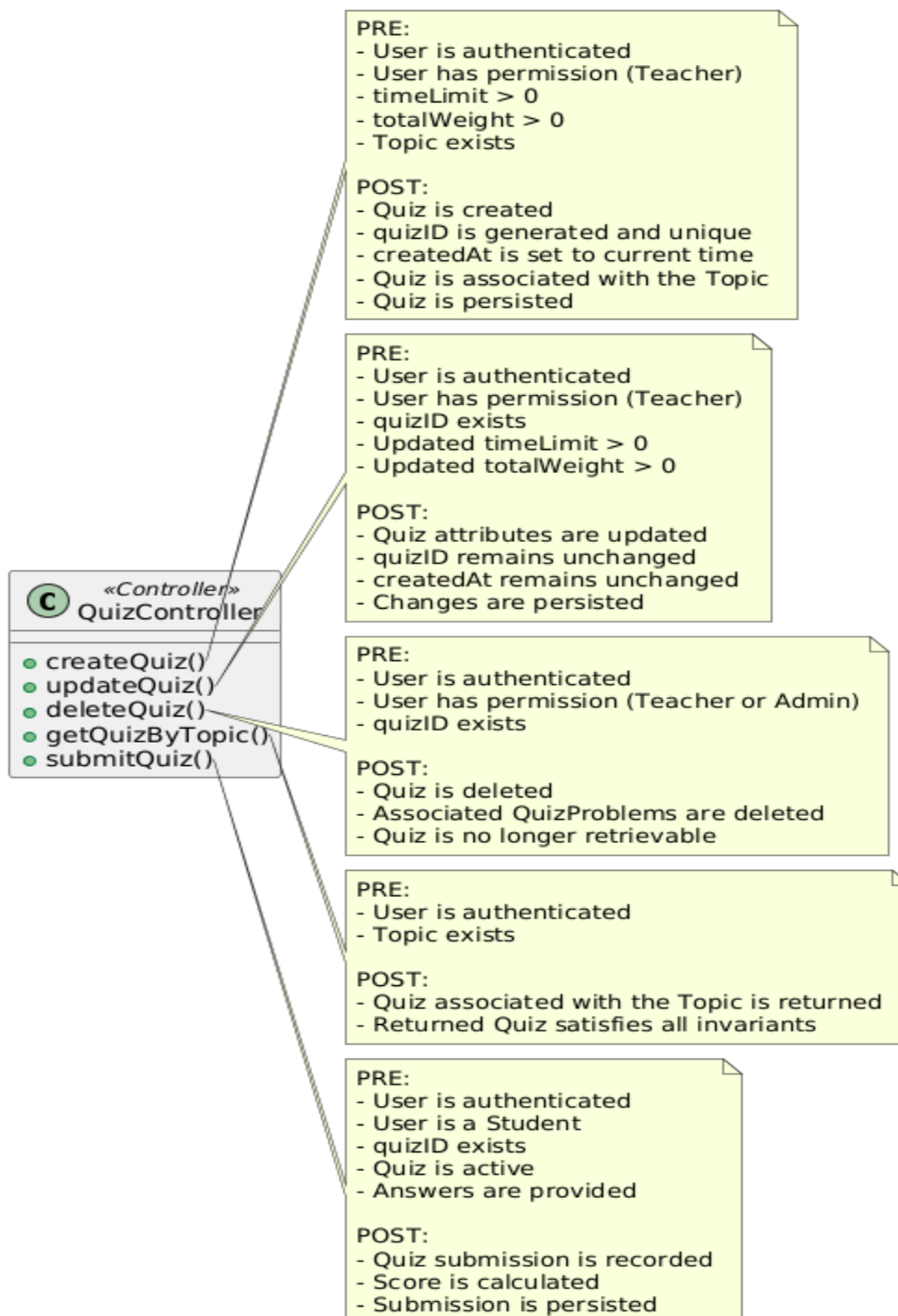


Figure 3-0-14 QuizController Class Detailed Diagram

Bookmark: this class is responsible for representing bookmarks made by students.

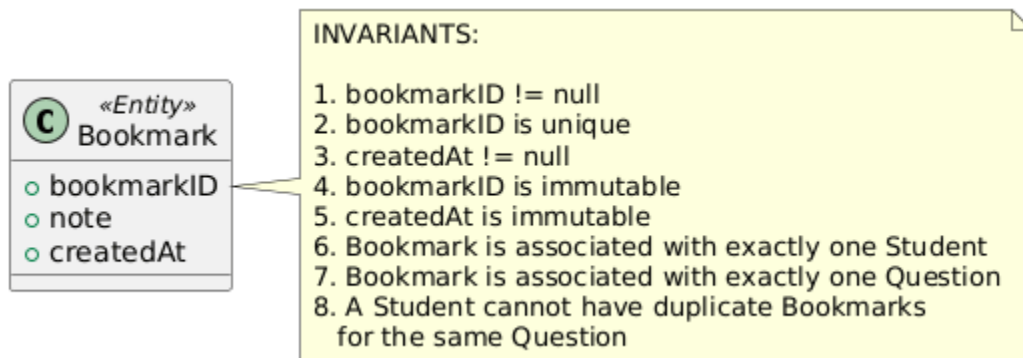


Figure 3-0-15 Bookmark Class Detailed Diagram

BookmarkController: this class is responsible for managing bookmarks.

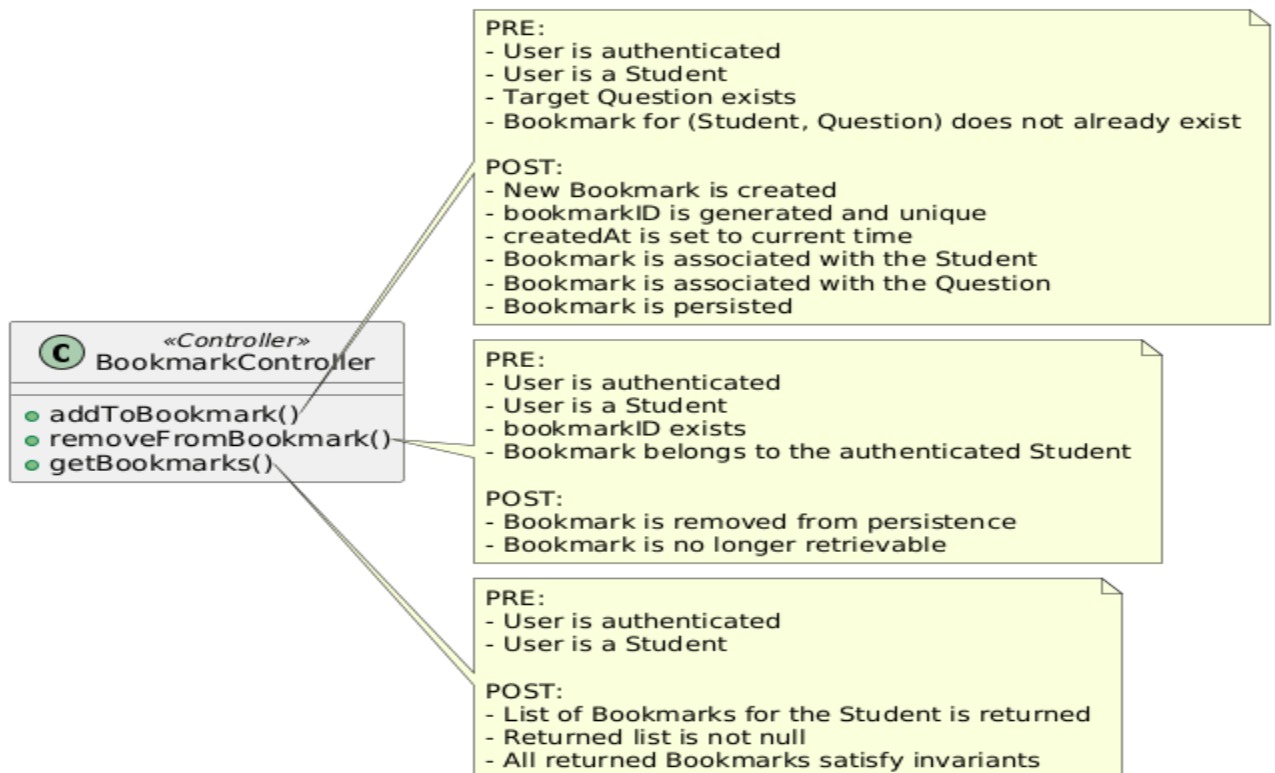


Figure 3-0-16 BookmarkController Class Detailed Diagram

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Notification: this class is responsible for representing notifications and maintaining their delivery and read-state consistency.

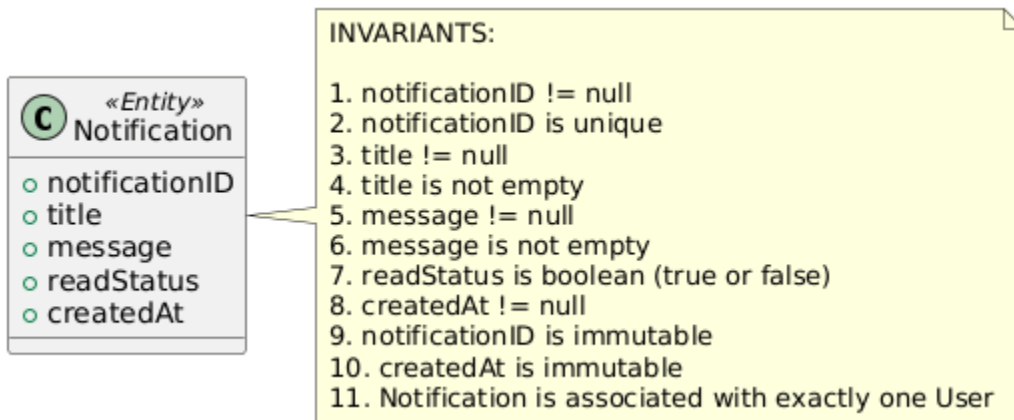


Figure 3-0-17 NotificationClass Detailed Diagram

NotificationController: this class is responsible for creating an environment that allows the system to send and manage notifications for users.

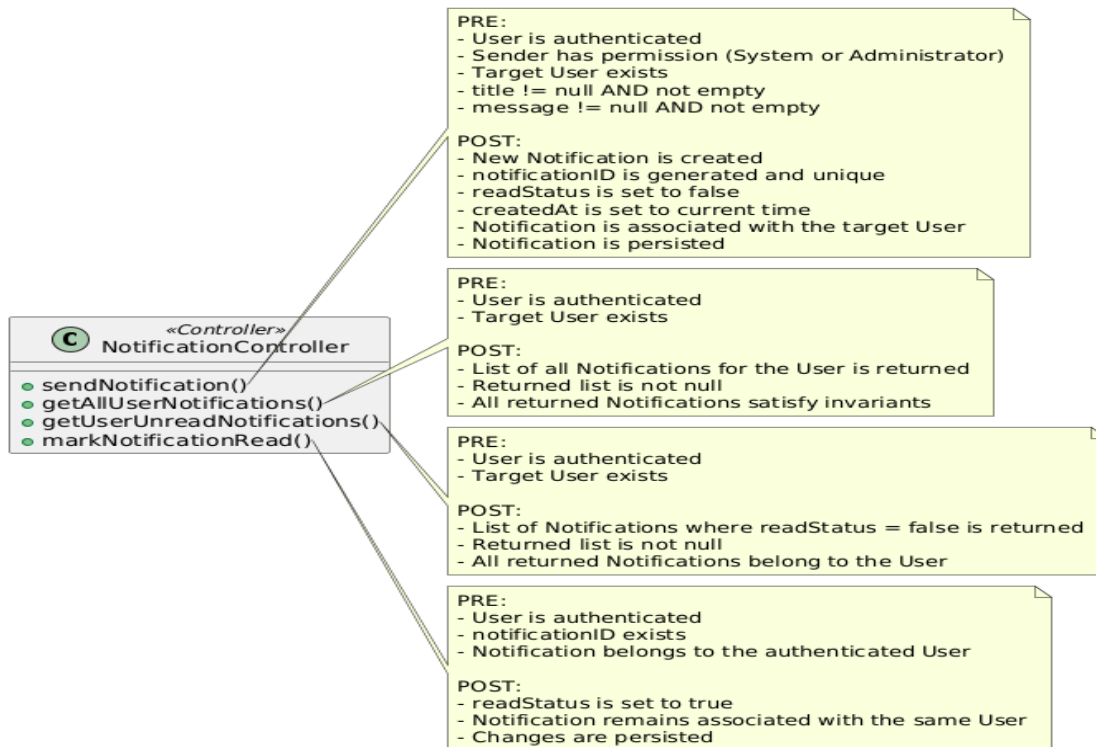


Figure 3-0-18 NotificationController Class Detailed Diagram

3.4 Packages

The system is organized into three main packages: Interface Package, Controller Package, and Storage Package. The Interface Package only communicates with the Controller Package, and controllers use the Storage Package for all persistent data operations.

Interface Package

The Interface Package contains all web UI components for Students, Teachers, and Administrators. It renders pages such as login/registration, subject and topic views, exercises and quizzes, progress dashboards, bookmarks, questions, and issue reporting screens, and calls controllers through HTTP endpoints.

Controller Package

The Controller Package implements the business logic and coordinates requests from the UI to the data layer. It includes controllers for authentication and user management, subject/chapter/topic and content management, exercises/quizzes/exam problems, bookmarks and progress tracking, questions and answers, issue handling, and notifications.

Storage Package

The Storage Package defines entities and data-access components mapped to the database. It stores users (Student, Teacher, Administrator), structure entities (Subject, Chapter, Topic), learning content (videos, concept pages, exercises, quizzes, previous exam problems), interaction data (bookmarks, questions, answers, issues, notifications), and progress records, and exposes repository operations used by controllers.

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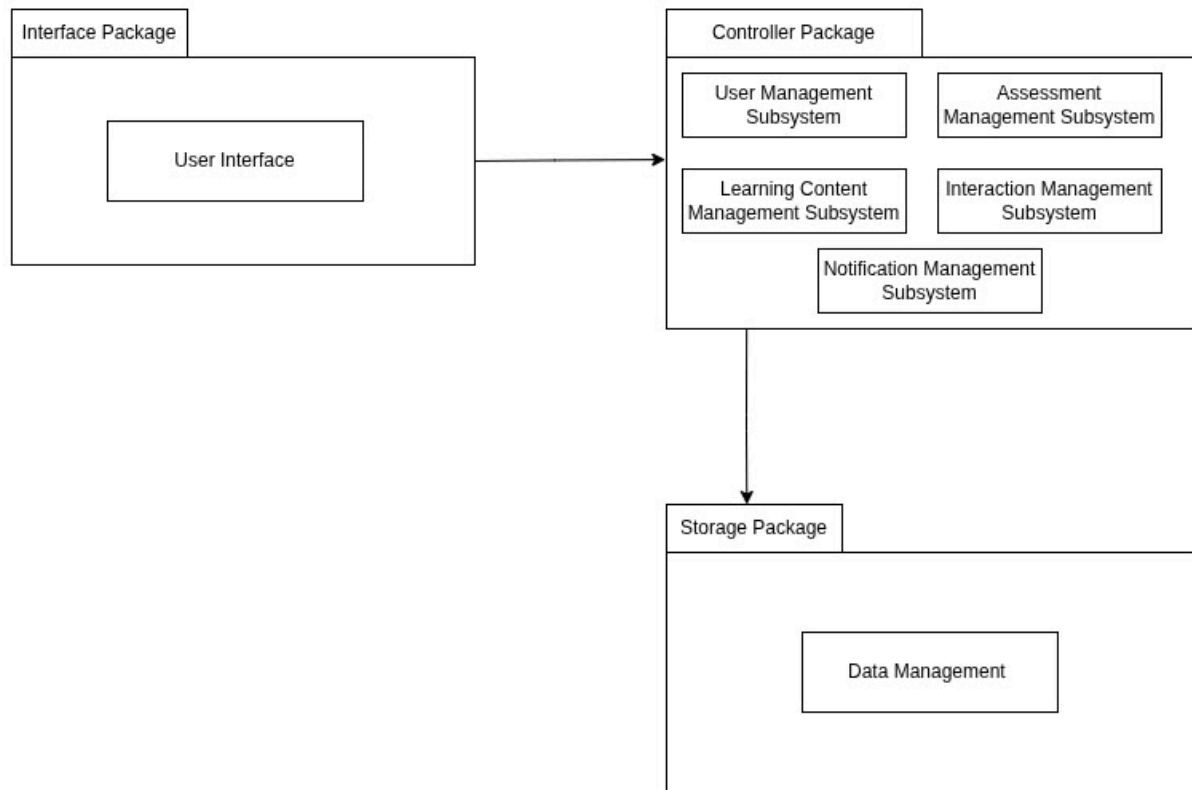


Figure 3-19 Package Diagram

CHAPTER FOUR - IMPLEMENTATION

CHAPTER FOUR - IMPLEMENTATION

4.1 Introduction

The implementation phase has mapped the system design document into actual code. This chapter consists of screenshots for selected user interface pages and code snippets of selected classes.

4.2 Screenshots

1. Sign In

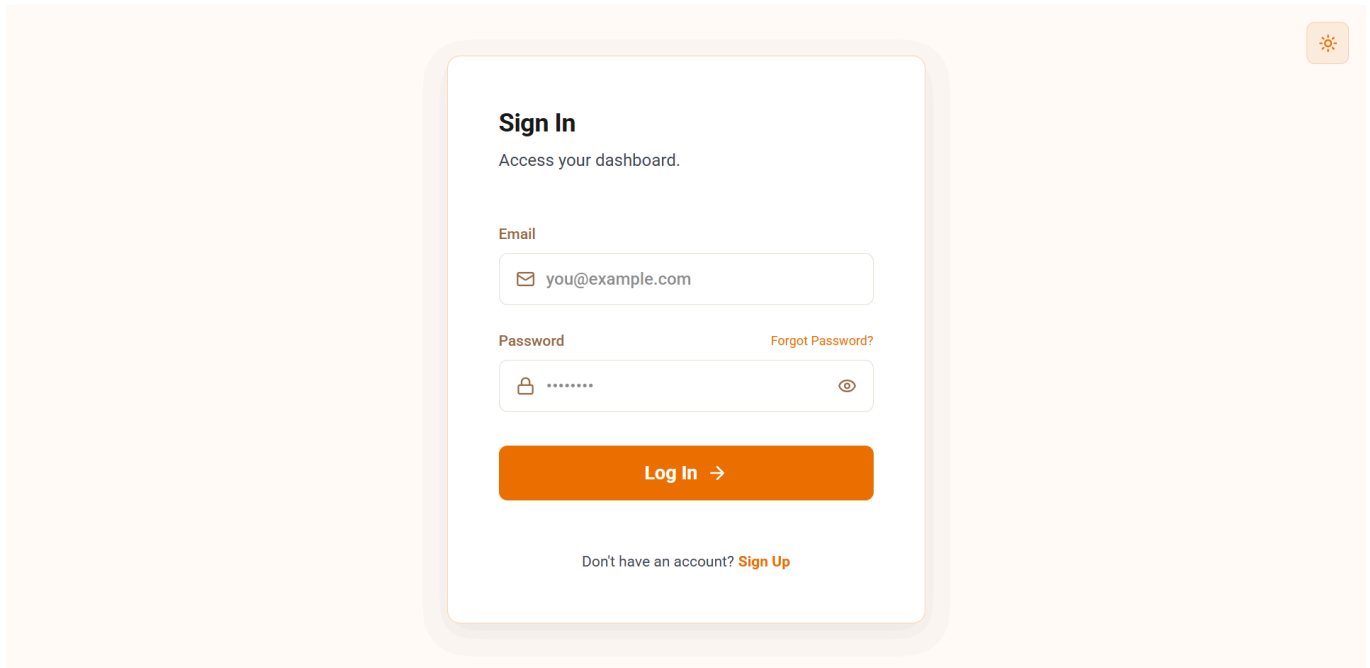


Figure 4-1 Screenshot for sign in page

2. Sign Up

The screenshot shows a 'Create Account' form with the title 'Step 1: Your Details' and a 'PROGRESS' indicator. The form contains the following fields: First Name (John), Last Name (Doe), Email (you@example.com), Confirm Email (you@example.com), Password (masked with dots), Confirm Password (masked with dots), and Phone Number (+251 9XX XXX XXXX). A large orange button at the bottom says 'Continue to Stream Selection ->'. Below the button, it says 'Already registered? Sign In'.

Figure 4-2 Screenshot for sign-up page

3. Stream Selection

The screenshot shows a 'Select your stream' page with the title 'STEP 02: STREAM SELECTION'. The main heading is 'Select your stream'. Below it, there is a section for 'Academic Stream' with two options: 'Natural Sciences' and 'Social Sciences'. At the bottom left, there is a 'Back' button with a left arrow. At the bottom right, there is a large orange button that says 'Complete Registration ->'.

Figure 4 - 3 Screenshot for stream selection page

CHAPTER FOUR - IMPLEMENTATION

4. User Profile

[← Back to Dashboard](#)

[Logout](#)

Profile Details

Manage your personal information.

First Name

 Daniel

Last Name

Berihun

Email (Disabled)

 premiumdet2026@gmail.com

Phone Number

 +251965312666

Save Changes 

Change Password

Keep your account secure.

Current Password





New Password





Confirm New Password





Update Password 

Academic Stream

Your current registered area of focus:

 Natural Science

[↻ Request Stream Change](#)

Note: Academic streams are locked once registered to ensure curriculum consistency. If you believe your stream was assigned incorrectly, please submit a formal request using the button above.

Figure 4-4 Screenshot for User Profile Page

139

5. Student Dashboard

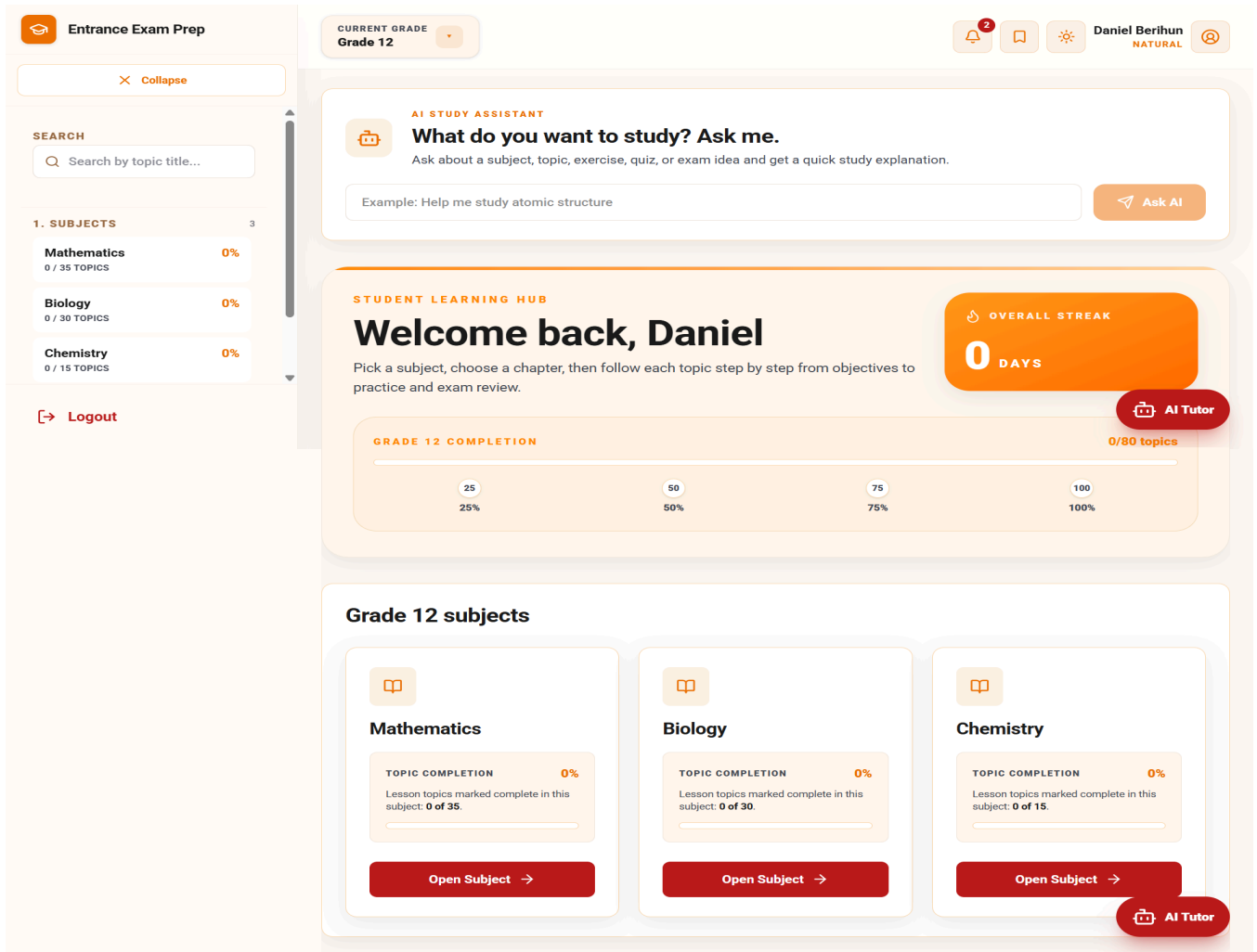


Figure 4-0-5 Screenshot for Student Dashboard Page

CHAPTER FOUR - IMPLEMENTATION

6. Chapter List

The screenshot displays a user interface for a mathematics curriculum. At the top, a navigation bar includes a home icon, a 'CURRENT GRADE' dropdown set to 'Grade 12', a notification bell with a red '2', a bookmark icon, a settings gear, and a user profile for 'Daniel Berihun NATURAL'. Below the navigation bar, a sidebar on the left contains a menu icon and a 'BACK TO SCHOLAR CONSOLE' link. The main content area is titled 'Curriculum: Mathematics'. It features a 'TOPIC COMPLETION' section showing '0%' completion (0 of 35 topics) and a 'TOTAL CHAPTERS' section showing '7'. The curriculum is organized into seven units, each with a unit icon, title, description, and a 'VIEW TOPICS' button. Units 2, 5, and 7 also include an 'AI Tutor' button. Unit 1: Sequences and series (Arithmetic and geometric patterns, partial sums, sigma notation, convergence preview). Unit 2: Limits and continuity (Informal limits of sequences and functions; continuity; one-sided limits). Unit 3: Derivatives (Derivative as limit; differentiation rules including chain rule). Unit 4: Applications of differential calculus (Monotonicity, extrema, optimization, concavity, related rates (intro)). Unit 5: Integral calculus (Antiderivatives, definite integrals, FTC, substitution, area between curves). Unit 6: Solid geometry and vectors in space ($\mathbb{R}^3$ coordinates, space vectors, dot & cross products, lines and planes (intro)). Unit 7: Proof and mathematical induction (Logic, direct proof, contradiction intro, induction).

Unit	Unit Title	Description	View Topics	AI Tutor
1	Unit 1: Sequences and series	Arithmetic and geometric patterns, partial sums, sigma notation, convergence preview.	VIEW TOPICS	
2	Unit 2: Limits and continuity	Informal limits of sequences and functions; continuity; one-sided limits.	VIEW TOPICS	AI Tutor
3	Unit 3: Derivatives	Derivative as limit; differentiation rules including chain rule.	VIEW TOPICS	
4	Unit 4: Applications of differential calculus	Monotonicity, extrema, optimization, concavity, related rates (intro).	VIEW TOPICS	
5	Unit 5: Integral calculus	Antiderivatives, definite integrals, FTC, substitution, area between curves.	VIEW TOPICS	AI Tutor
6	Unit 6: Solid geometry and vectors in space	$\mathbb{R}^3$ coordinates, space vectors, dot & cross products, lines and planes (intro).	VIEW TOPICS	
7	Unit 7: Proof and mathematical induction	Logic, direct proof, contradiction intro, induction.	VIEW TOPICS	AI Tutor

Figure 4-0-6 Screenshot for Chapter List page

7. Topic List



CURRENT GRADE
Grade 12



2





Daniel Berihun
NATURAL





< BACK TO SCHOLAR CONSOLE

Curriculum: Mathematics

TOPIC COMPLETION
Lesson topics marked complete across this subject (Mathematics): 0 of 35

0%

TOTAL CHAPTERS
7



Unit 1: Sequences and series
Arithmetic and geometric patterns, partial sums, sigma notation, convergence preview.

VIEW TOPICS



Unit 2: Limits and continuity
Informal limits of sequences and functions; continuity; one-sided limits.

VIEW TOPICS

 AI Tutor



Unit 3: Derivatives
Derivative as limit; differentiation rules including chain rule.

VIEW TOPICS



Unit 4: Applications of differential calculus
Monotonicity, extrema, optimization, concavity, related rates (intro).

VIEW TOPICS



Unit 5: Integral calculus
Antiderivatives, definite integrals, FTC, substitution, area between curves.

VIEW TOPICS

 AI Tutor



Unit 6: Solid geometry and vectors in space
 $\mathbb{R}^3$ coordinates, space vectors, dot & cross products, lines and planes (intro).

VIEW TOPICS



Unit 7: Proof and mathematical induction
Logic, direct proof, contradiction intro, induction.

VIEW TOPICS

 AI Tutor

Figure 4-0-7 Screenshot for Topic List

CHAPTER FOUR - IMPLEMENTATION

8. Topic Objective

The screenshot shows a user interface for a mathematics learning platform. At the top, there's a header with a graduation cap icon, 'CURRENT GRADE Grade 12', and user information 'Daniel Berihun NATURAL'. Below this is a navigation bar with a 'BACK TO TOPICS' link and a series of tabs: 'Objectives' (selected), 'Concept', 'Video', 'Exercise', 'Quiz', and 'Exam'. The main content area is titled 'MATHEMATICS • UNIT 5: INTEGRAL CALCULUS' and features the topic '5.1 Antiderivatives & indefinite integrals'. It includes a 'Mark Topic Complete' button and a 'Continue to Concept' button. A progress bar indicates 'Answer 7 more quiz questions before marking this topic complete.' Below this is a 'SUPPORT' section with 'Q&A' and 'Reports' links, and a red 'AI Tutor' button. A 'LEARNING OBJECTIVES' box contains the heading 'Expected outcomes for this topic' and a list of two objectives: 1. Reverse basic derivatives (powers, etc.) including +C awareness. 2. Check antiderivatives by differentiation. At the bottom, there's a 'Start' button, 'Step 1 of 6: Objectives', and another 'AI Tutor' button.

CURRENT GRADE
Grade 12

Daniel Berihun
NATURAL

< BACK TO TOPICS

1 Objectives 2 Concept 3 Video 4 Exercise 5 Quiz 6 Exam

MATHEMATICS • UNIT 5: INTEGRAL CALCULUS

5.1 Antiderivatives & indefinite integrals

Mark Topic Complete

Continue to Concept >

Answer 7 more quiz questions before marking this topic complete.

SUPPORT

Q&A Reports

AI Tutor

LEARNING OBJECTIVES

Expected outcomes for this topic

Upon successful completion of this unit, you should be able to demonstrate the following.

- 1 Reverse basic derivatives (powers, etc.) including +C awareness.
- 2 Check antiderivatives by differentiation.

< Start

Step 1 of 6: Objectives

AI Tutor

Figure 4-0-8 Screenshot for Topic List page

9. Exercise Page

CURRENT GRADE: Grade 12

 Daniel Berhoun NATURAL

BACK TO TOPICS
 1 Objectives
2 Concept
3 Videos
4 Exercise
5 Quiz
6 Exam

MATHEMATICS - UNIT 5: INTEGRAL CALCULUS

 Mark Topic Complete
 Continue to Quiz >

5.1 Antiderivatives & indefinite integrals

Answer 7 more quiz questions before marking this topic complete.

SUPPORT

 Q&A
 Reports
 AI Tutor

Practice exercises - 7 items

 Pick a choice, then press Check answer on the bottom-right of that question. Change your choice after a wrong attempt, then check again.

1 **Anti** EASY

$\int 2x \, dx =$

A $x^2 + C$

C $x + C$

B $2 + C$

D $2x^2 + C$

AI Tutor
 CHECK ANSWER

2 **Const** EASY

$\int 0 \, dx =$

A C

C x

B 0

D $x + C$

AI Tutor
 CHECK ANSWER

3 **Antiderivatives & indefinite — Refine partition** EASY

Refining a partition of $[a, b]$ usually means

A subdividing intervals so the mesh gets smaller

C replacing f by f'

B using fewer rectangles

D removing sample points

AI Tutor
 CHECK ANSWER

4 **Antiderivatives & indefinite — Constant on interval** EASY

If $f(x) = k$ is constant on $[a, b]$, then $\int_a^b k \, dx$ equals

A $k(b - a)$

C 0

B k

D $b \cdot k - a \cdot k$

AI Tutor
 CHECK ANSWER

5 **Antiderivatives & indefinite — FTC** EASY

$\int_a^b g' \, dx$ equals

A $g(b) - g(a)$

C $g'(b)$

B $g(a) - g(b)$

D 0 always

AI Tutor
 CHECK ANSWER

6 **Antiderivatives & indefinite — Additivity** MEDIUM

$\int_a^c f(x) \, dx = \int_a^b f(x) \, dx + \int_b^c f(x) \, dx$ holds (standard hypotheses) when

A b lies between a and c and f is integrable on $[a, c]$

C f is constant only

B $a = b$ always

D never

AI Tutor
 CHECK ANSWER

7 **Antiderivatives & indefinite — Mesh limit** MEDIUM

As mesh (max subinterval width) $\rightarrow 0$, Riemann sums for an integrable f approach

A the definite integral $\int_a^b f(x) \, dx$

C always ∞

B $f(a)$

D the number of subintervals

AI Tutor
 CHECK ANSWER

< Video
 Step 4 of 6: Exercise
AI Tutor

CHAPTER FOUR - IMPLEMENTATION

Figure 4-0-9 Screenshot for Exercise page

10. Admin Subject Management

The screenshot displays the 'Admin Subject Management' interface. On the left is a sidebar with the following navigation items: 'Entrance Exam Prep ADMIN', 'Dashboard', 'Users', 'Subjects' (highlighted), 'Issues', 'Discussions', and a 'Logout' button at the bottom. The main content area is titled 'Subjects' and features a 'Subject Management' section. This section includes a 'TOTAL SUBJECTS 9' counter and a '+ CREATE SUBJECT' button. Below this is a form for creating or editing a subject, with fields for 'Academic Grade' (a dropdown menu), 'Subject Name' (a text input with a placeholder 'e.g. Advanced Mathematics'), and 'Academic Stream' (a dropdown menu with 'General' selected). A 'CREATE SUBJECT' button is positioned to the right of the form. A search bar labeled 'Search subjects...' and two filter buttons, 'ALL GRADES' and 'ALL STREAMS', are located below the form. The subjects are organized into sections by grade level: 'GRADE 9', 'GRADE 10', 'GRADE 11', and 'GRADE 12'. Each grade section contains subject cards. For Grade 9, there is a 'Mathematics' card (labeled 'M') with a teacher assigned as 'Daniel Barhoun' and a 'Biology' card (labeled 'B') with no teacher assigned. For Grades 10, 11, and 12, there are 'Mathematics' (labeled 'M') and 'Biology' (labeled 'B') cards, all with no teacher assigned. Grade 12 also includes a 'Chemistry' card (labeled 'C') with no teacher assigned. Each card displays the subject name, a 'TEACHER' field with an 'Unassigned' status, and an 'UPDATE TEACHER' or 'ASSIGN TEACHER' button. The interface uses a clean, modern design with a light orange color scheme.

Figure 4-10 Screenshot for Admin Subject Management page

11. Admin User Management

Entrance Exam Prep
ADMIN

Dashboard

Users

Subjects

Issues

Discussions

Logout

Users

TOTAL USERS
14

Search users...

ALL ROLES

ALL STATUSES

USER	ROLE	ACCOUNT STATUS	ACTION
DB Daniel Berihun premiumdet2026@gmail.com	STUDENT	ACTIVE	
SH Surafiel H. surafiel@gmail.com	STUDENT	ACTIVE	
AA Aman Atalay amanatalay00@gmail.com	STUDENT	ACTIVE	
OK Oftanenus Kasa oftanenuskasa@gmail.com	STUDENT	ACTIVE	
HH Hamza Hassen hamzehassen@gmail.com	STUDENT	ACTIVE	
AT Abel Tesfaye abel@yopmail.com	STUDENT	ACTIVE	
YA Ye Berehaw Anbessa lessannoyingcrm@logoskal.com	STUDENT	ACTIVE	
bt biruk tafese cstafesebiruk23@gmail.com	STUDENT	ACTIVE	
AM Asnake MENGESHA asnakemengesha80@gmail.com	STUDENT	ACTIVE	
SS Social Student social.student@example.com	STUDENT	ACTIVE	

<

1

2

>

Figure 4-0-11 Screenshot for Admin User Management page

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12. Admin Issue Management

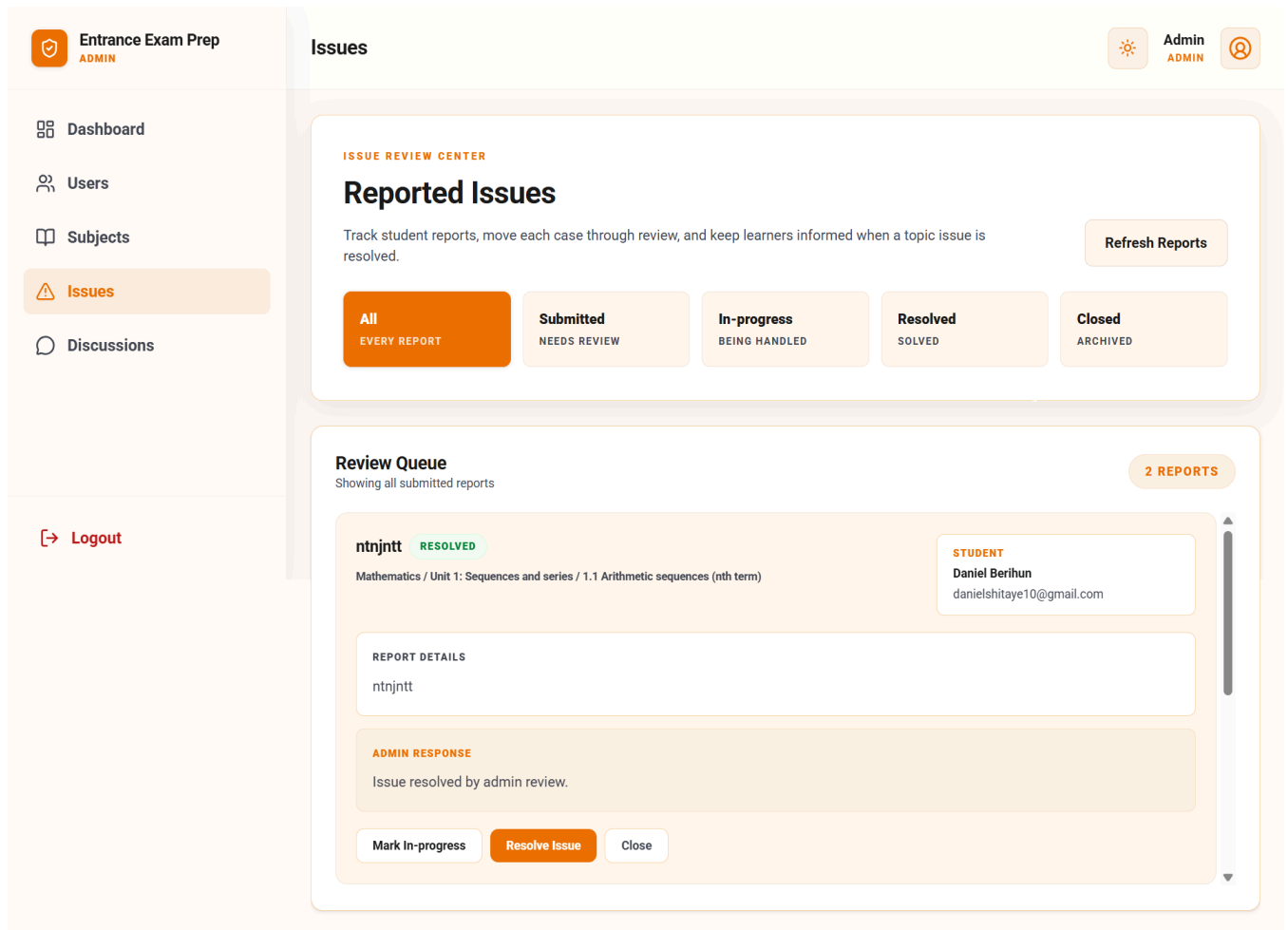


Figure 4-0-12 Admin Issue Management page

13. Teacher Dashboard

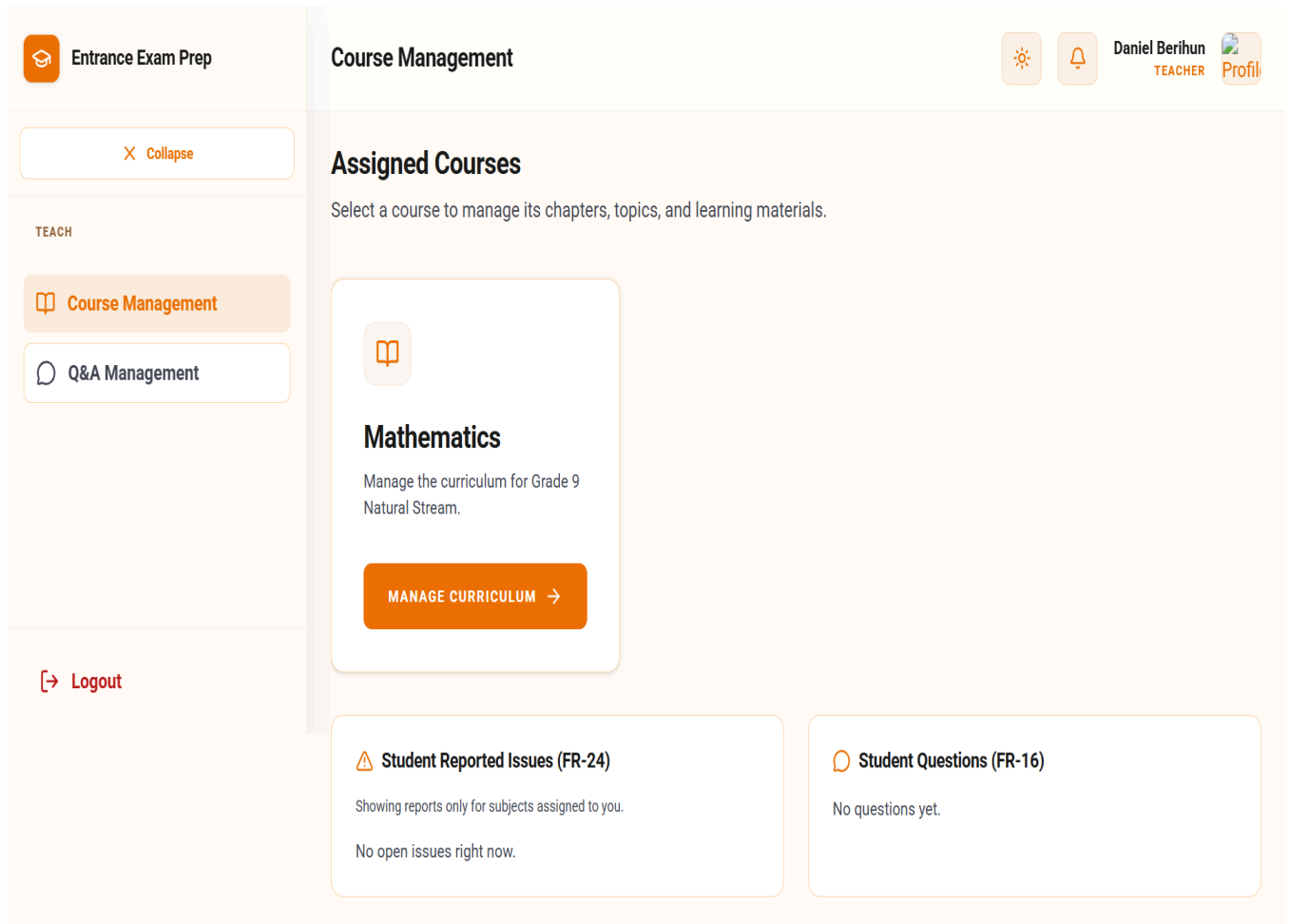


Figure 4-0-13 Screenshot for Teacher Dashboard page

CHAPTER FOUR - IMPLEMENTATION

14. Teacher Quiz Management

Course Management

BACK TO TOPICS

1 Objectives

2 Concept

3 Video

4 Exercise

5 Quiz

6 Exam

MATHEMATICS - UNIT 1: SETS

1.3 Union, intersection, and complement

SUPPORT

Q&A

BACK TO QUIZZES

Edit Question

Building: Unit 1 Quiz 3 - Operations

QUESTION TEXT

(A $\cup$ B) $\cap$ C is read as:

CHOICES & CORRECT ANSWER

A

Elements in A or B and in C

☒

B

Elements in A and B or C

☐

C

Elements only in C

☐

D

Elements in none of the sets

☐

ANSWER EXPLANATION (OPTIONAL)

Union first, then intersection with C.

CANCEL

UPDATE QUESTION

Quiz questions - 7 Items

1

(A $\cup$ B) $\cap$ C is read as:

A: Elements in A or B and in C

B: Elements in A and B or C

C: Elements only in C

D: Elements in none of the sets

2

If A $\cap$ B = $\emptyset$, then A and B are:

A: Equal

B: Disjoint

C: Subsets

D: Infinite

3

A $\cup \emptyset$ =

A: $\emptyset$

B: A

C: U

D: Undefined

4

A $\cap \emptyset$ =

A: A

B: $\emptyset$

C: U

D: A'

5

If A $\subset$ B, then A $\cup$ B =

A: A

B: B

C: $\emptyset$

D: A $\cap$ B

6

The complement of U is:

A: U

B: $\emptyset$

C: (U)

D: A

7

Which identity is always true?

A: A $\cap$ (B $\cup$ C) = (A $\cap$ B) $\cup$ (A $\cap$ C)

B: A $\cup$ (B $\cap$ C) = (A $\cup$ B) $\cap$ A

C: A $\cap$ A' = U

D: A $\cup$ A' = $\emptyset$

Exercise

Step 5 of 6: Quiz

Exam

Figure 4-0-14 Screenshot for Teacher Quiz Management page

15. Teacher Q&A Management

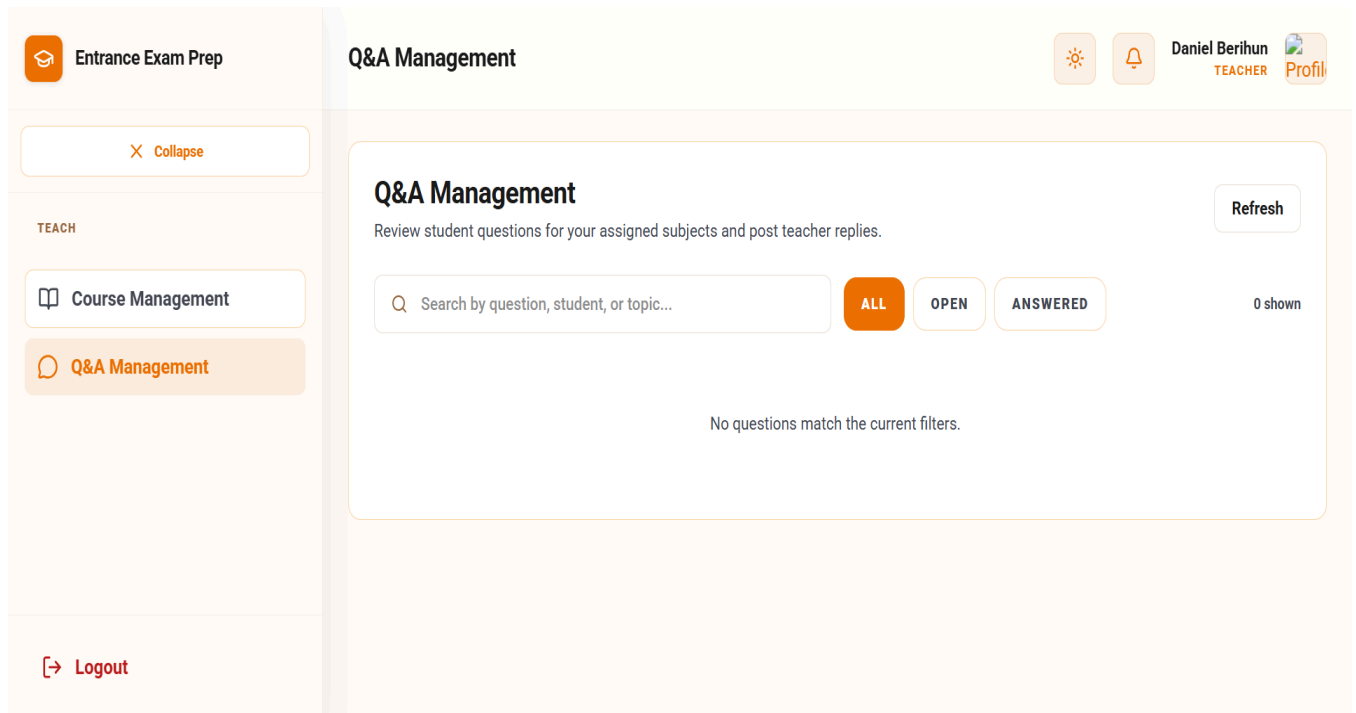


Figure 4-0-15 Screenshot for Q&A Management page

16. AI Tutor Chat-bot

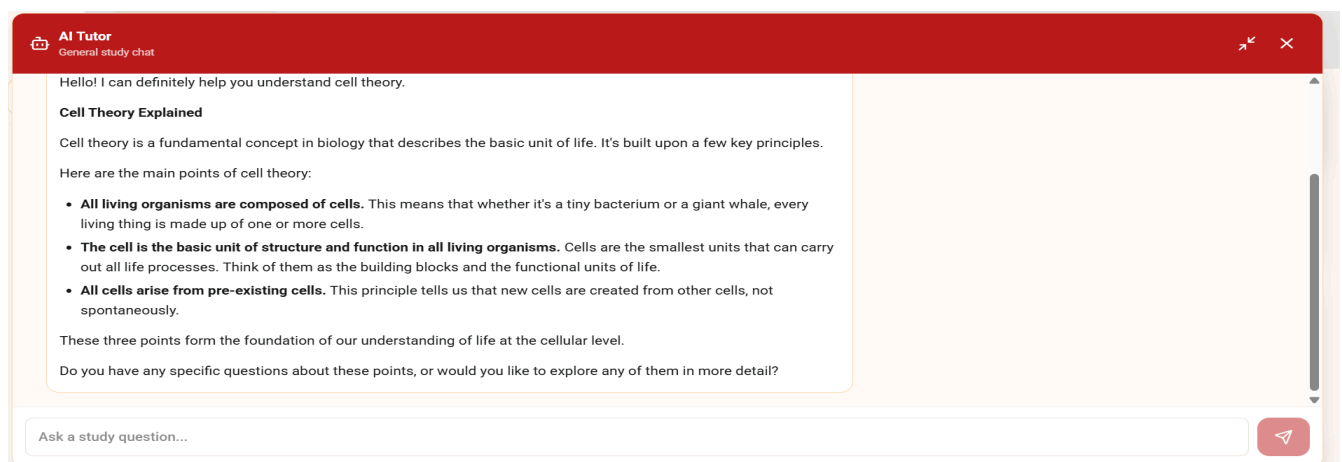


Figure 4-0-16 Screenshot for AI Chat-bot page

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4.3 Code Snippets

1. User Model Schema

```
const mongoose = require('mongoose');
const bcryptjs = require('bcryptjs');
const userSchema = new mongoose.Schema(
  {
    firstName: {
      type: String,
      required: [true, 'First name is required'],
      trim: true,
    },
    lastName: {
      type: String,
      required: [true, 'Last name is required'],
      trim: true,
    },
    email: {
      type: String,
      required: [true, 'Email is required'],
      unique: true,
      lowercase: true,
      match: [/^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\.\w{2,3})+$/, 'Please provide a valid email'],
    },
    password: {
```

```
type: String,
required: [true, 'Password is required'],
minlength: [6, 'Password must be at least 6 characters'],
select: false, // Don't include password by default
},
phoneNumber: {
  type: String,
  trim: true,
},
profileImage: {
  type: String, // URL to profile image
  default: null,
},
role: {
  type: String,
  enum: ['student', 'teacher', 'admin'],
  required: [true, 'Role is required'],
  default: 'student',
},
status: {
  type: String,
  enum: ['active', 'inactive', 'suspended'],
  default: 'active',
},
stream: {
  type: String,
  enum: ['Natural', 'Social', null],
  default: null,
},
gradeLevel: {
  type: String,
```

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```
enum: ['9', '10', '11', '12', null],
default: null,
},
isEmailVerified: {
  type: Boolean,
  default: false,
},
verificationCode: {
  type: String,
  default: null,
},
geminiApiKeyEncrypted: {
  type: String,
  default: null,
  select: false,
},
geminiApiKeyLast4: {
  type: String,
  default: null,
},
geminiApiKeyUpdatedAt: {
  type: Date,
  default: null,
},
},
{
  timestamps: { createdAt: 'created_at', updatedAt: 'updated_at' },
}
);
// Index for email
userSchema.index({ email: 1 });
```

2. Subject Model Schema

```
const mongoose = require('mongoose');
const { Schema } = mongoose;
const subjectSchema = new Schema(
  {
    subjectName: {
      type: String,
      required: [true, 'Subject name is required.'],
      trim: true,
    },
    subjectDescription: {
      type: String,
      trim: true,
    },
    gradeLevel: {
      type: String,
      required: [true, 'Grade level is required.'],
      trim: true,
    },
    teacher: {
      type: Schema.Types.ObjectId,
      ref: 'User', // Creates a reference to the User model
      required: false, // A teacher can be assigned later
    },
    stream: {
      type: String,
      enum: ['Natural', 'Social', 'General', null, ""],
      default: null,
    },
  },

```

CHAPTER FOUR - IMPLEMENTATION

```
    },  
    {  
const Subject = mongoose.model('Subject', subjectSchema);  
module.exports = Subject;
```

3. Authentication Controller Class

```
const { validationResult } = require('express-validator');  
const authService = require('../services/authService');  
const { getRedisClient } = require('../config/redis');  
  
class AuthController {  
  async register(req, res, next) {  
    try {  
      // Check for validation errors  
      const errors = validationResult(req);  
      if (!errors.isEmpty()) {  
        return res.status(400).json({ success: false, errors: errors.array() });  
      }  
  
      const {  
        firstName,  
        lastName,  
        email,  
        password,  
        phoneNumber,  
        role,  
        profileImage,  
        stream,  
        gradeLevel,  
      } = req.body || {};
```

```
const uploadedProfileImage = req.file
  ? `${req.protocol}://${req.get('host')}/uploads/profiles/${req.file.filename}`
  : undefined;
const result = await authService.registerUser({
  firstName,
  lastName,
  email,
  password,
  phoneNumber,
  role,
  stream,
  gradeLevel,
  profileImage: uploadedProfileImage || profileImage,
});
res.status(201).json({
  success: true,
  message: 'User registered successfully',
  data: result,
});
} catch (error) {
  next(error);
}
}
}
async login(req, res, next) {
  try {
    // Check for validation errors
    const errors = validationResult(req);
    if (!errors.isEmpty()) {
      return res.status(400).json({ success: false, errors: errors.array() });
    }
  }
```

CHAPTER FOUR - IMPLEMENTATION

```
const { email, password } = req.body;

const result = await authService.loginUser(email, password);

// Reset failed attempts on successful login
const redisClient = getRedisClient();
if (redisClient) {
  await redisClient.del(`login_attempts:${email}`);
}
res.status(200).json({
  success: true,
  message: 'Login successful',
  data: result,
});
} catch (error) {
  // Increment failed attempts on error (only if it's an authentication error)
  if (error.status === 401) {
    const { email } = req.body;
    const redisClient = getRedisClient();
    if (redisClient && email) {
      const key = `login_attempts:${email}`;
      try {
        const attempts = await redisClient.incr(key);
        if (attempts === 1) {
          await redisClient.expire(key, 30);
        }
      } catch (redisErr) {
        console.error('Redis error during rate limiting:', redisErr);
      }
    }
  }
}
```

```
    next(error);  
  }  
}
```

4. Subject Controller Class

```
const subjectService = require('../services/subjectService');  
const mongoose = require('mongoose');  
const { notifyStudentsOfSubjectUpdate } =  
require('../services/contentNotificationService');  
const appCache = require('../services/appCache');  
const isValidObjectId = (value) => mongoose.Types.ObjectId.isValid(value);  
  
class SubjectController {  
  async inviteAndAssignTeacherByEmail(req, res, next) {  
    try {  
      const { subjectId } = req.params;  
      const { email, firstName, lastName } = req.body;  
      if (!isValidObjectId(subjectId)) {  
        return res.status(400).json({ success: false, message: 'Invalid subject id format.'  
      });  
    }  
    if (!email) {  
      return res.status(400).json({ success: false, message: 'Email is required.' });  
    }  
    const result = await subjectService.inviteAndAssignTeacherByEmail(subjectId,  
email, firstName, lastName);  
    await appCache.invalidateSubjectRecord(subjectId);  
    res.status(200).json(result);  
  } catch (error) {  
    next(error);  
  }  
}
```


CHAPTER FOUR - IMPLEMENTATION

```
    }  
  }  
  async createSubject(req, res, next) {  
    try {  
      console.log('--- Creating Subject ---');  
      console.log('Body:', req.body);  
  
      if (!req.body?.subjectName || !req.body?.gradeLevel) {  
        console.warn('Missing required fields:', { name: req.body?.subjectName, grade:  
req.body?.gradeLevel });  
        return res.status(400).json({  
          success: false,  
          message: 'subjectName and gradeLevel are required.',  
        });  
      }  
  
      // Ensure gradeLevel is a string if the model expects it, though Mongoose handles  
this  
      const subject = await subjectService.createSubject(req.body);  
      await appCache.invalidateSubjectsCatalog();  
      await notifyStudentsOfSubjectUpdate(  
        subject._id,  
        'New subject available',  
        `A new subject "${subject.subjectName}" is now available for your grade.`  
      );  
      console.log('Subject created successfully:', subject._id);  
      res.status(201).json(subject);  
    } catch (error) {  
      console.error('Error creating subject:', error);  
      next(error);  
    }  
  }  
}
```

5. Exercise Controller Class

```

const Exercise = require('../models/Exercise');
const ExerciseProblem = require('../models/ExerciseProblem');
const Topic = require('../models/Topic');
const Chapter = require('../models/Chapter');
const Answer = require('../models/Answer');
const appCache = require('../services/appCache');
const asyncHandler = require('../middleware/async');
const ErrorResponse = require('../utils/errors');
const { notifyStudentsOfSubjectUpdate } =
  require('../services/contentNotificationService');
const getRequester = (req) => ({
  id: req.user?.id || req.user?._id?.toString(),
  role: req.user?.role,
});
const getSubjectIdForTopic = async (topicId) => {
  const topic = await Topic.findById(topicId).select('chapter topicName').lean();
  if (!topic?.chapter) {
    return null;
  }
  const chapter = await Chapter.findById(topic.chapter).select('subject').lean();
  return chapter?.subject || null;
};
exports.createExercise = asyncHandler(async (req, res, next) => {
  const { topic: topicId, title, description, question, options, correctAnswer, hint,
  difficulty } = req.body;
  const topic = await Topic.findById(topicId);
  if (!topic) {
    return next(new ErrorResponse(`Topic not found with id of ${topicId}`, 404));
  }

```

CHAPTER FOUR - IMPLEMENTATION

```
}

const exercise = await Exercise.create({
  topic: topicId,
  title,
  description,
  question,
  options,
  correctAnswer,
  hint,
  difficulty,
  createdBy: req.user.id,
});

const subjectId = await getSubjectIdForTopic(topicId);
if (subjectId) {
  await notifyStudentsOfSubjectUpdate(
    subjectId,
    'New exercise available',
    `A new practice exercise "${exercise.title}" was added.`
  );
}

res.status(201).json({
  success: true,
  data: exercise
});
})

exports.addExerciseProblem = asyncHandler(async (req, res, next) => {
  const { exerciseId } = req.params;
  req.body.exerciseId = exerciseId;

  const exercise = await Exercise.findById(exerciseId);
```

```
if (!exercise) {
  return next(new ErrorResponse(`Exercise not found with id of ${exerciseId}`,
404));
}

// Make sure user is exercise owner
if (exercise.createdBy.toString() !== req.user.id) {
  return next(new ErrorResponse(`User ${req.user.id} is not authorized to add a
problem to exercise ${exercise._id}`, 401));
}
const problem = await ExerciseProblem.create(req.body);
res.status(201).json({
  success: true,
  data: problem,
});
});
```

6. Sign In Page

```
import React, { useState, useContext } from 'react';
import { useNavigate, Link, useLocation } from 'react-router-dom';
import { useForm } from 'react-hook-form';
import { loginUser } from '../services/auth';
import { AuthContext } from '../context/AuthContext';
import ThemeToggle from '../components/ThemeToggle';
import { Eye, EyeOff, GraduationCap, ShieldCheck, Mail, Lock, ArrowRight,
BookOpen, CheckCircle } from 'lucide-react';
const Login = () => {
  const { login } = useContext(AuthContext);
  const navigate = useNavigate();
  const location = useLocation();
```

CHAPTER FOUR - IMPLEMENTATION

```
const [error, setError] = useState(null);
const successMessage = location.state?.message;
const [showPassword, setShowPassword] = useState(false);
const { register, handleSubmit, formState: { errors, isSubmitting } } = useForm();
const onSubmit = async (data) => {
  try {
    setError(null);
    const response = await loginUser(data);
    if (response && response.data && response.data.token) {
      const { user, token } = response.data;
      login(token, user)
      if (user.role === 'admin') {
        navigate('/admin');
      } else if (user.role === 'teacher') {
        navigate('/teacher');
      } else {
        navigate('/dashboard');
      }
    } else {
      setError('Login failed parsing response.');
```

credentials.');

```
    }
  } catch (err) {
    setError(err.response?.data?.message || 'Login failed. Please check your
    credentials.');
```

font-sans overflow-x-hidden">

```
    <div className="fixed top-3 right-3 sm:top-4 sm:right-4 z-50">
      <ThemeToggle />
```

```

</div>
<main className="flex-grow flex items-center justify-center py-8 sm:py-12 px-4
sm:px-6 w-full min-w-0">
  <div className="w-full max-w-[450px] min-w-0 bg-card rounded-xl border
border-outline-variant p-5 sm:p-stack-lg shadow-[0px_8px_24px_rgba(0,0,0,0.08)]">
    <div className="mb-stack-lg">
      <h3 className="text-xl sm:text-2xl font-semibold text-on-surface mb-2">Sign
In</h3>
      <p className="text-body-md text-on-surface-variant">Access your
dashboard.</p>
    </div>
    {error && (
      <div className="mb-6 p-4 bg-error-container text-on-error-container
rounded-lg text-sm flex items-center gap-3">
        <ShieldCheck size={20} />
        {error}
      </div>
    )}

    {successMessage && !error && (
      <div className="mb-6 p-4 bg-primary-container/10 text-primary-container
rounded-lg text-sm flex items-center gap-3">
        <CheckCircle size={20} />
        {successMessage}
      </div>
    )}
  <form className="space-y-stack-md" onSubmit={handleSubmit(onSubmit)}>
    <div>
      <label className="block text-sm font-medium text-on-surface mb-2"
htmlFor="email">Email</label>
      <div className="relative">

```

CHAPTER FOUR - IMPLEMENTATION

```
<input
  id="email"
  type="email"
  {...register('email', { required: 'Email is required' })}
  className="w-full px-4 py-3 pl-11 rounded-lg border border-outline/20
focus:border-primary-container focus:ring-1 focus:ring-primary-container
outline-none transition-all"
  placeholder="you@example.com"
/>
<Mail size={18} className="absolute left-4 top-1/2 -translate-y-1/2
text-outline" />
</div>
{errors.email && <p className="mt-1 text-xs
text-error">{errors.email.message}</p>}
</div>
<div>
  <div className="flex justify-between items-center mb-2">
    <label className="block text-sm font-medium text-on-surface"
htmlFor="password">Password</label>
    <Link to="/forgot-password" size="sm" className="text-xs
text-primary-container hover:underline">Forgot Password?</Link>
  </div>
  <div className="relative">
    <input
      id="password"
      type={showPassword ? 'text' : 'password'}
      {...register('password', { required: 'Password is required' })}
      className="w-full px-4 py-3 pl-11 rounded-lg border border-outline/20
focus:border-primary-container focus:ring-1 focus:ring-primary-container
outline-none transition-all"
      placeholder="••••••••"
```

```

/>
<Lock size={18} className="absolute left-4 top-1/2 -translate-y-1/2
text-outline" />
<button
  type="button"
  onClick={() => setShowPassword(!showPassword)}
  className="absolute right-4 top-1/2 -translate-y-1/2 text-outline
hover:text-on-surface transition-colors"
  >
    {showPassword ? <EyeOff size={18} /> : <Eye size={18} />}
  </button>
</div>
{errors.password && <p className="mt-1 text-xs
text-error">{errors.password.message}</p>}}
</div>

<div className="pt-2">
  <button
    type="submit"
    disabled={isSubmitting}
    className="w-full bg-primary-container text-on-primary py-3 px-6
rounded-lg font-semibold text-lg hover:brightness-110 active:opacity-80 transition-all
duration-200 flex items-center justify-center gap-2"
    >
    {isSubmitting ? 'Authenticating...' : 'Log In'}
    {!isSubmitting && <ArrowRight size={20} />}
  </button>
</div>
</form>

<div className="mt-stack-lg text-center">

```


CHAPTER FOUR - IMPLEMENTATION

```
<p className="text-sm text-on-surface-variant">
  Don't have an account?
  <Link to="/register" className="text-primary-container font-semibold
hover:underline ml-1">Sign Up</Link>
</p>
</div>
</div>
</main>
</div>
);
};
export default Login;
```

7. Student Dashboard Page

```
import React, { useEffect, useState, useContext } from 'react';
import { Link, useNavigate } from 'react-router-dom';
import {
  GraduationCap,
  Layout,
  User,
  LogOut,
  BookOpen,
  ArrowRight,
  CircleUserRound,
  Menu,
  X,
  ChevronRight,
  Bell,
  Bookmark,
  TriangleAlert,
  Search,
```

```
MessageSquarePlus,  
  Send  
} from 'lucide-react';  
import { AuthContext } from '../context/AuthContext';  
import api from '../services/api';  
import {  
  getUnreadNotifications,  
  markNotificationRead,  
  searchTopics,  
  createIssue,  
  getMyIssues,  
  getBookmarks,  
  removeBookmark,  
  askQuestion  
} from '../services/engagement';  
import { formatTopicTitleDisplay } from '../utils/formatTopicDisplayText';  
const gradeMatchesFilter = (subjectGrade, selectedGrade) => {  
  const g = String(subjectGrade ?? "").replace(/D/g, "");  
  const s = String(selectedGrade ?? "").replace(/D/g, "");  
  if (g && s) return g === s;  
  return String(subjectGrade) === String(selectedGrade);  
};  
  
const StudentDashboard = () => {  
  const { user, logout } = useContext(AuthContext);  
  const navigate = useNavigate();  
  const [subjects, setSubjects] = useState([]);  
  const [isLoading, setIsLoading] = useState(true);  
  const [isMobileMenuOpen, setIsMobileMenuOpen] = useState(false);  
  const [notifications, setNotifications] = useState([]);  
  const [bookmarks, setBookmarks] = useState([]);
```

CHAPTER FOUR - IMPLEMENTATION

```
const [myIssues, setMyIssues] = useState([]);
const [topicSearch, setTopicSearch] = useState("");
const [topicResults, setTopicResults] = useState([]);
const [issueForm, setIssueForm] = useState({ title: "", issueDescription: "", issueType:
'bug' });
const [questionForm, setQuestionForm] = useState({ topicId: "", questionText: "" });
const [feedback, setFeedback] = useState("");

useEffect(() => {
  const fetchSubjects = async () => {
    try {
      const response = await api.get('/subjects');
      const subjectsList = Array.isArray(response.data) ? response.data :
(response.data.data || []);

      const filtered = subjectsList.filter(s =>
        gradeMatchesFilter(s.gradeLevel, user.gradeLevel) &&
        (!s.stream || s.stream === user.stream)
      );
      setSubjects(filtered);
    } catch (err) {
      console.error('Failed to fetch subjects:', err);
    } finally {
      setIsLoading(false);
    }
  };

  if (user) {
    fetchSubjects();
  }
}, [user]);
```

```
useEffect(() => {
  const fetchEngagement = async () => {
    try {
      const [notifRes, bookmarkRes, issueRes] = await Promise.all([
        getUnreadNotifications(),
        getBookmarks(),
        getMyIssues()
      ]);
      setNotifications(notifRes?.data || []);
      setBookmarks(bookmarkRes?.data || []);
      setMyIssues(issueRes?.data || []);
    } catch (err) {
      console.error('Failed to fetch student engagement data', err);
    }
  };
  if (user?.role === 'student') {
    fetchEngagement();
  }
}, [user]);
```

```
useEffect(() => {
  const run = async () => {
    if (!topicSearch || topicSearch.trim().length < 2) {
      setTopicResults([]);
      return;
    }
    try {
      const res = await searchTopics(topicSearch.trim());
      const list = Array.isArray(res) ? res : (res?.data || []);
      setTopicResults(list.slice(0, 8));
    }
  };
  run();
}, [topicSearch]);
```

CHAPTER FOUR - IMPLEMENTATION

```
    } catch (err) {
      console.error('Topic search failed', err);
    }
  };
  const t = setTimeout(run, 250);
  return () => clearTimeout(t);
}, [topicSearch]);

const handleSelectSubject = (subjectId) => {
  navigate(`/curriculum/subject/${subjectId}/chapters`);
};

const SIDEBAR_ITEMS = [
  { key: 'dashboard', label: 'Dashboard', icon: <Layout size={20} />, path:
'/dashboard' },
  { key: 'curriculum', label: 'My Subjects', icon: <BookOpen size={20} />, path:
'/dashboard' }, // Same as dashboard for student
  { key: 'profile', label: 'Profile', icon: <User size={20} />, path: '/profile' },
];

const handleMarkRead = async (id) => {
  try {
    await markNotificationRead(id);
    setNotifications((prev) => prev.filter((n) => n._id !== id));
  } catch (err) {
    console.error(err);
  }
};

const handleRemoveBookmark = async (id) => {
  try {
```

```
    await removeBookmark(id);
    setBookmarks((prev) => prev.filter((b) => b._id !== id));
  } catch (err) {
    console.error(err);
  }
};

const handleOpenBookmark = (bookmark) => {
  if (!bookmark?.targetPath) return;
  navigate(bookmark.targetPath);
};

const handleIssueSubmit = async (e) => {
  e.preventDefault();
  if (!issueForm.title || !issueForm.issueDescription) return;
  try {
    const res = await createIssue(issueForm);
    setMyIssues((prev) => [res.data, ...prev]);
    setIssueForm({ title: '', issueDescription: '', issueType: 'bug' });
    setFeedback('Issue submitted successfully.');
```

```
  } catch (err) {
    setFeedback(err?.response?.data?.message || 'Failed to submit issue.');
```

```
  }
};
```

CHAPTER FIVE-TESTING

CHAPTER FIVE - TESTING

5.1 Introduction

Testing is the process of evaluating a system or its components to verify whether the implemented functionalities satisfy the specified requirements and expected behaviors. It is a systematic activity performed to identify errors, validate system performance, and ensure the reliability and quality of the developed software system [1].

In this project, testing was conducted to verify that the **“Ethiopia Entrance Exam Preparation Platform”** functions correctly according to the identified functional and non-functional requirements. The testing process focused on validating major system components such as authentication, subject management, content management, quizzes, progress tracking, bookmarking, notifications, and student–teacher interactions.

5.2 General Testing Strategy

During the testing phase of this project, black box testing techniques were applied to evaluate the functionality of the system without considering the internal implementation details of the source code.

The testing process mainly focused on validating whether the system features operate according to the specified requirements and expected outputs.

The following testing techniques were used to test the proposed system.

5.2.1 System Testing

System testing was conducted to verify the complete and integrated behavior of the platform. The implemented functionalities were tested against the use case models and functional requirements defined during the analysis phase. This testing ensured that important modules such as user authentication, subject management, quiz handling, bookmarking, question and answer interaction, notification delivery, and issue reporting work correctly and interact properly with one another.

5.2.2 Requirements Testing

A requirement testing was performed to validate whether the developed system satisfies the specified functional and non-functional requirements. The test cases were designed based on the identified use cases and scenarios to ensure that the system behavior matches the expected system requirements. This testing also helped identify missing functionalities, validation errors, and inconsistencies between the proposed design and the implemented system.

5.2.3 Usability Testing

Usability testing was performed to evaluate the user interface design, ease of navigation, and overall user experience of the system. Random users interacted with the platform to determine whether the interface was understandable, user-friendly, and easy to use without confusion. The testing also verified whether students, teachers, and administrators could accomplish their intended tasks efficiently, such as accessing learning materials, creating quizzes, managing subjects, and tracking progress.

5.3 Test case

Some selected test cases for the functional requirements are as follows:

Test Scenario name: Signup

Description: The functionality of Complainants' signup is covered in this scenario.

Test Scenario Name: Create Account

Table 5-1 Test case for Signup

#	Test case	Input Data	Expected Result	Actual Result	P	F
1	When empty values are entered in required registration fields	—	An error message “This field is required” is displayed	An error message “This field is required” is displayed		

CHAPTER FIVE-TESTING

2	When an invalid email format is entered	E.g., student@gmail	An error message “Invalid email format” is displayed	An error message “Invalid email format” is displayed		
3	When an already registered email address is entered	E.g., abebe@gmail.com	An error message “Email already exists” is displayed	An error message “Email already exists” is displayed		
4	When password length is less than the minimum required characters	E.g., 12345	An error message “Password must be at least 6 characters” is displayed	An error message “Password must be at least 6 characters” is displayed		
5	When the confirm password value does not match the password value	Password: abc123, Confirm Password: abc12	An error message “Passwords do not match” is displayed	An error message “Passwords do not match” is displayed		

Test Scenario name: login

Description: The functionality of user's login is covered in this scenario.

Test Scenario Name: Create Account

Table 5-2 Test case for login

	Test case	Input Data	Expected Result	Actual Result	P	F
1	When empty values are entered in email and password fields	—	An error message “This field is required” is displayed	An error message “This field is required” is displayed		
2	When an invalid email format is entered	E.g., abebe@gmail	An error message “Invalid email format” is displayed	An error message “Invalid email format” is displayed		
3	When a non-existing email address is entered	E.g., unknown@gmail.com	An error message “Account not found” is displayed	An error message “Account not found” is displayed		
4	When an incorrect password is entered	E.g., Password: 123abc	An error message “Invalid credentials” is displayed	An error message “Invalid credentials” is displayed		

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5	When a deactivated account attempts to login	Inactive student account	An error message “Account is inactive. Contact administrator” is displayed	An error message “Account is inactive. Contact administrator” is displayed		
6	When valid email and password are entered	Valid registered account credentials	The system authenticates the user and redirects to the appropriate dashboard	The system authenticates the user and redirects to the appropriate dashboard		
7	When password field length is below the minimum allowed value	E.g., 123	An error message “Password must be at least 6 characters” is displayed	An error message “Password must be at least 6 characters” is displayed		

Test Scenario name: Create Subject

Description: The functionality of creating a subject is covered in this test scenario.

Table 5-3 Test case for Create Subject

#	Test case	Input Data	Expected Result	Actual Result	P	F
1	When empty values are entered	—	An error message “This field is	An error message “This field is		

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	in required subject fields		required” is displayed	required” is displayed		
2	When an invalid grade level is entered	E.g., Grade 13	An error message “Grade level must be between 9 and 12” is displayed	An error message “Grade level must be between 9 and 12” is displayed		
3	When a duplicate subject title for the same grade is entered	Mathematics – Grade 12 (already exists)	An error message “Subject already exists” is displayed	An error message “Subject already exists” is displayed		
4	When the subject description exceeds the allowed length	Long text exceeding limit	An error message “Description exceeds maximum length” is displayed	An error message “Description exceeds maximum length” is displayed		
5	When valid subject information is entered	Subject: Biology, Grade: 11	The system creates the subject successfully and displays a confirmation message	The system creates the subject successfully and displays a confirmation message		
6	When the administrator cancels subject creation before submission	Click “Cancel” button	The system returns to the Manage Subjects page without saving data	The system returns to the Manage Subjects page without saving data		

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7	When unauthorized users attempt to create a subject	Student or Teacher account	The system denies access and displays an authorization error message	The system denies access and displays an authorization error message		
---	---	----------------------------	--	--	--	--

Test Scenario Name: Upload Subject Content

Description; Teachers can add and manage content under each topic including videos, concept pages, exercises, and previous-year exam questions.

Table 5-4 Test case for Uploading Subject content

#	Test case	Input Data	Expected Result	Actual Result	P	F
1	When empty values are submitted in required content fields	–	An error message “This field is required” is displayed	An error message “This field is required” is displayed		
2	When an unsupported file type is uploaded	E.g., .exe file	An error message “Unsupported file type” is displayed	An error message “Unsupported file type” is displayed		
3	When uploaded file size exceeds the allowed limit	Large video file exceeding limit	An error message “File size exceeds maximum limit” is displayed	An error message “File size exceeds maximum limit” is displayed		
4	When a teacher uploads content without selecting a subject topic	No topic selected	An error message “Please select a topic” is displayed	An error message “Please select a topic” is displayed		

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5	When invalid video URL format is entered	E.g., YouTube/video 123	An error message “Invalid video URL” is displayed	An error message “Invalid video URL” is displayed		
6	When valid content information and files are uploaded	Valid PDF, video URL, exercise content	The system uploads and stores the content successfully and displays a confirmation message	The system uploads and stores the content successfully and displays a confirmation message		
7	When an unauthorized user attempts to upload content	Student account	The system denies access and displays an authorization error message	The system denies access and displays an authorization error message		
8	When a teacher saves uploaded content as draft	Valid content with “Save as Draft” option	The system stores the content as draft and hides it from students	The system stores the content as draft and hides it from students		
9	When internet connection is interrupted during upload	Upload interrupted	The system displays “Upload failed. Please try again” message	The system displays “Upload failed. Please try again” message		
10	When content upload is completed successfully	Topic content submitted correctly	The uploaded content becomes visible under the selected topic	The uploaded content becomes visible under the selected topic		

CHAPTER FIVE-TESTING

Test Scenario name: Attempt Exercises, Quizzes, and Previous-Year Exam Problems

Description: The functionality of user's login is covered in this scenario.

Table 5-5 Test case for Attempt Exercises, Quizzes, and Previous-Year Exam Problems scenario

#	Test case	Input Data	Expected Result	Actual Result	P	F
1	When a student attempts a quiz without selecting an answer	Empty answer submission	An error message "Please answer all required questions" is displayed	An error message "Please answer all required questions" is displayed		
2	When a student submits valid answers for a quiz	Correctly answered quiz questions	The system evaluates the answers and displays the score	The system evaluates the answers and displays the score		
3	When incorrect answers are submitted	Wrong answer selections	The system displays the achieved score and feedback	The system displays the achieved score and feedback		
4	When the quiz timer expires before submission	Quiz time limit reached	The system automatically submits the quiz attempt	The system automatically submits the quiz attempt		
5	When there are no exercises or	Empty topic content	The system displays "No	The system displays "No		

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	quizzes under a topic		exercises or quizzes available”	exercises or quizzes available”		
6	When a student attempts a previous-year exam problem	Selected exam problem	The system displays the question and allows answer submission	The system displays the question and allows answer submission		
7	When a student successfully completes a quiz	Completed quiz attempt	The system updates the student’s progress and stores the score	The system updates the student’s progress and stores the score		
8	When a student refreshes the page during an active quiz attempt	Browser refresh during quiz	The system restores or preserves the current quiz state	The system restores or preserves the current quiz state		
9	When a student submits multiple attempts for the same quiz	Reattempt same quiz	The system records the new attempt according to quiz rules	The system records the new attempt according to quiz rules		
10	When an inactive student account attempts to access quizzes	Deactivated student account	The system denies access and displays an account status message	The system denies access and displays an account status message		

CHAPTER FIVE-TESTING

Test Scenario name: Ask Questions under Topic

Description: The functionality of Students Asking Questions under specific topics

Table 5-6 Test cases for Ask Questions under Topic

#	Test case	Input Data	Expected Result	Actual Result	P	F
1	When a student submits an empty question	–	An error message “Question field is required” is displayed	An error message “Question field is required” is displayed		
2	When a student submits a question successfully	Valid question text	The system saves and posts the question under the selected topic	The system saves and posts the question under the selected topic		
3	When a student submits a question exceeding the allowed character limit	Long question text exceeding limit	An error message “Question exceeds maximum length” is displayed	An error message “Question exceeds maximum length” is displayed		
4	When a student submits a question without selecting a topic	No topic selected	An error message “Please select a topic” is displayed	An error message “Please select a topic” is displayed		

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5	When a teacher receives a newly submitted question	Student submits valid question	The system sends a notification to the assigned teacher	The system sends a notification to the assigned teacher		
6	When an inactive student attempts to ask a question	Deactivated student account	The system denies access and displays an account status message	The system denies access and displays an account status message		
7	When internet connection is interrupted during question submission	Submission interrupted	The system displays “Failed to submit question. Please try again”	The system displays “Failed to submit question. Please try again”		
8	When a student edits the question before submission	Modified question text	The system updates the edited question content before posting	The system updates the edited question content before posting		
9	When unauthorized users attempt to access the ask question feature	Guest user	The system redirects the user to the login page	The system redirects the user to the login page		
10	When the question is submitted successfully	Valid topic-related question	The question becomes visible in the “Ask Question” section	The question becomes visible in the “Ask Question” section		

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